Question No : 1 of 52 Marks: 1 (Budgeted Time 1 Min)

The parameter of the chi- square distribution is.....

Edited By Mahar Afaq Safdar Muhammadi

Answer (Please select your correct option)

VuAnswers.com

- v
- $v-1$
- $v-2$
- $v-p$

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Question No : 2 of 52 Marks: 1 (Budgeted Time 1 Min)

The mean of the F-distribution is:

Answer (Please select your correct option)

VuAnswers.com

- $\frac{v_1}{v_1 - 2}$ $for v_1 \rangle 2$
- $\frac{v_2}{v_2 - 2}$ $for v_2 \rangle 2$
- $\frac{v_1}{v_1 - 2}$ $for v_1 \geq 2$
- $\frac{v_2}{v_2 - 2}$ $for v_1 \leq 2$

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Question No : 3 of 52 Marks: 1 (Budgeted Time 1 Min)

The F-distribution always ranges from:

Answer (Please select your correct option)

VuAnswers.com

- 0 to 1
- 0 to -∞
- -∞ to +∞
- 0 to +∞

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Question No : 4 of 52 Marks: 1 (Budgeted Time 1 Min)

An expected value of a random variable is equal to:

Answer (Please select your correct option)

VuAnswers.com

- Variance
- Mean
- Standard deviation
- Covariance

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Question No : 5 of 52 Marks: 1 (Budgeted Time 1 Min)

When f(x) is continuous probability function, then $P(X = 1)$ is:

Answer (Please select your correct option)

VuAnswers.com

- 1
- ∞
- $-\infty$
- 0

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Question No : 6 of 52 Marks: 1 (Budgeted Time 1 Min)

Rumour has reached the Trading Standards Officer that the manufacturer ABC is deliberately underfilling his cartons of orange juice. It is decided that a sample should be taken to check this claim. The stated contents on the carton are 100 ml on the average, then the alternative hypothesis is:

Answer (Please select your correct option)

VuAnswers.com

- $H_1 : \mu = 100$
- $H_1 : \mu > 100$
- $H_1 : \mu < 100$
- $H_1 : \mu \neq 100$

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Question No : 7 of 52 Marks: 1 (Budgeted Time 1 Min)

Which of the following is a characteristics of the normal distribution:

Answer (Please select your correct option)

VuAnswers.com

- It is a skewed distribution
- It is bell-shaped
- It is not asymptotic
- It is leptokurtic

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Question No : 8 of 52 Marks: 1 (Budgeted Time 1 Min)

For the given poisson distribution $P(X=1)=\frac{e^{-0.135}0.135^1}{1!}$ the mean value is :

Answer (Please select your correct option)

VuAnswers.com

- $e^{-0.135}$
- -0.135
- 0.135
- 1

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Question No : 9 of 52 Marks: 1 (Budgeted Time 1 Min)

In normal distribution β_2 = ... :

Answer (Please select your correct option)

VuAnswers.com

- 1
- 2
- 3
- 0

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Question No : 10 of 52 Marks: 1 (Budgeted Time 1 Min)

In normal distribution, the quartile deviation Q.D = ... :

Answer (Please select your correct option)

VuAnswers.com

- 0.5σ
- 0.75σ
- 0.7979σ
- 0.6745σ

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Question No : 11 of 52 Marks: 1 (Budgeted Time 1 Min)

A good way to get a small standard error is to use a _____.

logical quiez not sure

VuAnswers.com

Answer (Please select your correct option)

- Repeated sampling
- Small sample

mind it not sure

- Large sample
- Large population

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Question No : 12 of 52 Marks: 1 (Budgeted Time 1 Min)

The difference between the largest and the smallest data values is called the

xm-x0

Answer (Please select your correct option)

VuAnswers.com

- Variance
- Interquartile range
- Range
- Coefficient of variation

xm=max rane x0= min range

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Question No : 13 of 52 Marks: 1 (Budgeted Time 1 Min)

Which is appropriate average for finding the average speed of a car:

Answer (Please select your correct option)

VuAnswers.com

- Mean
- Geometric mean
- Harmonic mean
- Weighted mean

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Question No : 14 of 52 Marks: 1 (Budgeted Time 1 Min)

Which one is the formula of mid range:

Answer (Please select your correct option)

VuAnswers.com

- $x_m - x_0$
- $x_0 - x_m$
- $\frac{x_0 - x_m}{2}$
- $\frac{x_0 + x_m}{2}$

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Question No : 15 of 52 Marks: 1 (Budgeted Time 1 Min)

Which one of the following is a meso-kurtic curve?

Answer (Please select your correct option)

VuAnswers.com

- Negatively skewed
- Positively skewed
- J-shaped
- Normal

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Question No : 16 of 52 Marks: 1 (Budgeted Time 1 Min)

If you draw all possible samples from some population, calculate the mean for each of the sample and construct the probability distribution of the sample means, what would you have?

Answer (Please select your correct option)

VuAnswers.com

- A population distribution
- A sample distribution
- A sampling distribution
- A parameter distribution

Logical not sure

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Question No : 17 of 52 Marks: 1 (Budgeted Time 1 Min)

By definition f(y | x) = _____

Answer (Please select your correct option)

VuAnswers.com

- $f(y)$
- $f(x,y)$
- $\frac{f(x,y)}{h(x)}$
- $\frac{f(x,y)}{h(y)}$

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Question No : 18 of 52 Marks: 1 (Budgeted Time 1 Min)

The critical region for $H_1 : \mu > \mu_0$ *when* $\alpha = 0.01$ is:

not sure

Answer (Please select your correct option) VuAnswers.com

- $z > z_{0.01}$
- $|z| > z_{0.01}$
- $z < -z_{0.05}$
- $|z| > z_{0.05}$

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Question No : 19 of 52 Marks: 1 (Budgeted Time 1 Min)

For degree of freedom $\nu > 2$ the variance of t-distribution is always:

Answer (Please select your correct option)

VuAnswers.com

- Greater than zero
- Less than one
- Equal to one
- Greater than one

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Question No : 20 of 52 Marks: 1 (Budgeted Time 1 Min)

Ideally, the width of confidence interval should be:

Answer (Please select your correct option)

VuAnswers.com

- 0
- 1
- 99
- 100

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Question No : 21 of 52 Marks: 1 (Budgeted Time 1 Min)

If the sampling distribution of $\bar{X}$ is normal, we would expect 99% of the sample means to be within the interval:

Answer (Please select your correct option)

VuAnswers.com

- $\mu_{\bar{x}} \pm 2\sigma_{\bar{x}}$
- $\mu_{\bar{x}} \pm 1.96\sigma_{\bar{x}}$
- $\mu_{\bar{x}} \pm 2.58\sigma_{\bar{x}}$
- $\mu_{\bar{x}} \pm \sigma_{\bar{x}}$

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Question No : 22 of 52 Marks: 1 (Budgeted Time 1 Min)

Which one of the formula will be used to find out the confidence interval for μ, when population variance unknown and sample size is large?

Answer (Please select your correct option)

VuAnswers.com

- $\bar{x} \pm Z_{\alpha/2} \frac{s}{\sqrt{n}}$
- $\bar{x} \pm t_{\alpha/2(v)} \frac{s}{\sqrt{n}}$
- $\bar{x} \pm t_{\alpha/2(v)} \frac{\sigma}{\sqrt{n}}$
- $\bar{x} \pm Z_{\alpha/2} \frac{s}{\sqrt{n-1}}$

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Question No : 23 of 52 Marks: 1 (Budgeted Time 1 Min)

If $\bar{X}$ is the mean of the n observations, then which test statistic will be used to calculate the confidence limits of the population variance σ^2?

Answer (Please select your correct option)

VuAnswers.com

- Z-statistic
- T-statistic
- χ^2-statistics
- F-statistics

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Question No : 24 of 52 Marks: 1 (Budgeted Time 1 Min)

In Statistics, we have MSE which is abbreviation of:

Answer (Please select your correct option)

VuAnswers.com

- Mean square error
- Measured square error
- Medical screening exam
- Major sampling error

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Question No : 25 of 52 Marks: 1 (Budgeted Time 1 Min)

In the test of goodness of fit, the _____ is used as a test statistic.

Answer (Please select your correct option)

VuAnswers.com

- F
- t
- Z
- χ^2

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Question No : 26 of 52 Marks: 1 (Budgeted Time 1 Min)

If there are 8 treatments with 6 blocks in a randomized completed block design then what are the degrees of freedom for treatments?

v=n-1
v repersent the
degree of fredom

VuAnswers.com

Answer (Please select your correct option)

- 5
- 4
- 6
- 7

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Question No : 27 of 52 Marks: 1 (Budgeted Time 1 Min)

What factor determines the shape of the t-distribution?

Answer (Please select your correct option) VuAnswers.com

- Degree of freedom
- Critical value
- Frequency of data
- Probability

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Question No : 28 of 52 Marks: 1 (Budgeted Time 1 Min)

If X and Y are random variables, then $E(X-Y)$ is equal to:

Answer (Please select your correct option)

VuAnswers.com

- $E(X)+E(Y)$
- $E(X)-E(Y)$
- $X-E(Y)$
- $E(X)-Y$

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Question No : 29 of 52 Marks: 1 (Budgeted Time 1 Min)

Two continuous r.v.'s X and Y are said to be independent if and only if :

Answer (Please select your correct option)

VuAnswers.com

- f(x,y)= g(x) h(y)
- f(x,y) ≠ g(x) h(y)
- f(x,y)> g(x) h(y)
- f(x,y)<g(x) h(y)

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Question No : 30 of 52 Marks: 1 (Budgeted Time 1 Min)

The lottery tickets issued for the purpose of money-making follows a:

Answer (Please select your correct option)

VuAnswers.com

- Normal distribution
- Discrete uniform distribution
- Binomial distribution
- Hypergeometric distribution

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Question No : 31 of 52 Marks: 1 (Budgeted Time 1 Min)

Uniform distribution is defined by:

Answer (Please select your correct option)

VuAnswers.com

- Largest value
- Largest and smallest value
- Smallest value
- Central value

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Question No : 32 of 52 Marks: 1 (Budgeted Time 1 Min)

If an estimator gets closer to the population parameter by increasing sample size then it is known as:

Answer (Please select your correct option)

VuAnswers.com

- Consistent estimator
- Sufficient estimator
- Efficient estimator
- Unbiased estimator

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Question No : 33 of 52 Marks: 1 (Budgeted Time 1 Min)

Quantitative variable is further divided into:

Answer (Please select your correct option)

VuAnswers.com

- Continuous variable
- Discrete variable
- Continuous & Discrete variable
- None of the above

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Question No : 34 of 52 Marks: 1 (Budgeted Time 1 Min)

Color of the dress is the example of

Answer: (Please select your correct option)

VuAnswers.com

- Qualitative data
- Quantitative data
- Continuous data
- Discrete data

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Question No : 35 of 52 Marks: 1 (Budgeted Time 1 Min)

Which one is commonly called a bell shaped distribution?

Answer (Please select your correct option)

VuAnswers.com

- Symmetrical
- Bimodal
- Skewed
- U shaped

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Question No : 36 of 52 Marks: 1 (Budgeted Time 1 Min)

A fair coin is tossed three times, the probability that at least one head appear is:

Answer (Please select your correct option)

VuAnswers.com

- 1/2
- 1/8
- 6/8
- 7/8

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Question No : 37 of 52 Marks: 1 (Budgeted Time 1 Min)

The probability of simultaneous occurrence of two events is called:

Answer (Please select your correct option)

VuAnswers.com

- Subjective probability
- Conditional probability
- Joint probability
- Prior probability

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Question No : 38 of 52 Marks: 1 (Budgeted Time 1 Min)

What is the stem part of 243:

Answer (Please select your correct option)

VuAnswers.com

- 3
- 43
- 23
- 24

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Question No : 39 of 52 Marks: 1 (Budgeted Time 1 Min)

A numerical value used as a summary measure for a sample, such as sample mean, is known as a :

Answer (Please select your correct option)

VuAnswers.com

- Population Parameter
- Sample Parameter
- Sample Statistic
- Population mean

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Question No : 40 of 52 Marks: 1 (Budgeted Time 1 Min)

Given the series 1,2,1,1,2,2,2,2,3,4,5,3,2,3,1,4,2,3. Which one of the following is mode of the given seires:

Answer (Please select your correct option)

VuAnswers.com

- 4
- 3
  3
- 2
  3
  2
- 1
  3
  2*

repeted values is called

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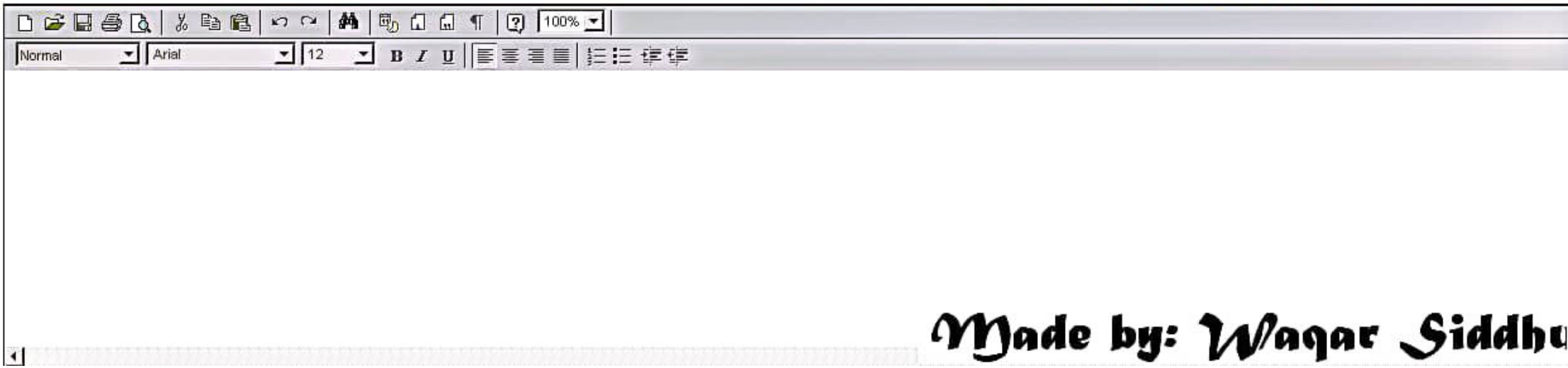
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Question No : 41 of 52 Marks: 2 (Budgeted Time 4 Min)

Name the measures of dispersion that are not based on all the values.

Answer (Please click here to Add Answer)

VuAnswers.com



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Question No : 42 of 52 Marks: 2 (Budgeted Time 4 Min)

When we use two-tailed test?

Answer (Please click here to Add Answer)

VuAnswers.com

Normal Arial 12

test :
A hypothesis test in which rejection of the null hypothesis occurs for values of the test statistic in either tail of the sampling distribution.

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Question No : 43 of 52 Marks: 2 (Budgeted Time 4 Min)

For a sample data n = 15, calculate $t_{\frac{\alpha(v)}{2}}$ for $\alpha = 0.10$.

Answer (Please click here to Add Answer)

VuAnswers.com

Normal Arial 12 100%

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Question No : 44 of 52 Marks: 2 (Budgeted Time 4 Min)

Suppose we want to determine the proportions of smokers and non smoker in a city? In this situation what type of distribution we can use?

Answer (Please click here to Add Answer)

VuAnswers.com

Normal Arial 12 B I U 100%

following proportions, we are dealing with a BINOMIAL situation:

- **Proportion of smokers in a city smoker → success, non-smokers → failure.**
- **Proportion of literates in a community → literacy rate, literate → success, illiterate → failure.**
- **Proportion of males in a city → *sex ratio*).**

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Question No : 45 of 52 Marks: 3 (Budgeted Time 6 Min)

If mean of a distribution is 20 and standard deviation is 2. Find out $\mu \pm 2\sigma$ limits by applying empirical rule. What percent of data will lies between these two limits?

Answer (Please click here to Add Answer)

VuAnswers.com

Normal Arial 12 B I U 100%

calculate app keroo hun

According to this empirical rule:

- **Approximately 68% of the measurements will fall within 1 standard deviation of the mean, i.e. within the interval (–X – S, –X + S)**
- **Approximately 95% of the measurements will fall within 2 standard deviations of the mean, i.e. within the interval (–X – 2S, –X + 2S).**
- **Approximately 100% (practically all) of the measurements will fall within 3 standard deviations of the mean, i.e. within the interval (–X – 3S, –X + 3S).**

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Question No : 46 of 52 Marks: 3 (Budgeted Time 6 Min)

Suppose X is a random variable having Poisson distribution with its parameter value 3, find value of P(X=1).

Answer (Please click here to Add Answer)

VuAnswers.com

Normal Arial 12 B I U 100%

The Poisson distribution has only one parameter μ > 0.

μ=3 X=1)

$$\lim_{\substack{n\to\infty \\ p\to 0}} b(x;n,p) = \frac{e^{-\mu}\mu^{x}}{x!}, \quad x = 0,1,2,\ldots,\infty$$

where e = 2.71828.

p is 0.05 or less,
&
n is 20 or more,

calculate easly now..



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Question No : 47 of 52 Marks: 3 (Budgeted Time 6 Min)

If X = 341, n = 634, p_0 = 0.50 then find the z-test statistic for proportion.

Answer (Please click here to Add Answer)

VuAnswers.com

Normal Arial 12 B I U 100%

Step 3:
Test statistic:

$$Z = \frac{X \pm \frac{1}{2} - n\,p_0}{\sqrt{n\,p_0(1-p_0)}}$$

Step 4:
Computation:

Here np0 = 634 (0.50) = 317
and X = 341
Hence X > np0 so use X - ½

$$\text{So} \quad Z = \frac{341 - \frac{1}{2} - 317}{\sqrt{634(0.50)(0.50)}} = \frac{23.5}{12.59}$$

$$= 1.87$$

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Question No : 48 of 52 Marks: 3 (Budgeted Time 6 Min)

A random sample of size n is drawn from normal population with mean 5 and variance σ^2
If s=2.5, $\bar{x}$ =7and t=3, then what is the values of n?

Answer (Please click here to Add Answer)

VuAnswers.com

Normal Arial 12 100%

The test-statistic to be used is

$$t = \frac{\overline{X} - \mu_0}{s/\sqrt{n}}$$

hare mue = 5...

ider uder lai aj ker laiana now. n ko

ans shid 1.92 ai ga aprox 2 i htink not sure :) becoz stat ae :p

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Question No : 49 of 52 Marks: 5 (Budgeted Time 10 Min)

A random variable X is normally distributed with $\mu = 50$ *and* $\sigma^2 = 25$. Find the probability of X larger than 54.

Answer (Please click here to Add Answer)

VuAnswers.com

The test-statistic to be used is

$$Z = \frac{\overline{X} - \mu_0}{\text{sigma}}$$

hare give sigma sqr, take under root

sigma= 5 mue alsi give and X= 54 we find X>54

(54-50)/5= 4/5

its half ques see these type of ques 70 % slove now.. 30 % see table etc

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Question No : 50 of 52 Marks: 5 (Budgeted Time 10 Min)

Find the coefficient of variation (C.V) for the following price of a commodity.
Price (X): 8, 13, 18, 23, 30

Answer (Please click here to Add Answer)

VuAnswers.com

Normal Arial 12 100%

COEFFICIENT OF VARIATION

$$C.V. = \frac{S}{\bar{X}} \times 100$$

Shortcut Formula for Ungroup data

$$S = \sqrt{\left\{\frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2\right\}}$$

mean ka farmula laga laian X bar ki jaga.... =X/n
small n is 5

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Question No : 51 of 52 Marks: 5 (Budgeted Time 10 Min)

The given data is $n = 1150, x = 450, p = 0.39, H_0 : p_0 = 0.3, \alpha = 0.01$

Test the stated hypothesis. (*Use table value of* $z = \pm 2.58$)

Answer (Please click here to Add Answer)

VuAnswers.com



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Question No : 52 of 52 Marks: 5 (Budgeted Time 10 Min)

Given, $n_1 = n_2 = 16, s_1^2 = 50, s_2^2 = 16$, Construct a 90% confidence interval for the variance ratio σ_1^2 / σ_2^2 .

Answer (Please click here to Add Answer)

VuAnswers.com

Normal Arial 12 100%

regards i have no idea abou F test and this ques in the F test...
remmber me in ur pryaer after 3 hr slove 1 paper.. :)

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Question No : 1 of 52 Marks: 1 (Budgeted Time 1 Min)

P (A or B) =P (A) +P (B), then A and B are:

Edited By Mahar Afaq Safdar Muhammadi

Answer (Please select your correct option) VuAnswers.com

- ● Mutually exclusive events
- ○ Independent events
- ○ Exhaustive events
- ○ Equally likely events

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Question No : 2 of 52 Marks: 1 (Budgeted Time 1 Min)

First moment about origin is always equals to:

VuAnswers.com

Answer (Please select your correct option)

- Mean
- Variance
- Standard Deviation
- Zero

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Question No : 3 of 52 Marks: 1 (Budgeted Time 1 Min)

$E(4X + 5) =$_____

Answer (Please select your correct option)

VuAnswers.com

- 16 E (X)
- 16 E (X) + 5
- 12 E (X)
- 4 E (X) + 5

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Question No : 4 of 52 Marks: 1 (Budgeted Time 1 Min)

When two coins are tossed the probability of at least one head is:

VuAnswers.com

Answer (Please select your correct option)

- 1/4
- 3/4
- 2/4
- 4/4

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Question No : 5 of 52 Marks: 1 (Budgeted Time 1 Min)

If σ^2 is unknown, then we use Z-test if the sample size is:

Answer (Please select your correct option)

VuAnswers.com

- $n \geq 30$
- n<30
- n=25
- n=20

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Question No : 6 of 52 Marks: 1 (Budgeted Time 1 Min)

When a coin is tossed 3 times, the probability of 3 tails is:

VuAnswers.com

Answer (Please select your correct option)

- (●) 1/8
- ( ) 2/4
- ( ) 3/8
- ( ) 2/8

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Question No : 7 of 52 Marks: 1 (Budgeted Time 1 Min)

The F-distribution has …… parameter.

VuAnswers.com

Answer (Please select your correct option)

- One
- No
- Two
- Three

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Question No : 8 of 52 Marks: 1 (Budgeted Time 1 Min)

Which one of the following provides the basis for hypothesis testing?

Answer (Please select your correct option)

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- Null hypothesis
- Alternative hypothesis
- Critical value
- Test-statistic

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Question No : 9 of 52 Marks: 1 (Budgeted Time 1 Min)

Rumour has reached the Trading Standards Officer that the manufacturer ABC is deliberately underfilling his cartons of orange juice. It is decided that a sample should be taken to check this claim. The stated contents on the carton are 100 ml on the average, then the alternative hypothesis is:

VuAnswers.com

Answer (Please select your correct option)

- $H_1 : \mu = 100$
- $H_1 : \mu > 100$
- $H_1 : \mu < 100$
- $H_1 : \mu \neq 100$

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Question No : 10 of 52 Marks: 1 (Budgeted Time 1 Min)

Rumour has reached the Trading Standards Officer that the manufacturer ABC is deliberately underfilling his cartons of orange juice. It is decided that a sample should be taken to check this claim. The stated contents on the carton are 100 ml on the average, then the null hypothesis is:

Answer (Please select your correct option)

VuAnswers.com

- ● $H_0 : \mu = 100$
- ○ $H_0 : \mu > 100$
- ○ $H_0 : \mu < 100$
- ○ $H_0 : \mu \neq 100$

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Question No : 11 of 52 Marks: 1 (Budgeted Time 1 Min)

By definition $f(x_i \mid y_j) =$ _____

Answer (Please select your correct option)

VuAnswers.com

- ● $\frac{f(x_i, y_j)}{h(y_j)}$
- ○ $\frac{f(x_i, y_j)}{h(x_i)}$
- ○ $f(x_i, y_j)$
- ○ $f(y_j)$

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Question No : 12 of 52 Marks: 1 (Budgeted Time 1 Min)

The test statistic used in analysis of variance procedure follow the :

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Answer (Please select your correct option)

- χ^2 -distribution.
- T-distribution.
- Z-distribution.
- F-distribution.

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Question No : 13 of 52 Marks: 1 (Budgeted Time 1 Min)

Which one of the following is the most common example of a situation for which the main parameter of interest is a population proportion?

Answer (Please select your correct option)

VuAnswers.com

- An observational study
- A normal experiment
-
- A binomial experiment

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Question No : 14 of 52 Marks: 1 (Budgeted Time 1 Min)

An estimator which has the smallest standard error among all unbiased estimators fulfills the property of _____.

Answer (Please select your correct option)

VuAnswers.com

- Unbiasedness
- Efficiency
- Consistency
- Sufficiency

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Question No : 14 of 52 Marks: 1 (Budgeted Time 1 Min)

An estimator which has the smallest standard error among all unbiased estimators fulfills the property of _____.

VuAnswers.com

Answer (Please select your correct option)

- Unbiasedness
- Efficiency
- Consistency
- Sufficiency

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Question No : 15 of 52 Marks: 1 (Budgeted Time 1 Min)

Which of the following can never be taken as the probability of an event?

Answer (Please select your correct option)

VuAnswers.com

- 1
- 0
- 0.5
- -0.5

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Question No : 16 of 52 Marks: 1 (Budgeted Time 1 Min)

A set is any well-defined collection of

Answer (Please select your correct option)

VuAnswers.com

- Positive Objects
- Negative Objects
- Same Objects
- Distinct Objects

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Question No : 17 of 52 Marks: 1 (Budgeted Time 1 Min)

Measure of dispersion is used to calculate the:

Answer (Please select your correct option)

VuAnswers.com

- Central value
- Highest value
- Lowest value
- Scattered value

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Question No : 18 of 52 Marks: 1 (Budgeted Time 1 Min)

If X and Y are independent variables then Var (X-Y) = _____

Answer (Please select your correct option)

VuAnswers.com

- Var(X) - Var(Y)
- Var(X) + Var(Y)
- Var(X+Y)
- Var(X) ×Var(Y)

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Question No : 19 of 52 Marks: 1 (Budgeted Time 1 Min)

How the standard error is decreased :

Answer (Please select your correct option)

VuAnswers.com

- By decreasing the sample size
- By decreasing the mean
- By increasing the standard deviation
- By increasing the sample size

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Question No : 20 of 52 Marks: 1 (Budgeted Time 1 Min)

The total number of samples when sampling is done with replacement is equal to:

Answer (Please select your correct option)

VuAnswers.com

- ● N^n
- ○ C_n^N
- ○ $\frac{N-n}{N-1}$
- ○ 1

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Question No : 21 of 52 Marks: 1 (Budgeted Time 1 Min)

If you draw all possible samples from some population, calculate the mean for each of the sample and construct the probability distribution of the sample means, what would you have?

Answer (Please select your correct option)

VuAnswers.com

- A population distribution
- A sample distribution
- A sampling distribution
- A parameter distribution

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Question No : 22 of 52 Marks: 1 (Budgeted Time 1 Min)

The conditional probability function $f(x \mid 1)$ = _____

Answer (Please select your correct option)

VuAnswers.com

- $f(1,1)$
- $f(x,1)$
- $\frac{f(x,1)}{h(1)}$
- $\frac{f(x,1)}{h(x)}$

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Question No : 23 of 52 Marks: 1 (Budgeted Time 1 Min)

For $\alpha = 0.01$, the critical values of z for two tailed test are equal to:

Answer (Please select your correct option)

VuAnswers.com

- -2.58 and +2.58
- -2.33 and +2.33
- -1.645 and +1.645
- -1.96 and + 1.96

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Question No : 24 of 52 Marks: 1 (Budgeted Time 1 Min)

A deserving player is not selected in the team is an example of:

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Answer (Please select your correct option)

- Type I error
- Type II error
- Correct decision
- No information regarding this

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Question No : 25 of 52 Marks: 1 (Budgeted Time 1 Min)

"A point estimate plus/minus a few times the standard error of that estimate". This statement represents:

Answer (Please select your correct option)

VuAnswers.com

- Confidence interval
- Critical region
- Acceptance region
- Critical value

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Question No : 26 of 52 Marks: 1 (Budgeted Time 1 Min)

The proportion of males in Pakistan is at least 0.48, the alternative hypothesis H_1 is

VuAnswers.com

Answer (Please select your correct option)

- $P \leq 0.48$
- $P = 0.48$
- $P < 0.48$
- $P \geq 0.48$

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Question No : 27 of 52 Marks: 1 (Budgeted Time 1 Min)

If $\bar{X}$ is the mean of the n observations, then which test statistic will be used to calculate the confidence limits of the population variance σ^2?

VuAnswers.com

Answer (Please select your correct option)

- Z-statistic
- T-statistic
- χ^2-statistics

not sure

- F-statistics

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Question No : 28 of 52 Marks: 1 (Budgeted Time 1 Min)

To find the confidence interval for the ratio of two variances, we use

Answer (Please select your correct option)

VuAnswers.com

- (●) F-Distribution
- ( ) Z-Distribution
- ( ) Chi-square-Distribution
- ( ) t-Distribution

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Question No : 29 of 52 Marks: 1 (Budgeted Time 1 Min)

The Chi- Square distribution is continuous distribution ranging from:

VuAnswers.com

Answer (Please select your correct option)

- $-\infty \leq \chi^2 \leq \infty$
- $-\infty \leq \chi^2 \leq 1$
- $-\infty \leq \chi^2 \leq 0$
- $0 \leq \chi^2 \leq \infty$

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Question No : 30 of 52 Marks: 1 (Budgeted Time 1 Min)

In Statistics, we have MSE which is abbreviation of:

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Answer (Please select your correct option)

- (●) Mean square error
- ( ) Measured square error
- ( ) Medical screening exam
- ( ) Major sampling error

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Question No : 31 of 52 Marks: 1 (Budgeted Time 1 Min)

In a binomial experiment the total number of trials are:

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Answer (Please select your correct option)

- Fixed in advance
- Changeable according to situation
- Unpredictable
- Not independent

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Question No : 32 of 52 Marks: 1 (Budgeted Time 1 Min)

The lottery tickets issued for the purpose of money-making follows a:

Answer (Please select your correct option)

VuAnswers.com

- Normal distribution
- Discrete uniform distribution
- Binomial distribution
- Hypergeometric distribution

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Question No : 33 of 52 Marks: 1 (Budgeted Time 1 Min)

Which of the following value could not represent a coefficient of correlation?

Answer (Please select your correct option)

VuAnswers.com

- r = 0.99
- r = 1.09
- r = -0.73
- r = -1

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Question No : 34 of 52 Marks: 1 (Budgeted Time 1 Min)

In a one way ANOVA test there are 5 observations in each of three treatments. The degrees of freedom for the treatments is:

VuAnswers.com

Answer (Please select your correct option)

- 5
- 3
- 1
- 2

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Question No : 35 of 52 Marks: 1 (Budgeted Time 1 Min)

If P(B|A) = 0.25 and $P(A \cap B) = 0.20$, then P(A) =

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Answer (Please select your correct option)

- 0.05
- 0.80
- 0.95
- 0.75

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Question No : 36 of 52 Marks: 1 (Budgeted Time 1 Min)

If a random variable X denotes the number of heads when three distinct coins are tossed, the X assumed the values:

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Answer (Please select your correct option)

- (●) 0,1,2,3
- ( ) 1,3,3,1
- ( ) 1, 2, 3
- ( ) 3, 2

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Question No : 37 of 52 Marks: 1 (Budgeted Time 1 Min)

When f(x) is continuous probability function, then $P(X = 2)$ is:

VuAnswers.com

Answer (Please select your correct option)

- 1
- 0.5
- 0
- 0.25

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Question No : 38 of 52 Marks: 1 (Budgeted Time 1 Min)

Atmosphere pressure is the example of:

VuAnswers.com

Answer (Please select your correct option)

- Constant
- Qualitative variable
- Quantitative variable
- None of the above

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Question No : 39 of 52 Marks: 1 (Budgeted Time 1 Min)

Which of the following scale has true zero point?

Answer (Please select your correct option)

VuAnswers.com

- (●) Ratio Scale
- ( ) Interval scale
- ( ) Nominal scale
- ( ) Ordinal scale

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Question No : 40 of 52 Marks: 1 (Budgeted Time 1 Min)

Given the series 1,2,1,1,2,2,2,2,3,4,5,3,2,3,1,4,2,3. Which one of the following is mode of the given seires:

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Answer (Please select your correct option)

- 4
- 3
  3
- 2
  3
  2
- 1
  3
  2*

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Question No : 41 of 52 Marks: 2 (Budgeted Time 4 Min)

Describe the Poisson distribution as the limiting form of the binomial distribution.

Answer (Please click here to Add Answer)

VuAnswers.com



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Question No : 42 of 52 Marks: 2 (Budgeted Time 4 Min)

Define Disjoint Sets.

Answer (Please click here to Add Answer)

VuAnswers.com

Normal Arial 12 100%

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Question No : 43 of 52 Marks: 2 (Budgeted Time 4 Min)

What is acceptance region?

Answer (Please click here to Add Answer)

VuAnswers.com

Normal Arial 12 100%

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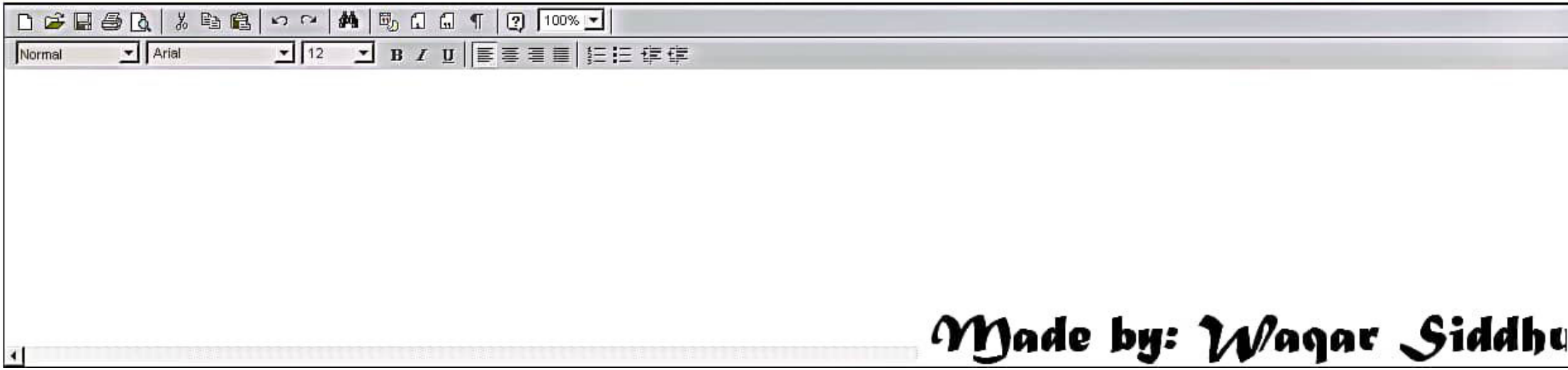
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Question No : 44 of 52 Marks: 2 (Budgeted Time 4 Min)

If there are K treatments and R rows in a Randomized Complete Block Ddesign then calculate the total number of experimental units used.

Answer (Please click here to Add Answer)

VuAnswers.com



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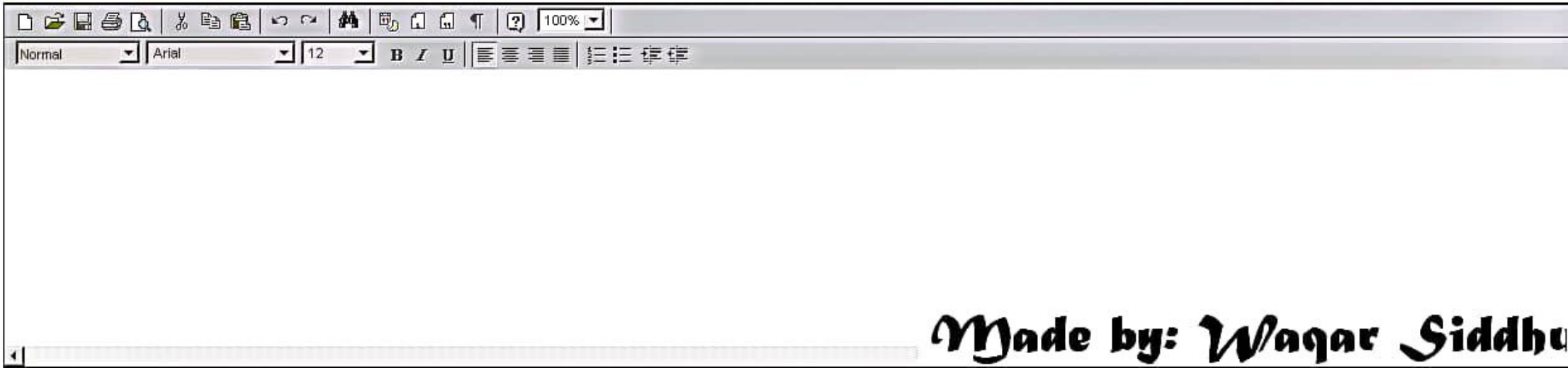
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Question No : 45 of 52 Marks: 3 (Budgeted Time 6 Min)

In how many ways a three-person committee can be formed from a group of ten persons? (Use the formula)

Answer (Please click here to Add Answer)

VuAnswers.com



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Question No : 46 of 52 Marks: 3 (Budgeted Time 6 Min)

From the given data, calculate mean and standard deviation of sampling distribution of mean if the sampling is done with replacement.

$N = 120, n = 64, \mu = 50, \sigma = 2$

Answer (Please click here to Add Answer)

VuAnswers.com

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Question No : 47 of 52 Marks: 3 (Budgeted Time 6 Min)

Construct 90% confidence interval for the difference in means $\mu_1 - \mu_2$ in case of paired observations, where

$\overline{d} = 1.8$, $s_d = 1.32$, $t_{0.05(9)} = 1.833$

Answer (Please click here to Add Answer)

VuAnswers.com

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Question No : 48 of 52 Marks: 3 (Budgeted Time 6 Min)

For a data X : $N(\mu,\sigma^2)$. Two unbiased estimators T_1 and T_2 have following variances

$$V(T_1)=\frac{11\sigma^2}{25}$$

$$V(T_2)=\frac{9\sigma^2}{25}$$

Answer (Please click here to Add Answer)

VuAnswers.com

Normal Arial 12 100%

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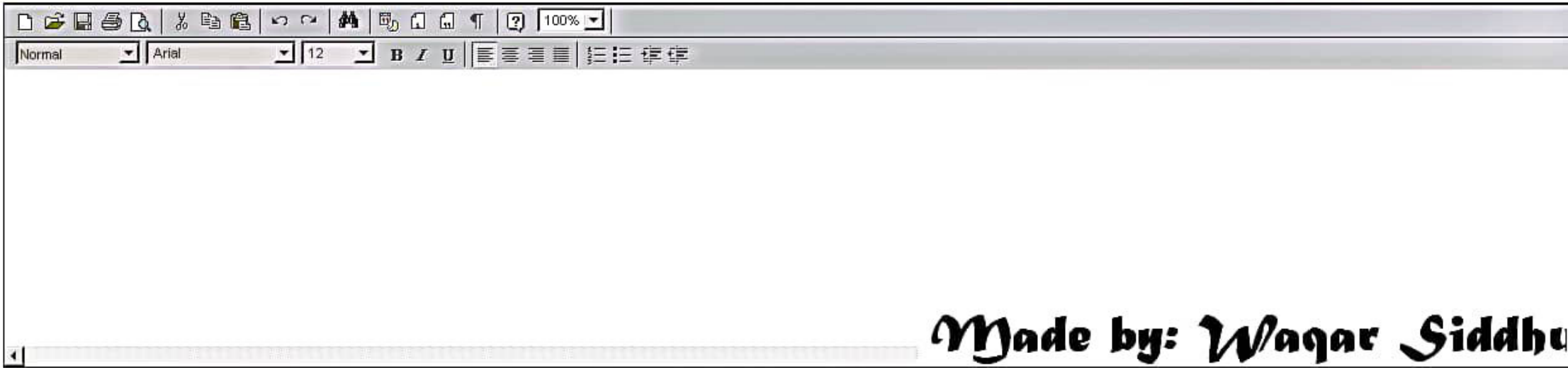
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Question No : 49 of 52 Marks: 5 (Budgeted Time 10 Min)

A man travels on car from Lahore to karachi on Motor way in 8 stages of equal intervals. The speed of the car in the various stages was observed to be 110, 116, 120, 114, 115,112, 120, 117 kilometers per hour. Find the average speed at which the car travels.

Answer (Please click here to Add Answer)

VuAnswers.com



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Question No : 50 of 52 **Marks: 5 (Budgeted Time 10 Min)**

A random sample of size three is drawn without replacement from the population consisting of four numbers 4, 5, 5, 7. Sampling distribution of sample means is calculated as below;

Sample Means $(\bar{X})$	$f(\bar{X})$
14/3	1/4
16/3	2/4
17/3	1/4

Answer (Please click here to Add Answer)

VuAnswers.com

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Question No : 51 of 52 Marks: 5 (Budgeted Time 10 Min)

If $n = 1150, x = 450, p = 0.39, H_0 : p_0 = 0.5, and\ \alpha = 0.05$

Test the stated hypothesis.

(*Use table value of* $z = \pm 1.96$)

Answer (Please click here to Add Answer)

VuAnswers.com

Normal Arial 12 100%

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Question No : 52 of 52 **Marks: 5 (Budgeted Time 10 Min)**

A personal manager is interested in trying to determine whether absenteeism is greater on some specific day of the week or not? His records for the past year show the following sample distributions.

Day of the week:	Monday	Tuesday	Wednesday	Thursday	Friday
No. of absentees:	66	57	54	48	75

Test goodness of fit.

Where test statistic $chi-square = \sum \frac{(o_i - e_i)^2}{e_i} = 7.50$

Answer (Please click here to Add Answer)

VuAnswers.com

Normal Arial 12 B I U 100%

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Var(4X + 5) =_____

Edited By Mahar Afaq Safdar Muhammadi

Answer (Please select your correct option)

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- 16 Var (X)
- 16 Var (X) + 5
- 4 Var (X) + 5
- 12 Var (X)

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When f(x) is continuous probability function, then $P(X = 1)$ is:

Answer (Please select your correct option)

VuAnswers.com

- 1
- ∞
- $-\infty$
- 0

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The hyper geometric random variable is a(an):

Answer (Please select your correct option)

VuAnswers.com

- Continuous variable
- Discrete variable
- Undefined
- Independent variable

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The degrees of freedom for a t-test with sample size 'n' is:

Answer (Please select your correct option)

- n-1
- n+1
- n-2
- n+2

VuAnswers.com

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Rumour has reached the Trading Standards Officer that the manufacturer ABC is deliberately underfilling his cartons of orange juice. It is decided that a sample should be taken to check this claim. The stated contents on the carton are 100 ml on the average, then the null hypothesis is:

Answer (Please select your correct option)

VuAnswers.com

- $H_0 : \mu = 100$
- $H_0 : \mu > 100$
- $H_0 : \mu < 100$
- $H_0 : \mu \neq 100$

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Assuming that following is a probability distribution, then what is the value of 'a':

X	1	2	3
P(X)	0.1	a	0.1

Answer (Please select your correct option)

VuAnswers.com

- 0.6
- 0.8
- 0.2
- 0.4

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In normal distribution β_1 = ... :

Answer (Please select your correct option)

- 0
- 1
- 2
- 3

VuAnswers.com

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A good way to get a small standard error is to use a _____.

Answer (Please select your correct option)

VuAnswers.com

- Repeated sampling
- Small sample
- Large sample
- Large population

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A randomly selected sample of 400 students at university was asked whether or not they will participate in politics. Forty-six percent of the 400 student surveyed answered "yes". Which one of the following statement about number 46% is correct?

Answer (Please select your correct option)

VuAnswers.com

- It is a sample statistic.
- It is a population parameter.
- It is a margin of error.
- It is a standard error. not sure

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If you draw all possible samples from some population, calculate the mean for each of the sample and construct the probability distribution of the sample means, what would you have?

Answer (Please select your correct option)

VuAnswers.com

- A population distribution
- A sample distribution
- A sampling distribution
- A parameter distribution

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The conditional probability function $f(x \mid 1)$ = _____

Answer (Please select your correct option)

VuAnswers.com

- $f(1,1)$
- $f(x,1)$
- $\frac{f(x,1)}{h(1)}$
- $\frac{f(x,1)}{h(x)}$

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If f (x, y) is bivariate probability density function of continuous random variables X and Y then marginal density function of y i.e. $h(y)$ is:

Answer (Please select your correct option)

VuAnswers.com

- $\int_{-\infty}^{\infty} f(x,y)\,dx$
- $\int_{-\infty}^{\infty} f(x,y)\,dy$
- $\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)dx\,dy$
- $\int_{a}^{b}\int_{c}^{d} f(x,y)\,dy\,dx$

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_____ is a range of numbers inferred from the sample that has a certain probability of including the population parameter over the long run.

Answer (Please select your correct option)

VuAnswers.com

- Hypothesis
- Lower limit
- Confidence interval **not sure**
- Probability limit

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How many parameter(s) are in t-distribution?

Answer (Please select your correct option)

VuAnswers.com

- 0
- 1
- 2
- 3

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A judge can acquit a guilty person is the example of:

Answer (Please select your correct option)

VuAnswers.com

- Type I error
- Type II error
- Correct decision **not sure**
- No information regarding this

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If the sampling distribution of $\bar{X}$ is normal, we would expect 99% of the sample means to be within the interval.

Answer (Please select your correct option)

VuAnswers.com

- $\mu_{\bar{x}} \pm 2\sigma_{\bar{x}}$
- $\mu_{\bar{x}} \pm 1.96\sigma_{\bar{x}}$
- $\mu_{\bar{x}} \pm 2.58\sigma_{\bar{x}}$
- $\mu_{\bar{x}} \pm \sigma_{\bar{x}}$

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The proportion of males in Pakistan is at least 0.48, the alternative hypothesis H_1 is

Answer (Please select your correct option)

VuAnswers.com

- $P \leq 0.48$
- $P = 0.48$
- $P < 0.48$ **not sure**
- $P \geq 0.48$

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If mean of χ^2 distribution is k then variance will be:

Answer (Please select your correct option)

VuAnswers.com

- k^2
- 2k
- 1/k
- k

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Test Statistics χ^2 is equal to:

Answer (Please select your correct option)

VuAnswers.com

- $\frac{n^2 S^2}{\sigma_0^2}$
- $\frac{nS^2}{\sigma_0^2}$
- $\frac{S^2}{2\sigma_0^2}$
- $\frac{n^2 S^2}{2\sigma_0^2}$

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F- distribution is .....

Answer (Please select your correct option)

VuAnswers.com

- Positively skewed
- Negatively skewed
- Normal
- Symmetrical

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The Chi- Square distribution is continuous distribution ranging from:

Answer (Please select your correct option)

VuAnswers.com

- $-\infty \leq \chi^2 \leq \infty$
- $-\infty \leq \chi^2 \leq 1$
- $-\infty \leq \chi^2 \leq 0$
- $0 \leq \chi^2 \leq \infty$

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In Statistics, we have MSE which is abbreviation of:

Answer (Please select your correct option)

- Mean square error
- Measured square error
- Medical screening exam
- Major sampling error

VuAnswers.com

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The LSD test is applied when the null hypothesis is:

Answer (Please select your correct option)

- Rejected
- Accepted
- Finalized
- Acknowledged

VuAnswers.com

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As the degree of freedom increases, the t-distribution tends to coincide with:

Answer (Please select your correct option)

VuAnswers.com

- Binomial distribution
- Uniform distribution
- Hypergeometric distribution
- Normal distribution

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In a binomial experiment the total number of trials are:

Answer (Please select your correct option)

- Fixed in advance
- Changeable according to situation
- Unpredictable
- Not independent

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What is the probability of drawing a red-queen card from a well shuffled pack of 52 playing cards?

Answer (Please select your correct option)

VuAnswers.com

- 4/52
- 2/52
- 13/52
- 26/52

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The probability of drawing a spade card is:

Answer (Please select your correct option)

VuAnswers.com

- 1/52
- 4/52
- 13/52
- 26/52

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Which of the following is true for the binomial distribution b(x; n, p):

Answer (Please select your correct option)

VuAnswers.com

- mean > variance
- mean < variance
- mean = variance
- mean= standard deviation

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When f(x) is continuous probability function, then $P(X = 2)$ is:

Answer (Please select your correct option)

VuAnswers.com

- 1
- 0.5
- 0
- 0.25

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For any two estimators T1 and T2, if VAR(T1) < VAR(T2), then T1 is:

Answer (Please select your correct option)

VuAnswers.com

- Unbaised
- Sufficient
- Efficient
- Consistent

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In construction of a histogram, what would be taken along X-axis?

Answer (Please select your correct option)

VuAnswers.com

- Mid points
- Class limits
- Class interval
- Class boundaries

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If you connect the mid-points of rectangles in a histogram by a series of lines that also touches the x-axis from both ends, what will you get?

Answer (Please select your correct option)

VuAnswers.com

- Ogive
- Frequency polygon
- Frequency curve
- Historigram

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Tabulation is the process of arranging data into:

Answer (Please select your correct option)

VuAnswers.com

- Different classes
- Rows
- Columns
- Rows and Columns

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Which type of the curve is represented by the following shape?



Answer (Please select your correct option)

- Negatively skewed curve
- Bell shape curve
- semi-symmetrical curve
- Positively skewed curve

VuAnswers.com

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If A = (1, 2, 3, 4, 5, 10) and B = (1, 3, 5) then B⊂A means:

Answer (Please select your correct option)

VuAnswers.com

- A is less than B
- A is contained in B
- B is contained in A.
- B is less than A

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What is the mode in the word STATISTICS:

The Modeis defined as that value which occurs most frequently ina set of data i.e. it indicates the most common result.

Answer (Please select your correct option)

VuAnswers.com

- S
- T
- I
- S and T

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In a symmetrical distribution, if $Q_1 = 20$, $Q_2 = 30$ *then* Q_3 *is* :

Answer (Please select your correct option)

VuAnswers.com

- 50
- 40
- 30
- 10

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Which one of the following is not included in measures of central tendency:

Answer (Please select your correct option)

- Quartile Deviation
- Harmonic Mean
- Geometric Mean
- Arithmetic Mean

VuAnswers.com

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The sum of deviations from mean is:

Answer (Please select your correct option)

VuAnswers.com

- Maximum
- Minimum
- Zero
- Undefined

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Mean deviation is always:

Answer (Please select your correct option)

VuAnswers.com

- Less than Standard Deviation
- Greater than Standard Deviation
- Greater or equal to Standard Deviation
- Less or equal to Standard Deviation

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In how many branches, estimation can be divided? Also write down the names of the branches.

Answer (Please click here to Add Answer)

VuAnswers.com



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A Sample has following values.
101, 99, 111, 199, 150, 140
Find out the maximum likelihood estimate of population mean.

Answer (Please click here to Add Answer)

VuAnswers.com

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Write down critical region for the following hypothesis.

$H_0 : \sigma^2 = 20$

$H_1 : \sigma^2 < 20$

Where, $\alpha = 0.01$ and n=10

Answer (Please click here to Add Answer)

VuAnswers.com

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What is the concept of extrapolation in regression analysis?

Answer (Please click here to Add Answer)

VuAnswers.com



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If Z is a standard normal random variable with mean 0 and variance 1, then find the Lower quartile.

Answer (Please click here to Add Answer)

VuAnswers.com

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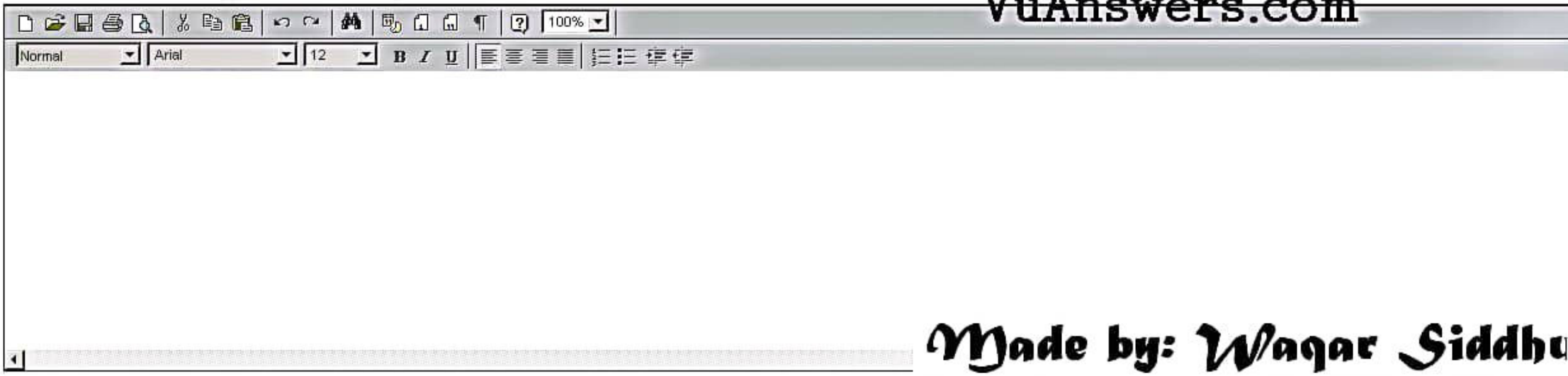
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For the following table we want to test the independence of smoking pattern and marital status. What will be the degrees of freedom for the chi-square test of independence?

Marital Status	Smoking Pattern			Total
	Total self-restraint	Only at times	Regular Smoker	
Single	67	213	74	354

Answer (Please click here to Add Answer)

VuAnswers.com



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Find the proportion for the X_1 and X_2, Where X_1 =300 with sample size= n_1 = 400 and X_2 = 200 with sample size = n_2 =300.

Answer (Please click here to Add Answer)

VuAnswers.com

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Calculate class boundaries from the following data.

Age Group	f
20-29	2
30-39	4
40-49	5

Answer (Please click here to Add Answer)

VuAnswers.com



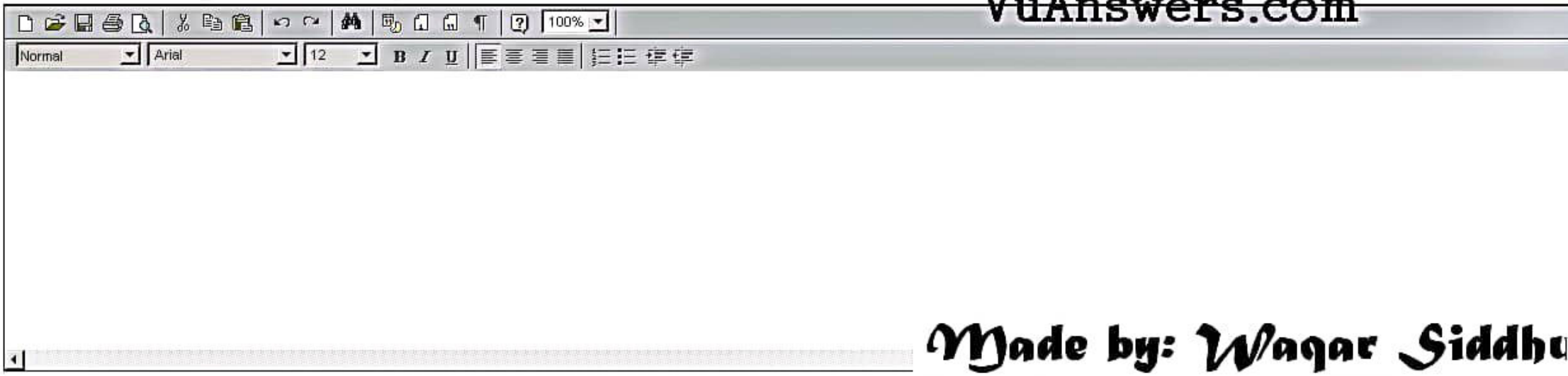
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A certain type of storage battery lasts on the average 3.0 years, with a standard deviation of 0.5 year. Assuming that the battery lives are normally distributed, find the probability that a given battery will last less than 2.3 years.

Answer (Please click here to Add Answer)

VuAnswers.com



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For the Given information, what is your conclusion in testing the indicated null hypothesis?

$n_1 = 13, n_2 = 41, s_1^2 = 6.3, s_2^2 = 15.6$, $H_0 : \sigma_1^2 = \sigma_2^2$ and $H_1 : \sigma_2^2 > \sigma_1^2$

Answer (Please click here to Add Answer)

VuAnswers.com

Normal | Arial | 12 | B I U | 100%

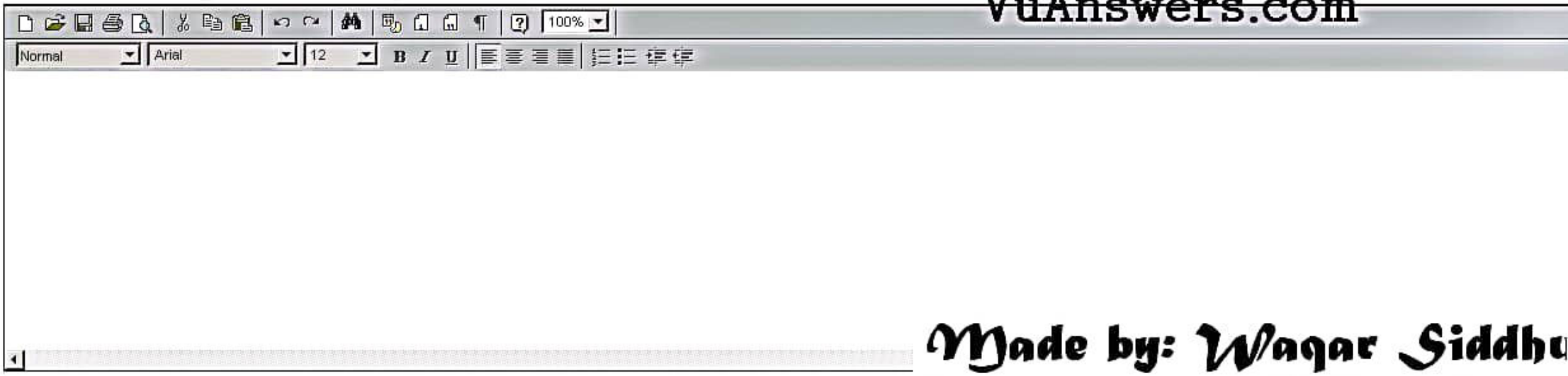
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In a random sample of 200 persons having their lunch at the University cafeteria on meatless day it was observed that 30 percent preferred vegetable dishes.Find 95% confidence interval for p.

Answer (Please click here to Add Answer)

VuAnswers.com



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Answer the following questions:

i. Which method has been used for colleting the data in the following case?
"An investigator collects the information personally from the individuals concerned".

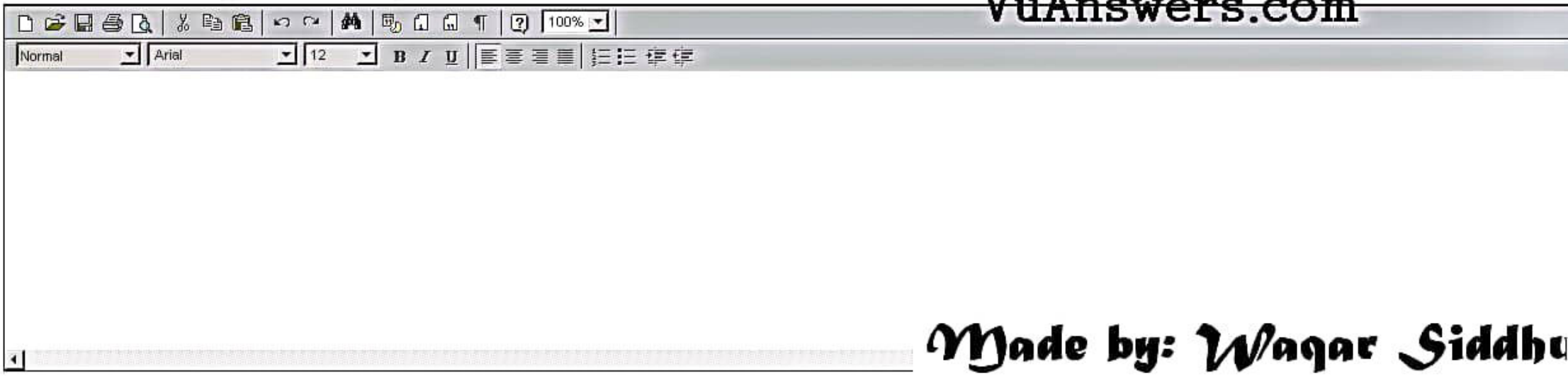
i. Find the sampling error when the population mean is 15.785 and sample mean is 15.6.

ii. Write down any two types of nonrandom sampling

iii. What is the singular of strata?

Answer (Please click here to Add Answer)

VuAnswers.com



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Question No : 1 of 52 Marks: 1 (Budgeted Time 1 Min)

When each outcome of a sample space has equal chance to occur as any other, the outcomes are called:

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Answer (Please select your correct option)

VuAnswers.com

- Mutually exclusive
- Equally likely
- Not mutually exclusive
- Exhaustive

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When each outcome of a sample space has equal chance to occur as any other, the outcomes are called:

Answer (Please select your correct option)

- Mutually exclusive
- Equally likely
- Not mutually exclusive
- Exhaustive

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Start Time: 9:48 AM

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An expected value of a random variable is equal to:

Answer (Please select your correct option)

- Variance
- Mean
- Standard deviation
- Covariance

VuAnswers.com

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When f(x) is continuous probability function, then $P(X = 1)$ is:

Answer (Please select your correct option)

- 1
- ∞
- $-\infty$
- 0

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The sum of deviations is zero, when deviations are taken from:

Answer (Please select your correct option)

- Mean
- Median
- Mode
- H.M

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The distribution function F(x) is equal to

Answer (Please select your correct option)

- $P(X = x)$
- $P(X \leq x)$
- $P(X \geq x)$
- $P(X > x)$

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In a one-way ANOVA:

Answer (Please select your correct option)

- The interaction term has (c - 1)(n - 1) degrees of freedom
- An interaction term is given
- An interaction effect can be tested
- There is no interaction term

VuAnswers.com

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The degrees of freedom for a t-test with sample size 'n' is:

Answer (Please select your correct option)

- n-1
- n+1
- n-2
- n+2

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Rumour has reached the Trading Standards Officer that the manufacturer ABC is deliberately underfilling his cartons of orange juice. It is decided that a sample should be taken to check this claim. The stated contents on the carton are 100 ml on the average, then the null hypothesis is:

Answer (Please select your correct option)

- $H_0 : \mu = 100$
- $H_0 : \mu > 100$
- $H_0 : \mu < 100$
- $H_0 : \mu \neq 100$

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When c is a constant, then E(c) is:

Answer (Please select your correct option)

- 0
- 1
- c
- -c

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The combined distribution of more than two random variables is:

Answer (Please select your correct option)

- Univariate distribution
- Joint distribution
- Marginal distribution
- Bivariate distribution

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The test statistic used in analysis of variance procedure follow the :

Answer (Please select your correct option)

- χ^2 -distribution.
- T-distribution.
- Z-distribution.
- F-distribution.

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In normal distribution β_2 = ... :

Answer (Please select your correct option)

- 1
- 2
- 3
- 0

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In normal distribution, the quartile deviation Q.D = ... :

Answer (Please select your correct option)

- 0.5σ
- 0.75σ
- 0.7979σ
- 0.6745σ

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In a symmetrical distribution, the coefficient of skewness is equal to :

Answer (Please select your correct option)

- -1
- +1
- 0
- 2

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The average which is defined as the reciprocal of the arithmetic mean of the reciprocals of the values is called:

Answer (Please select your correct option)

- Geometric Mean
- Harmonic Mean
- Mode
- Median

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Which measure of dispersion is used to compare the variation of two data sets?

Answer (Please select your correct option)

- Coefficient of variation
- Coefficient of comparison
- Mean deviation
- Standard deviation

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If S.D$(X) = 5$ then S.D$(\frac{2X+5}{2})$ = _____

Answer (Please select your correct option)

- 5
- 10
- 15
- 7.5

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The deviation of a distribution from symmetry is called:

Answer (Please select your correct option)

- Kurtosis
- Skewness
- Dispersion
- Flatness

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The conditional probability function $f(x \mid 1)$ = _____

Answer (Please select your correct option)

- $f(1,1)$
- $f(x,1)$
- $\frac{f(x,1)}{h(1)}$
- $\frac{f(x,1)}{h(x)}$

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Which one is the correct formula to find the desired sample size?

Answer (Please select your correct option)

- $n = \left(\frac{Z_{\alpha/2} \cdot \sigma}{e} \right)^2$
- $n = \left(\frac{Z_{\alpha/2} \cdot \sqrt{\sigma}}{e} \right)^2$
- $n = \left(\frac{Z_{\alpha/2} \cdot \bar{X}}{e} \right)^2$
- $n = \frac{Z_{\alpha/2} \cdot \sigma}{e}$

VuAnswers.com

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Start Time: 9:48 AM

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A deserving player is not selected in the team is an example of:

Answer (Please select your correct option)

- Type I error
- Type II error
- Correct decision
- No information regarding this

VuAnswers.com

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A judge can acquit a guilty person is the example of:

Answer (Please select your correct option)

- Type I error
- Type II error
- Correct decision
- No information regarding this

VuAnswers.com

not sure

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Ideally, the width of confidence interval should be:

Answer (Please select your correct option)

- 0
- 1
- 99
- 100

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If the sampling distribution of $\bar{X}$ is normal, we would expect 99% of the sample means to be within the interval:

Answer (Please select your correct option)

- $\mu_{\bar{x}} \pm 2\sigma_{\bar{x}}$
- $\mu_{\bar{x}} \pm 1.96\sigma_{\bar{x}}$
- $\mu_{\bar{x}} \pm 2.58\sigma_{\bar{x}}$
- $\mu_{\bar{x}} \pm \sigma_{\bar{x}}$

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If mean of χ^2 distribution is k then variance will be:

Answer (Please select your correct option)

- k^2
- 2k
- 1/k
- k

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Mean of the F-distribution is possible only, when

Answer (Please select your correct option)

- $\nu_1 > 2$
- $\nu_2 > 2$
- $\nu_1 < 2$
- $\nu_2 < 2$

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In Statistics, we have MSE which is abbreviation of:

Answer (Please select your correct option)

- Mean square error
- Measured square error
- Medical screening exam
- Major sampling error

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What is the graphical shape of the chi-square distribution?

Answer (Please select your correct option)

- Positively skewed
- Negatively skewed
- Uniformly distributed
- Normally distributed

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As the degree of freedom increases, the t-distribution tends to coincide with:

Answer (Please select your correct option)

- Binomial distribution
- Uniform distribution
- Hypergeometric distribution
- Normal distribution

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If X and Y are independent variables, then E (XY) is:

Answer (Please select your correct option)

- E(XX)
- E(X).E(Y)
- X.E(Y)
- Y.E(X)

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What are the number of ways in which four books can be arranged on a shelf?

Answer (Please select your correct option)

- 4
- 6
- 12
- 24

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The parameters of the binomial distribution b(x; n, p) are:

Answer (Please select your correct option)

- x & n
- x & p
- n & p
- x, n & p

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Hypergeometric probability distribution has :

Answer (Please select your correct option)

- (n, k) parameter
- (N) parameter
- (N, n, N-k) parameter
- (N ,n, k) parameter

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The variance of the hypergeometric probability distribution is:

Answer (Please select your correct option)

- $\sigma^2 = n\frac{k}{N}$
- $\sigma^2 = n\frac{k}{N}\frac{N-k}{N}\frac{N-n}{N-1}$,
- $\sigma^2 = npq$
- $\sigma^2 = N\frac{n}{K}$

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When f(x) is continuous probability function for $1 < X < 5$, then $P(X < 1)$ is:

Answer (Please select your correct option)

- 0
- 0.25
- 0.5
- 1

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For any two estimators T1 and T2, if VAR(T1) < VAR(T2), then T1 is:

Answer (Please select your correct option)

- Unbaised
- Sufficient
- Efficient
- Consistent

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If an estimator gets closer to the population parameter by increasing sample size then it is known as:

Answer (Please select your correct option)

- Consistent estimator
- Sufficient estimator
- Efficient estimator
- Unbiased estimator

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Which of the following comes first to make frequency distribution.

Answer (Please select your correct option)

- Number of Groups
- Class interval
- Range
- Tally marks

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What curve shape would you expect for the distribution of death rates of population of all age groups?

Answer (Please select your correct option)

- Symmetrical curve
- Skewed to the right
- Skewed to the left
- U shape curve

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Which one is the measure of central tendency:

Answer (Please select your correct option)

- Variation of the distribution
- Average of the distribution
- Scatterness of the distribution
- Dispersion of the distribution

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What does quartile deviation measure in a distribution?

Answer (Please click here to Add Answer)

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41

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In a random sample of 1000 women in a locality, 224 women said that they use Aerial washing powder. What is the point estimator and point estimate of the proportion of the women who use Aerial washing powder?

Answer (Please click here to Add Answer)

VuAnswers.com

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42

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What is the impact of level of confidence (1-α) on the value of $Z_{\alpha/2}$?

Answer (Please click here to Add Answer)

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Formulate the hypothesis in case of goodness of fit test.

Answer (Please click here to Add Answer)

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44

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Why the median is suitable average for the below data set? Explain.

Monthly income (in rupees)	No. of workers
Less than 2000/-	100
2000/-to 2999/-	300

Answer (Please click here to Add Answer)

VuAnswers.com

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45

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(in rupees)	
Less than 2000/-	100
2000/-to 2999/-	300
3000/- to 3999/-	500
4000/- to 4999/-	250
5000/- and above	50

Answer (Please click here to Add Answer)

VuAnswers.com

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45

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If Z is a standard normal random variable with mean 0 and variance 1, then find the Lower quartile.

Answer (Please click here to Add Answer)

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46

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Write down critical region for the following hypothesis at 5% level of significance.

$H_0 : \mu = 75$

$H_1 : \mu < 75$

Answer (Please click here to Add Answer)

VuAnswers.com

Normal | Arial | 12 | B I U | 100%

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47

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The manager of a bottling plant is anxious to reduce the variability in net weight of fruit bottled. Over a long period, the standard deviation has been 15.2 gm. A new machine is introduced and the net weights (in grams) in 10 randomly selected bottles (all of the same nominal weight) are 987, 966, 955, 977, 981, 967, 975, 980, 953, and 972.
State null and alternative hypothesis that machine has a better performance? Also write down test-statistics about the hypothesis.

Answer (Please click here to Add Answer)

VuAnswers.com

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Find the coefficient of standard deviation from the following data:
Life in Hours (X): 130, 150, 180, 250, 345

Answer (Please click here to Add Answer)

VuAnswers.com

Normal | Arial | 12 | 100%

Start Time: 9:48 AM

49

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What is the probability that a poker hand of 5 cards contain (i) exactly 2 aces (ii) exactly 1 ace?
(Use hypergeometric distribution)

Answer (Please click here to Add Answer)

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Normal | Arial | 12 | 100%

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If n=22, $\sum\left(d-\bar{d}\right)^2 = 270, \bar{d} = 4$

Then, using a 0.05 level of significance level, test the hypothesis $H_0 : \mu_d = 0$ against $H_1 : \mu_d \neq 0$

Answer (Please click here to Add Answer)

VuAnswers.com

Normal | Arial | 12 | B I U | 100%

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51

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A company launched new layout of its website. After a survey, 62 of 115 visitors liked the new layout while 59 of 135 visitors liked the old layout of the website. Company claims that new layout did not improve the visitor's liking about the website.

By using the critical value $Z < -Z_{0.05} = -1.645$, verify the company's claim:

$H_o: P_1 - P_2 \geq 0$

$H_1: P_1 - P_2 < 0$

Answer (Please click here to Add Answer)

VuAnswers.com

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52

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Question No : 1 of 52 Marks: 1 (Budgeted Time 1 Min)

The parameter of the chi- square distribution is.....

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Answer (Please select your correct option) VuAnswers.com

- v
- $v-1$
- $v-2$
- v-p

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Question No : 2 of 52 Marks: 1 (Budgeted Time 1 Min)

The value of χ^2 can never be :

Answer (Please select your correct option)

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- Zero
- Less than 1
- Greater than 1
- Negative

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Question No : 3 of 52 Marks: 1 (Budgeted Time 1 Min)

The probability of an event always lies between:

Answer (Please select your correct option)

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- 0 and -∞
- -1 and +1
- -∞ and + -∞
- 0 and 1

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Question No : 4 of 52 Marks: 1 (Budgeted Time 1 Min)

The number of parameters in a Poisson distribution is (are):

Answer (Please select your correct option)

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- 1
- 0
- 2
- 3

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Question No : 5 of 52 Marks: 1 (Budgeted Time 1 Min)

$E(4X + 5) =$_____

Answer (Please select your correct option)

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- 16 E (X)
- 16 E (X) + 5
- 12 E (X)
- 4 E (X) + 5

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Question No : 6 of 52 Marks: 1 (Budgeted Time 1 Min)

In a one-way ANOVA:

Answer (Please select your correct option)

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- The interaction term has (c - 1)(n - 1) degrees of freedom
- An interaction term is given
- An interaction effect can be tested
- There is no interaction term

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Question No : 7 of 52 Marks: 1 (Budgeted Time 1 Min)

The degrees of freedom for a t-test with sample size 14 is:

Answer (Please select your correct option)

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- 14
- 13
- 7
- 0

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Question No : 8 of 52 Marks: 1 (Budgeted Time 1 Min)

The degrees of freedom for a t-test with sample size 6 is:

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Answer (Please select your correct option)

- 1
- 3
- 5
- 7

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Question No : 9 of 52 Marks: 1 (Budgeted Time 1 Min)

Which of the following is a characteristics of the normal distribution:

Answer (Please select your correct option)

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- It is a skewed distribution
- It is bell-shaped
- It is not asymptotic
- It is leptokurtic

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Question No : 10 of 52 Marks: 1 (Budgeted Time 1 Min)

In normal distribution β_1 = ... :

Answer (Please select your correct option)

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- 0
- 1
- 2
- 3

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Question No : 11 of 52 Marks: 1 (Budgeted Time 1 Min)

In normal distribution β_2 = ... :

Answer (Please select your correct option)

VuAnswers.com

- 1
- 2
- 3
- 0

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Question No : 12 of 52 Marks: 1 (Budgeted Time 1 Min)

A good way to get a small standard error is to use a _____.

Answer (Please select your correct option)

VuAnswers.com

- Repeated sampling
- Small sample
- Large sample
- Large population

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Question No : 13 of 52 Marks: 1 (Budgeted Time 1 Min)

Which of the following is a measure of absolute dispersion?

Answer (Please select your correct option)

VuAnswers.com

- Skewness
- Mean Deviation
- Coefficient of variation
- Kurtosis

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Question No : 14 of 52 Marks: 1 (Budgeted Time 1 Min)

An automobile is running, during the first 60 Km, at the rate of 10 Km/hr. During the second 60 Km at the rate of 30Km/hr, while during the third 60 Km its speed was 40 Km/hr. What method is more appropriate to calculate the average speed?

Answer (Please select your correct option)

VuAnswers.com

- Median
- Arithematic mean
- Harmonic mean
- Geometric mean

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Question No : 15 of 52 Marks: 1 (Budgeted Time 1 Min)

If $S.D(X) = 5$ then $S.D(\frac{2X+5}{2}) =$ _____

Answer (Please select your correct option)

VuAnswers.com

- 5
- 10
- 15
- 7.5

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Question No : 16 of 52 Marks: 1 (Budgeted Time 1 Min)

For a particular data set the Pearson's coefficient of skewness is greater then zero. What will be the shape of distribution?

Answer (Please select your correct option)

VuAnswers.com

- Negatively skewed
- J-shaped
- Symmetrical
- Positively skewed

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Question No : 17 of 52 Marks: 1 (Budgeted Time 1 Min)

The total number of samples when sampling is done without replacement is equal to:

Answer (Please select your correct option)

VuAnswers.com

- N^n
- C_n^N
- $\frac{N-n}{N-1}$
- 1

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Question No : 18 of 52 Marks: 1 (Budgeted Time 1 Min)

Which of the following is correct option to find $P(X + Y \leq 1)$?

Answer (Please select your correct option)

VuAnswers.com

- f(0, 0) + f(0, 1) + f(1, 2)
- f(2, 0) + f(0, 1) + f(1, 0)
- f(0, 0) + f(1, 1) + f(1, 0)
- f(0, 0) + f(0, 1) + f(1, 0)

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Question No : 19 of 52 Marks: 1 (Budgeted Time 1 Min)

The conditional probability function $f(x \mid 1)$ = _____

Answer (Please select your correct option)

VuAnswers.com

- $f(1,1)$
- $f(x,1)$
- $\frac{f(x,1)}{h(1)}$
- $\frac{f(x,1)}{h(x)}$

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Question No : 20 of 52 **Marks: 1 (Budgeted Time 1 Min)**

If f (x, y) is bivariate probability density function of continuous random variables X and Y then marginal density function of y i.e. $h(y)$ is:

Answer (Please select your correct option)

VuAnswers.com

- $\int_{-\infty}^{\infty} f(x,y)\,dx$
- $\int_{-\infty}^{\infty} f(x,y)\,dy$
- $\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\,dx\,dy$
- $\int_{a}^{b}\int_{c}^{d} f(x,y)\,dy\,dx$

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Question No : 21 of 52 Marks: 1 (Budgeted Time 1 Min)

_____ is a range of numbers inferred from the sample that has a certain probability of including the population parameter over the long run.

Answer (Please select your correct option)

VuAnswers.com

- Hypothesis
- Lower limit
- Confidence interval not sure
- Probability limit

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Question No : 22 of 52 Marks: 1 (Budgeted Time 1 Min)

By definition f(y | x) = _____

Answer (Please select your correct option)

VuAnswers.com

- $f(y)$
- $f(x,y)$
- $\frac{f(x,y)}{h(x)}$
- $\frac{f(x,y)}{h(y)}$

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Question No : 23 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Which of the following is an unbiased estimator of population σ^2 ?

Answer (Please select your correct option)

VuAnswers.com

- $\frac{\sum(x-\bar{x})^2}{n}$
- $\frac{\sum(x-\bar{x})^2}{n-2}$
- $\frac{\sum(x-\bar{x})^2}{n(n-1)}$
- $\frac{\sum(x-\bar{x})^2}{n-1}$

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Question No : 24 of 52 Marks: 1 (Budgeted Time 1 Min)

If a significance level of 1% is used rather than 5%, the null hypothesis is:

Answer (Please select your correct option)

VuAnswers.com

- More likely to be rejected
- Less likely to be rejected
- Just as likely to be rejected
- None of the above

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Question No : 25 of 52 Marks: 1 (Budgeted Time 1 Min)

For $\alpha = 0.01$, the critical values of z for two tailed test are equal to:

Answer (Please select your correct option)

VuAnswers.com

- -2.58 and +2.58
- -2.33 and +2.33
- -1.645 and +1.645
- -1.96 and + 1.96

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Question No : 26 of 52 Marks: 1 (Budgeted Time 1 Min)

A judge can acquit a guilty person is the example of:

Answer (Please select your correct option)

VuAnswers.com

- Type I error
- Type II error
- Correct decision
- No information regarding this

idea

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Question No : 27 of 52 Marks: 1 (Budgeted Time 1 Min)

The proportion of males in Pakistan is at least 0.48, the alternative hypothesis H_1 is

Answer (Please select your correct option)

VuAnswers.com

- $P \leq 0.48$
- $P = 0.48$
- $P < 0.48$ idea
- $P \geq 0.48$

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Question No : 28 of 52 Marks: 1 (Budgeted Time 1 Min)

If mean of χ^2 distribution is k then variance will be:

Answer (Please select your correct option)

VuAnswers.com

- k^2
- 2k
- 1/k
- k

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Question No : 29 of 52 Marks: 1 (Budgeted Time 1 Min)

What is the graphical shape of the chi-square distribution?

Answer (Please select your correct option)

VuAnswers.com

- Positively skewed
- Negatively skewed
- Uniformly distributed
- Normally distributed

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Question No : 30 of 52 Marks: 1 (Budgeted Time 1 Min)

What factor determines the shape of the t-distribution?

Answer (Please select your correct option)

VuAnswers.com

- Degree of freedom
- Critical value
- Frequency of data
- Probability

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Question No : 31 of 52 Marks: 1 (Budgeted Time 1 Min)

The covariance of a random variable with itself is:

Answer (Please select your correct option)

VuAnswers.com

- Zero
- One
- Its variance
- Its correlation

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Question No : 32 of 52 Marks: 1 (Budgeted Time 1 Min)

Hypergeometric probability distribution has :

Answer (Please select your correct option)

VuAnswers.com

- (n, k) parameter
- (N) parameter
- (N, n, N-k) parameter
- (N ,n, k) parameter

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Question No : 33 of 52 Marks: 1 (Budgeted Time 1 Min)

For any two estimators T1 and T2, if VAR(T1) < VAR(T2), then T1 is:

Answer (Please select your correct option)

VuAnswers.com

- Consistent
- Unbaised
- Sufficient
- Efficient

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Question No : 34 of 52 Marks: 1 (Budgeted Time 1 Min)

Atmosphere pressure is the example of

Answer (Please select your correct option)

VuAnswers.com

- Constant
- Qualitative variable
- Quantitative variable
- None of the above

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Question No : 35 of 52 Marks: 1 (Budgeted Time 1 Min)

Which type of the curve is represented by the following shape?



Answer (Please select your correct option)

VuAnswers.com

- Bell shape curve
- semi-symmetrical curve
- Positively skewed curve
- Negatively skewed curve

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Question No : 36 of 52 Marks: 1 (Budgeted Time 1 Min)

Which one of the following is not a type of frequency curve?

Answer (Please select your correct option)

VuAnswers.com

- The symmetrical frequency curve
- The extremely skewed frequency curve
- The U-shaped frequency curve
- Frequency polygon

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Question No : 37 of 52 Marks: 1 (Budgeted Time 1 Min)

For positively skewed distribution
Mean.....Median.....Mode:

Answer (Please select your correct option)

VuAnswers.com

- ≠
- =
- <
- >

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Question No : 38 of 52 Marks: 1 (Budgeted Time 1 Min)

What is ' f_m ' in the formula of mode:

Answer (Please select your correct option)

VuAnswers.com

- First Frequency
- Last Frequency
- Maximum Frequency
- Minimum Frequency

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Question No : 39 of 52 Marks: 1 (Budgeted Time 1 Min)

Which one is the measure of central tendency:

Answer (Please select your correct option)

VuAnswers.com

- Variation of the distribution
- Average of the distribution
- Scatterness of the distribution
- Dispersion of the distribution

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Question No : 40 of 52 Marks: 1 (Budgeted Time 1 Min)

In a set of 10 values all the values are 5, what will be the P_{50}?

Answer (Please select your correct option)

VuAnswers.com

- 2
- 5 i think
- 10
- 20

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Question No : 41 of 52 Marks: 2 (Budgeted Time 4 Min)

Write down the name of methods/techniques that are used to represent the quantitative discrete data.

Answer (Please click here to Add Answer)

VuAnswers.com

Normal Arial 12 100%

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Question No : 42 of 52 Marks: 2 (Budgeted Time 4 Min)

A coin is tossed 900 times and heads appear 490 times. State the null and alternative hypotheses to show that the coin is unbiased.

Answer (Please click here to Add Answer)

VuAnswers.com

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Question No : 43 of 52 Marks: 2 (Budgeted Time 4 Min)

Write down the names of two types of experimental designs.

Answer (Please click here to Add Answer)

VuAnswers.com

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Question No : 44 of 52 Marks: 2 (Budgeted Time 4 Min)

What are the mean and variance of binomial distribution?

Answer (Please click here to Add Answer)

VuAnswers.com

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Question No : 45 of 52 Marks: 3 (Budgeted Time 6 Min)

Find the probability of drawing a white ball from a bag containing 4 red, 8 black and 3 white balls.

Answer (Please click here to Add Answer)

VuAnswers.com



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Question No : 46 of 52 Marks: 3 (Budgeted Time 6 Min)

Write down the properties of sampling distribution of proportion $\hat{p}$, when sampling is performed *without* replacement.

Answer (Please click here to Add Answer)

VuAnswers.com

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Question No : 47 of 52 Marks: 3 (Budgeted Time 6 Min)

If $X = 255, n = 500, p_0 = 0.60$ then find the z-test statistic for proportion.

Answer (Please click here to Add Answer)

VuAnswers.com

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Question No : 48 of 52 Marks: 3 (Budgeted Time 6 Min)

What is the impact of degrees of freedom on chi- square distribution?

Answer (Please click here to Add Answer)

VuAnswers.com

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Question No : 49 of 52 Marks: 5 (Budgeted Time 10 Min)

Find quartile deviation for the data given below:
18, 53, 45, 28, 39, 29, 23, 40 and 21

Answer (Please click here to Add Answer)

VuAnswers.com

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Question No : 50 of 52 Marks: 5 (Budgeted Time 10 Min)

A population consists of N=5 values 1, 2, 3, 5, 6.A sample size of n=3 is selected from the population without replacement, calculate sampling distribution of sample proportions for even numbers.

Answer (Please click here to Add Answer)

VuAnswers.com



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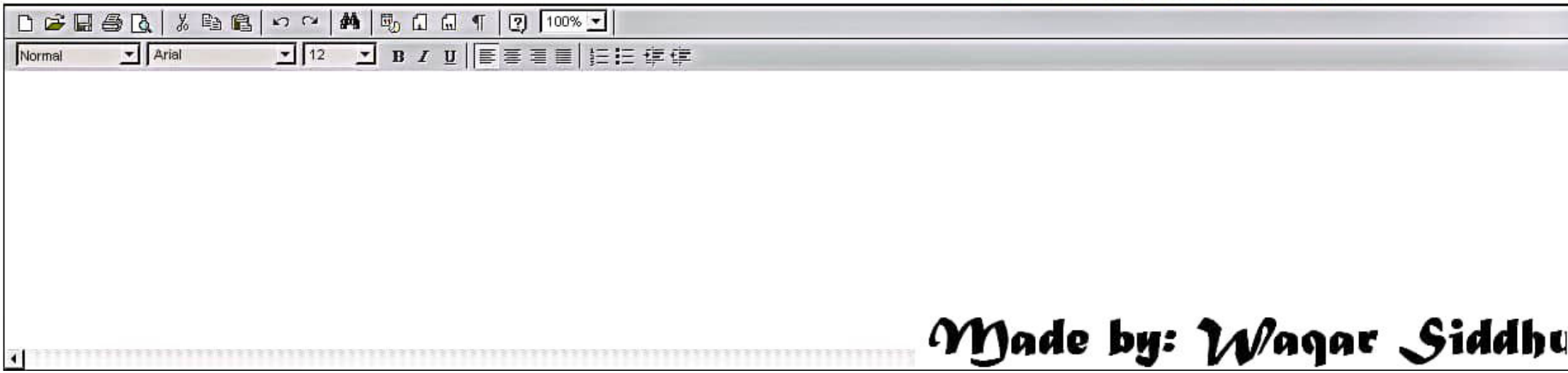
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Question No : 51 of 52 Marks: 5 (Budgeted Time 10 Min)

Write down the testing procedure in case of goodness of fit test.

Answer (Please click here to Add Answer)

VuAnswers.com



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Question No : 52 of 52 **Marks: 5 (Budgeted Time 10 Min)**

If we have n=634 and $\hat{p}$ =0.459, where $Z_{\frac{0.01}{2}}$=2.58, then find the 99% confidence interval for population proportion.

Answer (Please click here to Add Answer)

VuAnswers.com



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Question No : 1 of 52 Marks: 1 (Budgeted Time 1 Min)

The parameter of the chi- square distribution is.....

Edited By Mahar Afaq Safdar Muhammadi

Answer (Please select your correct option) VuAnswers.com

- v
- $v-1$
- $v-2$
- $v-p$

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Question No : 2 of 52 Marks: 1 (Budgeted Time 1 Min)

The value of χ^2 can never be :

Answer (Please select your correct option)

VuAnswers.com

- Zero
- Less than 1
- Greater than 1
- Negative

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Question No : 3 of 52 Marks: 1 (Budgeted Time 1 Min)

Analysis of variance is a procedure that enables us to test the equality of several:

Answer (Please select your correct option)

VuAnswers.com

- Variances
- Means
- Proportions
- Groups

ANALYSIS OF VARIANCE (ANOVA)

It is a procedure which enables us to test the hypothesis of equality of several population means

(i.e.

H0 : μ1 = μ2 = μ3 = …… = μk

against

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Question No : 4 of 52 Marks: 1 (Budgeted Time 1 Min)

When two coins are tossed the probability of at most one head is:

VuAnswers.com

Answer (Please select your correct option)

- 1/4
- 2/4
- 3/4
- 4/4

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Question No : 5 of 52 Marks: 1 (Budgeted Time 1 Min)

The critical region for $H_1 : \mu < \mu_0$ *when* $\alpha = 0.05$ is:

Answer (Please select your correct option)

VuAnswers.com

- $z > z_{0.05}$
- $|z| > z_{0.10}$
- $z < -z_{0.05}$
- $|z| > z_{0.05}$

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Question No : 6 of 52 Marks: 1 (Budgeted Time 1 Min)

A discrete probability function f(x) is always:

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Answer (Please select your correct option)

- Non-negative
- Negative
- One
- Zero

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Question No : 7 of 52 Marks: 1 (Budgeted Time 1 Min)

When f(x) is continuous probability function, then P(X = 1) is:

Answer (Please select your correct option)

VuAnswers.com

- 1
- ∞
- −∞
- 0

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Question No : 8 of 52 Marks: 1 (Budgeted Time 1 Min)

A random variable can be generated:

Answer (Please select your correct option)

VuAnswers.com

- Manually
- Mechanically

Not sure.. :) concpt.. :)

- Manually & Mechanically
- Mathematically

numerical quantity whose value is determined by the outcome of a random experiment is called a random variable.

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Question No : 9 of 52 Marks: 1 (Budgeted Time 1 Min)

The curve of the F- distribution depends upon:

Answer (Please select your correct option)

VuAnswers.com

- Degrees of freedom
- Standard deviation
- Mean
- Variance

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Question No : 10 of 52 Marks: 1 (Budgeted Time 1 Min)

Assuming that following is a probability distribution, then what is the value of 'a':

X	1	2	3
P(X)	0.1	a	0.1

Answer (Please select your correct option)

VuAnswers.com

- 0.6
- 0.8
- 0.2
- 0.4

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Question No : 11 of 52 Marks: 1 (Budgeted Time 1 Min)

The mode value from raw data can be obtained by the help of

Answer (Please select your correct option)

VuAnswers.com

- Dot plot
- Stem and leaf plot
- Bar chart
- None of these

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Question No : 12 of 52 Marks: 1 (Budgeted Time 1 Min)

In normal distribution, the quartile deviation Q.D = ... :

Answer (Please select your correct option)

VuAnswers.com

- 0.5σ
- 0.75σ
- 0.7979σ
- 0.6745σ

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Question No : 13 of 52 Marks: 1 (Budgeted Time 1 Min)

A randomly selected sample of 400 students at university was asked whether or not they will participate in politics. Forty-six percent of the 400 student surveyed answered "yes". Which one of the following statement about number 46% is correct?

VuAnswers.com

Answer (Please select your correct option)

- It is a sample statistic.
- It is a population parameter.
- It is a margin of error.
- It is a standard error. not sure

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Question No : 14 of 52 Marks: 1 (Budgeted Time 1 Min)

A student solved 25 questions from first 50 questions of a book . The probability that he will solve the remaining all questions is:

Answer (Please select your correct option)

VuAnswers.com

- 0.25
- 0.5
- 1
- 0

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Question No : 15 of 52 Marks: 1 (Budgeted Time 1 Min)

The average which is defined as the reciprocal of the arithmetic mean of the reciprocals of the values is called:

Answer (Please select your correct option)

VuAnswers.com

- Geometric Mean
- Harmonic Mean
- Mode
- Median

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Question No : 16 of 52 Marks: 1 (Budgeted Time 1 Min)

For the independent events A and B if P (A) = 0.25, P (B) =0.40 then P (A∩B) =.....

Answer (Please select your correct option)

VuAnswers.com

- 0.65
- 0.1
- 0.50
- 0.15

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Question No : 17 of 52 Marks: 1 (Budgeted Time 1 Min)

The total number of samples when sampling is done with replacement is equal to:

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Answer (Please select your correct option)

- N^n
- C_n^N
- $\frac{N-n}{N-1}$
- 1

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Question No : 18 of 52 Marks: 1 (Budgeted Time 1 Min)

Which of the following is correct option to find $P(X + Y \leq 1)$?

Answer (Please select your correct option)

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- f(0, 0) + f(0, 1) + f(1, 2)
- f(2, 0) + f(0, 1) + f(1, 0)
- f(0, 0) + f(1, 1) + f(1, 0)
- f(0, 0) + f(0, 1) + f(1, 0)

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Question No : 19 of 52 Marks: 1 (Budgeted Time 1 Min)

The conditional probability function $f(x \mid 1)$ = _____

Answer (Please select your correct option)

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- $f(1,1)$
- $f(x,1)$
- $\frac{f(x,1)}{h(1)}$
- $\frac{f(x,1)}{h(x)}$

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Question No : 20 of 52 Marks: 1 (Budgeted Time 1 Min)

In interval estimation, we always get:

Answer (Please select your correct option)

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- Range of values
- Zero
- Single value
- Two values

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Question No : 21 of 52 Marks: 1 (Budgeted Time 1 Min)

Which of the following is an unbiased estimator of population σ^2 ?

Answer (Please select your correct option)

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- $\frac{\sum(x-\bar{x})^2}{n}$
- $\frac{\sum(x-\bar{x})^2}{n-2}$
- $\frac{\sum(x-\bar{x})^2}{n(n-1)}$
- $\frac{\sum(x-\bar{x})^2}{n-1}$

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Question No : 22 of 52 Marks: 1 (Budgeted Time 1 Min)

How many parameter(s) are in t-distribution?

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Answer (Please select your correct option)

- 0
- 1
- 2
- 3

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Question No : 23 of 52 Marks: 1 (Budgeted Time 1 Min)

A deserving player is not selected in the team is an example of:

Answer (Please select your correct option)

VuAnswers.com

- Type I error
- Type II error
- Correct decision
- No information regarding this

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Question No : 24 of 52 Marks: 1 (Budgeted Time 1 Min)

After an interval is constructed, then what is the probability of occurrence of the parameter in it?

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Answer (Please select your correct option)

- Zero
- One
- Fifty
- Either one or zero

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Question No : 25 of 52 Marks: 1 (Budgeted Time 1 Min)

If $\bar{X}$ is the mean of the n observations, then which test statistic will be used to calculate the confidence limits of the population variance σ^2?

Answer (Please select your correct option)

VuAnswers.com

- Z-statistic
- T-statistic
- χ^2-statistics
- F-statistics

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Question No : 26 of 52 Marks: 1 (Budgeted Time 1 Min)

If mean of χ^2 distribution is k then variance will be:

Answer (Please select your correct option)

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- k^2
- 2k
- 1/k
- k

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Question No : 27 of 52 Marks: 1 (Budgeted Time 1 Min)

What is the graphical shape of the chi-square distribution?

Answer (Please select your correct option)

VuAnswers.com

- Positively skewed
- Negatively skewed
- Uniformly distributed
- Normally distributed

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Question No : 28 of 52 Marks: 1 (Budgeted Time 1 Min)

What factor determines the shape of the t-distribution?

Answer (Please select your correct option)

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- Degree of freedom
- Critical value
- Frequency of data
- Probability

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Question No : 29 of 52 Marks: 1 (Budgeted Time 1 Min)

If X and Y are random variables, then $E(X-Y)$ is equal to:

Answer (Please select your correct option)

VuAnswers.com

- $E(X)+E(Y)$
- $E(X)-E(Y)$
- $X-E(Y)$
- $E(X)-Y$

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Question No : 30 of 52 Marks: 1 (Budgeted Time 1 Min)

In the regression line Y= a + bX, the non-random variable is:

Answer (Please select your correct option)

VuAnswers.com

not sure

- Y
- X
- Both X and Y
- Neither X nor Y

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Question No : 31 of 52 Marks: 1 (Budgeted Time 1 Min)

The parameters of the binomial distribution b(x; n, p) are:

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Answer (Please select your correct option)

- x & n
- x & p
- n & p
- x, n & p

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Question No : 32 of 52 Marks: 1 (Budgeted Time 1 Min)

Which of the following is true for the binomial distribution b(x; n, p):

Answer (Please select your correct option)

VuAnswers.com

- mean > variance
- mean < variance
- mean = variance
- mean= standard deviation

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Question No : 33 of 52 Marks: 1 (Budgeted Time 1 Min)

Hypergeometric probability distribution has :

VuAnswers.com

Answer (Please select your correct option)

- (n, k) parameter
- (N) parameter
- (N, n, N-k) parameter
- (N ,n, k) parameter

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Question No : 34 of 52 Marks: 1 (Budgeted Time 1 Min)

The variance of the hypergeometric probability distribution is:

VuAnswers.com

Answer (Please select your correct option)

- $\sigma^2 = n\frac{k}{N}$
- $\sigma^2 = n\frac{k}{N}\frac{N-k}{N}\frac{N-n}{N-1}$,
- $\sigma^2 = npq$
- $\sigma^2 = N\frac{n}{K}$

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Question No : 35 of 52 Marks: 1 (Budgeted Time 1 Min)

Tabulation is the process of arranging data into:

VuAnswers.com

Answer (Please select your correct option)

- Different classes
- Rows
- Columns
- Rows and Columns

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Question No : 36 of 52 Marks: 1 (Budgeted Time 1 Min)

A simple bar chart consists of:

VuAnswers.com

Answer (Please select your correct option)

- Vertical bars
- Horizontal bars
- Vertical or horizontal bars
- Multiple bars

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Question No : 37 of 52 Marks: 1 (Budgeted Time 1 Min)

Which one is the formula to calculate the approximate value of class interval?

Answer (Please select your correct option)

VuAnswers.com

- Minimum value/Range
- Maximum value/No. of classes
- No. of classes/Range
- Range/No. of classes

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Question No : 38 of 52 Marks: 1 (Budgeted Time 1 Min)

Which one of the following is equal to the 2nd quartile:

Answer (Please select your correct option)

VuAnswers.com

- P_{33}
- D_3
- Median
- Mode

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Question No : 39 of 52 Marks: 1 (Budgeted Time 1 Min)

For the independent events A and B if P (A) = 0.15, P (B) =0.50 then P (A∩B) =.....

Answer (Please select your correct option)

VuAnswers.com

- 0.50
- 0.075
- 0.125
- 0.060

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Question No : 40 of 52 Marks: 1 (Budgeted Time 1 Min)

Which one of the following measure is not based on all the observations?

Answer (Please select your correct option)

VuAnswers.com

- Arithmetic Mean
- Geometric Mean
- Harmonic Mean
- Mode

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Question No : 41 of 52 Marks: 2 (Budgeted Time 4 Min)

Is there any difference between distribution function and cumulative distribution function of discrete r.v. X?

Answer (Please click here to Add Answer)

VuAnswers.com

Normal Arial 12 B I U 100%

The distribution function of a random variable X, denoted by F(x), is defined by F(x) = P(X < x).
The function F(x) gives the probability of the event that X takes a value LESS THAN OR EQUAL TO a specified value x. The distribution function is abbreviated to *d.f.* and is also called the *cumulative distribution function (cdf)* as it is the cumulative probability function of the random variable X from the smallest value upto a *specific* value x.
Let us illustrate this concept with the help of the same example that we have been considering — that of the probability distribution of the sum of the dots when two fair dice are thrown.

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Question No : 42 of 52 Marks: 2 (Budgeted Time 4 Min)

What is meant by sampling distribution of sample proportion?

Answer (Please click here to Add Answer)

VuAnswers.com

The probability distribution of any statistic (such as the mean, the standard deviation, the proportion of successes in a sample, etc.) is known as its sampling distribution.

The sample proportion has different values in different samples.

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Question No : 43 of 52 Marks: 2 (Budgeted Time 4 Min)

If an automobile is driven on the average no more than 16000 Km per year then formulate the null and alternative hypothesis.

Answer (Please click here to Add Answer)

VuAnswers.com



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Question No : 44 of 52 Marks: 2 (Budgeted Time 4 Min)

While applying the chi-square goodness of fit test, if an expected frequency in any category is less than 5, then what we do to solve this problem.

Answer (Please click here to Add Answer)

VuAnswers.com



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Question No : 45 of 52 Marks: 3 (Budgeted Time 6 Min)

Write a short note on Quota Sampling.

Answer (Please click here to Add Answer)

VuAnswers.com

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Question No : 46 of 52 Marks: 3 (Budgeted Time 6 Min)

Write down the properties of hypergeometric distribution.

Answer (Please click here to Add Answer)

VuAnswers.com

Normal Arial 12 100%

The probability of success changes on each trial.
- The successive trials are not independent.
- The experiment is repeated a fixed number of times.

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Question No : 47 of 52 Marks: 3 (Budgeted Time 6 Min)

Write down critical region for the following hypothesis at 5% level of significance.

$H_0 : \mu = 75$

$H_1 : \mu < 75$

Answer (Please click here to Add Answer)

VuAnswers.com

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Question No : 48 of 52 **Marks: 3 (Budgeted Time 6 Min)**

For the following table we want to test the independence of smoking pattern and marital status. What will be the degrees of freedom for the chi-square test of independence?

Marital Status	Smoking Pattern			Total
	Total self-restraint	Only at times	Regular Smoker	
Single	67	213	74	354

Answer (Please click here to Add Answer)

VuAnswers.com

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Question No : 48 of 52 Marks: 3 (Budgeted Time 6 Min)

Married	411	633	129	1173
Widowed	85	51	7	143
Divorced	27	60	15	102
Total	590	957	225	1772

Answer (Please click here to Add Answer)

VuAnswers.com

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Question No : 49 of 52 **Marks: 5 (Budgeted Time 10 Min)**

Answer the following question using the given stem and leaf plot of the ages of the people who attended a musical event.

Stem	Leaf
1	2 2 3 3 4 4 5 5 5 5 5 5 6 6 6 6 6 6 7 7 7 7 8 8 9 9
2	3 3 4 8
3	6

Answer (Please click here to Add Answer)

VuAnswers.com

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Question No : 49 of 52 Marks: 5 (Budgeted Time 10 Min)

2	3 3 4 8
3	6
4	5 8

a) How many people attended the event?
b) What is the age of youngest attendee?

Answer (Please click here to Add Answer)

VuAnswers.com

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Question No : 49 of 52 Marks: 5 (Budgeted Time 10 Min)

a) How many people attended the event?
b) What is the age of youngest attendee?
c) What was the age of oldest person attending event?
d) Which age group was more widely represented?
e) How many people were above 30?

Answer (Please click here to Add Answer)

VuAnswers.com

Normal Arial 12 B I U 100%

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Question No : 50 of 52 Marks: 5 (Budgeted Time 10 Min)

If $\bar{X}$ is normally distributed with $\mu_{\bar{x}} = \mu = 24000$ and $\sigma_{\bar{x}} = 412.20$

then find out the $P(\bar{X} > 24500)$.

Answer (Please click here to Add Answer)

VuAnswers.com



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Question No : 51 of 52 Marks: 5 (Budgeted Time 10 Min)

If we have $\bar{X} = 15$ and $\sum_{i=1}^{10}(X_i - \bar{X})^2 = 56$

Then, find the 90% confidence interval for population variance (σ^2).

Answer (Please click here to Add Answer)

VuAnswers.com

Normal Arial 12 100%

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Question No : 52 of 52 Marks: 5 (Budgeted Time 10 Min)

A research worker wishes to estimate the mean of a population using a sample sufficiently large that the probability will be 0.95 that the sample mean will not differ from the true mean by more than 25 percent of the standard deviation. How large a sample should be taken?

Answer (Please click here to Add Answer)

VuAnswers.com



Edited By Mahar Afaq Safdar Muhammadi

دعاؤں میں یاد رکھیے گا۔ جزاک اللہ

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FINAL TERM EXAMINATION
Spring 2012
STA301- Statistics and Probability

Question No: 1 (Marks: 1) - Please choose one
For a particular data the value of Pearson's coefficient of skewness is greater then zero. What will be the shape of distribution?

► Negatively skewed
► J-shaped
► Symmetrical
► **Positively skewed (Page 109)**

Question No: 2 (Marks: 1) - Please choose one
In measures of relative dispersion unit of measurement is:

► Changed
► **Vanish**
► Does not changed
► Dependent

Question No: 3 (Marks: 1) - Please choose one
The F-distribution always ranges from:

► 0 to 1
► 0 to -∞
► -∞ to +∞
► **0 to +∞ (Page 312)**

دنیا میں سب سے مشکل کام اپنی اصلاح اور سب سے آسان کام دوسروں پر نکتہ چینی کرنا ہے

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Question No: 4 (Marks: 1) - Please choose one
In chi-square test of independence the degrees of freedom are:
► **n - p**
► n - p-1
► n - p- 2
► n – 2

Question No: 5 (Marks: 1) - Please choose one
The Chi- Square distribution is continuous distribution ranging from:
► $-\infty \leq \chi 2 \leq \infty$
► $-\infty \leq \chi 2 \leq 1$
► $-\infty \leq \chi 2 \leq 0$
► **$0 \leq \chi 2 \leq \infty$ (Page 307)**

Question No: 6 (Marks: 1) - Please choose one
If X and Y are random variables, then $E(X-Y)$ is equal to:

► $E(X)+E(Y)$
► $E(X)-E(Y)$ **(Page 202)**
► $X-E(Y)$
► $E(X)-Y$

Question No: 7 (Marks: 1) - Please choose one
If ŷ is the predicted value for a given x-value and b is the y-intercept then the equation of a regression line for an independent variable x and a dependent variable y is:

► **ŷ = mx + b, where m = slope (Page 121)**
► x = ŷ + mb, where m = slope
► ŷ = x/m + b, where m = slope
► ŷ = x + mb, where m = slope

Question No: 8 (Marks: 1) - Please choose one
The location of the critical region depends upon:

► Null hypothesis
► **Alternative hypothesis (Page 281)**
► Value of alpha
► Value of test-statistic

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FINAL TERM EXAMINATION
Spring 2012
STA301- Statistics and Probability

Question No: 1 (Marks: 1) - Please choose one
The t-Distribution is……. Spread out then the standard normal Distribution.
► Less
►**More (Page 293)**
►Equally
►Not

Question No: 2 (Marks: 1) - Please choose one
To find the confidence internal for the ratio of two variances we use:
►**F-Distribution (Page 311)**
►Z-Distribution
►Chi-Square Distribution
►T-Distribution

Question No: 3 (Marks: 1) - Please choose one
How many percent of values are less then 4th deciles in a symmetric distribution.
►14
►24
►4
►**40 (Page 70)**

Question No: 4 (Marks: 1) - Please choose one
The combined distribution of more than two random variables is:
►Bivariate Distribution
►Marginal Distribution
►**Joint Distribution (Page 194)**
►Univariate Distribution

Question No: 5 (Marks: 1) - Please choose one
The degrees of freedom for a T-test with sample size 14 is:
►14
►**13 (Page 341)**
►**7**
►0



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Question No: 6 (Marks: 1) - Please choose one
Which of the falling is true for the binomial distribution b(x: n, p):
►**Mean > Variance (Page 214)**
►Mean < Variance
►Mean = Variance
►Mean = Standard Deviation

Question No: 7 (Marks: 1) - Please choose one
What is *m f* in the formula of mode?
►first frequency
►last frequency
►middle frequency
►**highest frequency (Page 54)**

FINAL TERM EXAMINATION
Fall 2011
STA301- Statistics and Probability

Question No: 1 (Marks: 1) - Please choose one
The parameters of the binomial distribution b(x; n, p) are:

► x & n
► x & p
► **n & p (Page 212)**
► x, n & p

Question No: 2 (Marks: 1) - Please choose one
Which of the following is true for the Poisson distribution:

► mean > variance
► mean < variance
► **mean = variance (Page 223)**
►mean= standard deviation

اللہ کا خوف سب سے بڑی دانائی ہے

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Question No: 3 (Marks: 1) - Please choose one
If a significance level of 1% is used rather than 5%, the null hypothesis is:

- ► More likely to be rejected
- ► **Less likely to be rejected** Click here for detail
- ► Just as likely to be rejected
- ► None of the above

Question No: 4 (Marks: 1) - Please choose one
The variance of the chi-square distribution is:

- ► $2v$ **(Page 307)**
- ► $v-1$
- ► $v-2$
- ► v

Question No: 5 (Marks: 1) - Please choose one
The degrees of freedom for a t-test with sample size 10 is:

- ►5
- ► 8
- ► **9 (Page 298)**
- ► 10

FINAL TERM EXAMINATION
Spring 2010
STA301- Statistics and Probability (Session - 4)

Question No: 1 (Marks: 1) - Please choose one
The value of χ^2 can never be :

- ► Zero–
- ► Less than 1
- ► Greater than 1
- ► **Negative (Page 307)**

ایماندار کو غصہ دیر سے آتا ہے اور جلدی دور ہو جاتا ہے

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Question No: 2 (Marks: 1) - Please choose one
The mean of the F-distribution is:

- $\frac{v_1}{v_1 - 2} \quad for v_1 \rangle 2$
- $\frac{v_2}{v_2 - 2} \quad for v_2 \rangle 2$ **(Page 312)**
- $\frac{v_1}{v_1 - 2} \quad for v_1 \geq 2$
- $\frac{v_2}{v_2 - 2} \quad for v_1 \leq 2$

Question No: 3 (Marks: 1) - Please choose one
The F-distribution always ranges from:

- 0 to 1
- 0 to -∞
- -∞ to +∞
- **0 to +∞ (Page 312) rep**

Question No: 4 (Marks: 1) - Please choose one
The total number of samples when sampling is done with replacement :

- N^n **(Page 237)**
- C_n^N
- $\frac{N-n}{N-1}$
- 1

زندگی میں کامیابی کا یہی راز ہے کہ پریشانیوں سے پریشان مت بنو

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Question No: 5 (Marks: 1) - Please choose one
ANOVA was introduced by :

- Helmert
- Pearson
- **R.A Fisher (Page 320)**
- Francis

Question No: 6 (Marks: 1) - Please choose one
The test statistic used in analysis of variance procedure follow the ..... distribution.:

- χ^2
- T
- Z
- **F (Page 326)**

Question No: 7 (Marks: 1) - Please choose one
For testing of hypothesis about population proportion , we use:

- **Z-test (Page 292)**
- t-Test
- Both Z & T-test
- F test

Question No: 8 (Marks: 1) - Please choose one
If X and Y are random variables, then $E(X-Y)$ is equal to:

- $E(X)+E(Y)$
- $E(X)-E(Y)$ **(Page 202) rep**
- $X-E(Y)$
- $E(X)-Y$

Question No: 9 (Marks: 1) - Please choose one
A die is rolled. What is the probability that the number rolled is greater than 2 and even:

- 1/2
- **1/3 2/6 = 1/3**
- 2/3
- 5/6

Hint: - number that is greater than 2 and is even = 4,6 i.e only 2 numbers out of 6
Therefore the probability 2/6 = 1/3

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Question No: 10 (Marks: 1) - Please choose one

The probability of drawing a king of spade from a pack of 52 cards is:

- ► 1/4
- ► 1/13
- ► 1/26
- ► **1/52**

Question No: 11 (Marks: 1) - Please choose one

An estimator T is said to be unbiased estimator of θ if

- ► **E (T) = θ (Page 258)**
- ► E (T) =T
- ► E (T) =0
- ► E (T) =1

Question No: 12 (Marks: 1) - Please choose one

From point estimation, we always get:

- ► **Single value (Page 257)**
- ► Two values
- ► Range of values
- ► Zero

Question No: 13 (Marks: 1) - Please choose one

The best unbiased estimator for population variance σ^2 is:

- ► Sample mean
- ► Sample median
- ► Sample proportion
- ► **Sample variance (Page 260)**

Question No: 14 (Marks: 1) - Please choose one

When c is a constant, then E(c) is:

- ► 0
- ► 1
- ► **c (Page 180)**
- ► -c



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Question No: 15 (Marks: 1) - Please choose one
Var(4X + 5) =_____

- ► 16 Var (X)
- ► **16 Var (X) + 5 (correct)**
- ► 4 Var (X) + 5
- ► 12 Var (X)

Question No: 16 (Marks: 1) - Please choose one
When f(x) is continuous probability function, then P(X = 1) is:

- ► 1
- ► ∞
- ► $-\infty$
- ► **0 (Page 188)**

Question No: 17 (Marks: 1) - Please choose one
The hyper geometric random variable is a(an):

- ► Continuous variable
- ► **Discrete variable** Click here for detail
- ► Undefined
- ► Independent variable

Question No: 18 (Marks: 1) - Please choose one
From a sample of 200 people were asked whether they like a particular product. Fifty said 'yes' and remain said 'no', assuming ‘yes’ means a success, which of the following is correct?

- ► Sample proportion p=0.33
- ► **Sample proportion p=0.25 (Page 245)**
- ► Population proportion p= 0.33
- ► Population proportion p=0.25

Question No: 19 (Marks: 1) - Please choose one
In any data set, what percent of values fall in the interval $Median \pm Q.D$?

- ► **50 per cent (Page 84)**
- ► 68.5 per cent
- ► 95.4 per cent
- ► 99 per cent

جھوٹ انسان اور ایمان دونوں کا دشمن ہے

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Question No: 20 (Marks: 1) - Please choose one

$$\sum_{i=1}^{5}(X_i - 20) = 0, then\ \bar{X} = ...$$

If

- **0 (Page 258)**
- 20
- 5
- 25

Question No: 21 (Marks: 1) - Please choose one

The height of a student is 60 inches. This is an example of .....?

- **Continuous data (correct)**
- Qualitative data
- Categorical data
- Discrete data

Question No: 22 (Marks: 1) - Please choose one

In Statistics, we have MSE which is abbreviation of……

- **Mean square error (Page 330)**
- Measured square error
- Medical screening exam
- Major sampling error

Question No: 23 (Marks: 1) - Please choose one

Which one is the formula of mid range:

- $x_m - x_0$
- $x_0 - x_m$
- $\frac{x_0 - x_m}{2}$
- $\frac{x_0 + x_m}{2}$ **(Page 80)**

عقل مند کہتا ہے میں کچھ نہیں جانتا جبکہ بے وقوف کہتا ہے کہ میں سب کچھ جانتا ہوں

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Question No: 24 (Marks: 1) - Please choose one
The deviation of a distribution from symmetry is called:

► Kurtosis
► **Skewness**
► Dispersion
► Flatness

Question No: 25 (Marks: 1) - Please choose one
If E is an impossible event, then P(E) is:

► 1
► 2
► **0 (Page 146)**
► 0.5

Question No: 26 (Marks: 1) - Please choose one
If a data set has the even number of observations, the median :
► Is the average value of the two middle items
► Can not be determined
► must be equal to the mean
► **Is the average value of the two middle items when all items are arranged in ascending order**
Click here for detail

Question No: 27 (Marks: 1) - Please choose one
For the Poisson distribution $P(X=1)=\frac{e^{-0.135}0.135^{1}}{1!}$ the mean value is :
► 2
► 5
►10
► **0.135 (Page 222)**

Question No: 28 (Marks: 1) - Please choose one
In testing of hypothesis, we always begin it with assuming that:

► **Null hypothesis is true (Page 277)**
► Alternative hypothesis is true
► Sample size is large
► Population is normal

خود کو تمھیں سے بڑھ کر کوئی اچھا مشورہ نہیں دے سکتا

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Question No: 29 (Marks: 1) - Please choose one

Variance of the t-distribution is given by the formula:

$$\sigma^2 = \sqrt{\frac{\nu}{\nu - 2}}$$

►

$$\sigma^2 = \frac{\nu^2}{\nu - 2}$$

►

$$\sigma^2 = \frac{\nu}{\nu - 1}$$

►

$$\sigma^2 = \frac{\nu}{\nu - 2}$$ **(Page 293)**

►

Question No: 30 (Marks: 1) - Please choose one

If a continuous probability distribution has $\beta_2 = 2.14$ then what will be peakedness of the distribution?

► **Platykurtic (Page 119)**
► Mesokurtic
► Leptokutic
► Moderately skewed

FINAL TERM EXAMINATION
Spring 2010
STA301- Statistics and Probability (Session - 4)

Question No: 1 (Marks: 1) - Please choose one

When each outcome of a sample space has equal chance to occur as any other, the outcomes are called:

► Mutually exclusive
► **Equally likely (Page 117)**
► Not mutually exclusive
► Exhaustive

جو شخص ناکامیوں سے ڈر کر بھاگتا ہے کامیابی اُس سے ڈر کر بھاگتی ہے

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Question No: 2 (Marks: 1) - Please choose one
The mean of the F-distribution is:

- $\frac{v_1}{v_1 - 2} \quad for v_1 \rangle 2$
- $\frac{v_2}{v_2 - 2} \quad for v_2 \rangle 2$ **(Page 312)**
- $\frac{v_1}{v_1 - 2} \quad for v_1 \geq 2$
- $\frac{v_2}{v_2 - 2} \quad for v_1 \leq 2$
-

Question No: 3 (Marks: 1) - Please choose one
The LSD test is applied only if the null hypothesis is:

- **Rejected (Page 331)**
- Accepted
- No conclusion
- Acknowledged

Question No: 4 (Marks: 1) - Please choose one
Analysis of variance is a procedure that enables us to test the equality of several:

- Variances
- **Means (Page 320)**
- Proportions
- Groups

Question No: 5 (Marks: 1) - Please choose one
ANOVA was introduced by :

- Helmert
- Pearson
- **R.A Fisher (Page 320) rep**
- Francis

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Question No: 6 (Marks: 1) - Please choose one
For testing of hypothesis about population proportion , we use:

- ► **Z-test (Page 292) rep**
- ► t-Test
- ► Both Z & T-test
- ► F test

Question No: 7 (Marks: 1) - Please choose one
If a random variable X denotes the number of heads when three distinct coins are tossed, the X assumed the values:

- ► **0,1,2,3**
- ► 1,3,3,1
- ► 1, 2, 3
- ► 3, 2

Question No: 8 (Marks: 1) - Please choose one
If X and Y are independent variables, then E (XY) is:

- ► E(XX)
- ► **E(X).E(Y) (Page 202)**
- ► X.E(Y)
- ► Y.E(X)

Question No: 9 (Marks: 1) - Please choose one
The parameters of the binomial distribution b(x; n, p) are:

- ► x & n
- ► x & p
- ► **n & p (Page 212) rep**
- ► x, n & p

Question No: 10 (Marks: 1) - Please choose one
If P (E) is the probability that an event will occur, which of the following must be false:

- ► **P(E)= - 1**
- ► P(E)=1
- ► P(E)=1/2
- ► P(E)=1/3

جو لوگوں کے سامنے فخر کرتا ہے وہ لوگوں کی نظروں سے گر جاتا ہے

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Question No: 11 (Marks: 1) - Please choose one

An estimator T is said to be unbiased estimator of θ if

- **E (T) = θ (Page 258) rep**
- E (T) =T
- E (T) =0
- E (T) =1

Question No: 12 (Marks: 1) - Please choose one

The best unbiased estimator for population variance σ^2 is:

- **Sample mean (Page 260) rep**
- Sample median
- Sample proportion
- Sample variance

Question No: 13 (Marks: 1) - Please choose one

$$S^2 = \frac{\sum(x-\bar{x})^2}{n}$$

The sample variance is:

- Unbiased estimator of σ^2
- **Biased estimator of σ^2 (Page 260)**
- Unbiased estimator of μ
- None of these

Question No: 14 (Marks: 1) - Please choose one

When c is a constant, then E(c) is:

- 0
- 1
- **c (Page 180) rep**
- -c

عقل مند اپنے عیب خود دیکھتا ہے اور بیوقوفوں کے عیب دنیا دیکھتی ہے

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Question No: 15 (Marks: 1) - Please choose one
If f (x, y) is bivariate probability density function of continuous r.v.'s X and Y then

$g(x)$ is:

- $\int_{-\infty}^{\infty} f(x,y)\,dx$
- $\int_{-\infty}^{\infty} f(x,y)\,dy$
- $\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\,dx\,dy$ **(Page 199)**
- $\int_{a}^{b}\int_{c}^{d} f(x,y)\,dy\,dx$
-

Question No: 16 (Marks: 1) - Please choose one
The analysis of variance technique is a method for :

- Comparing F distributions
- **Comparing three or more means (Page 320)**
- Measuring sampling error
- Comparing variances

Question No: 17 (Marks: 1) - Please choose one
The continuity correction factor is used when:

- The sample size is at least 5
- Both *nP* and *n (1-P)* are at least 30
- **A continuous distribution is used to approximate a discrete distribution** Click here for detail
- The standard normal distribution is applied

Question No: 18 (Marks: 1) - Please choose one
Stem and leaf is more informative when data is :

- Equal to 100
- Greater Than 100
- **Less than 100** click here for detail
- In all situations

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Question No: 19 (Marks: 1) - Please choose one
The branch of Statistics that is concerned with the procedures and methodology for obtaining valid conclusions is called:

- Descriptive Statistics
- Advance Statistics
- **Inferential Statistics (Page 237)**
- Sampled Statistics

Question No: 20 (Marks: 1) - Please choose one
Which of the following is a systematic arrangement of data into rows and columns?

- Classification
- **Tabulation**
- Bar chart
- Component bar chart

Question No: 21 (Marks: 1) - Please choose one
In normal distribution Q.D =

- 0.5σ
- 0.75σ
- 0.7979σ
- 0.6745σ Click here for detail

Question No: 22 (Marks: 1) - Please choose one
In normal distribution $\beta_2 =$

- 1
- 2
- **3 (Page 119)**
- 0

Question No: 23 (Marks: 1) - Please choose one
If you connect the mid-points of rectangles in a histogram by a series of lines that also touches the x-axis from both ends, what will you get?

- Ogive
- Frequency polygon
- **Frequency curve (Page 38)**
- Historigram

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Question No: 24 (Marks: 1) - Please choose one
Which one of the following statements is true regarding a population?

- **It must be a large number of values (Page 12)**
- It must refer to people
- It is a collection of individuals, objects, or measurements
- It is small part of whole

Question No: 25 (Marks: 1) - Please choose one
When $Q_1 = 2$ *and* $Q_3 = 4$,what is the value of Median, if the distribution is symmetrical:

- 1
- **2**
- 3
- 4

Question No: 26 (Marks: 1) - Please choose one
In a simple linear regression model, if it is assumed that the intercept parameter is equal to zero, then:

- The regression line will pass through the origin
- The regression line will pass through the point (0,10).
- The regression line will pass through the point (0,-10).
- **The slope of the line will also be equal to 0.**

Question No: 27 (Marks: 1) - Please choose one
The degrees of freedom for a t-test with sample size 10 is:

- 5
- 8
- **9 (Page 298) rep**
- 10

Question No: 28 (Marks: 1) - Please choose one
In testing of hypothesis, we always begin it with assuming that:

- **Null hypothesis is true (Page 277) rep**
- Alternative hypothesis is true
- Sample size is large
- Population is normal

بدصورت چہرہ بدصورت دماغ سے بہتر ہے

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Question No: 29 (Marks: 1) - Please choose one
A failing student is passed by an examiner is an example of:

- Type I error
- **Type II error**
- Correct decision
- No information regarding student exams

Question No: 30 (Marks: 1) - Please choose one
How to find $P(X+Y \le 1)$?

- f(0, 0) + f(0, 1) + f(1, 2)
- f(2, 0) + f(0, 1) + f(1, 0)
- f(0, 0) + f(1, 1) + f(1, 0)
- f(0, 0) + f(0, 1) + f(1, 0)

FINAL TERM EXAMINATION
Spring 2010
STA301- Statistics and Probability (Session - 3)

Question No: 1 (Marks: 1) - Please choose one
The total number of samples when sampling is done with replacement :

- N^n **(Page 237) rep**
- C_n^N
- $\frac{N-n}{N-1}$
-
- 1

عقل مند آدمی اس وقت تک نہیں بولتا جب تک خاموشی نہیں ہو جاتی

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Question No: 2 (Marks: 1) - Please choose one
The test statistic used in analysis of variance procedure follow the ..... distribution.:

- ► χ^2
- ► T
- ► Z
- ► **F (Page 326) rep**

Question No: 3 (Marks: 1) - Please choose one
If a random variable X denotes the number of heads when three distinct coins are tossed, the X assumed the values:

- ► **0,1,2,3 rep**
- ► 1,3,3,1
- ► 1, 2, 3
- ► 3, 2

Question No: 4 (Marks: 1) - Please choose one
If X and Y are independent variables, then E (XY) is:

- ► E(XX)
- ► **E(X).E(Y) (Page 202) rep**
- ► X.E(Y)
- ► Y.E(X)

Question No: 5 (Marks: 1) - Please choose one

$$S^2 = \frac{\sum(x-\bar{x})^2}{n}$$

The sample variance is:

- ► Unbiased estimator of σ^2
- ► **Biased estimator of σ^2 (Page 260) rep**
- ► Unbiased estimator of μ
- ► None of these

انسان دکھ نہیں دیتے بلکہ انسانوں سے وابستہ امیدیں دکھ دیتی ہیں

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Question No: 6 (Marks: 1) - Please choose one
When c is a constant, then E(c) is:

- ► 0
- ► 1
- ► **c (Page 180) rep**
- ► -c

Question No: 7 (Marks: 1) - Please choose one
When the random variable X and Y are independent, its co-variance is:

- ► One
- ► Negative
- ► **Zero (Page 207)**
- ► Positive

Question No: 8 (Marks: 1) - Please choose one
When f (x, y) is bivariate probability density function of continuous r.v.'s X and Y, then

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\,dx\,dy$$

is equal to:

- ► **1 (Page 199)**
- ► 0
- ► -1
- ► ∞

Question No: 9 (Marks: 1) - Please choose one
Which dispersion is calculated from all the observations?

- ► **Standard deviation**
- ► Quartile deviation
- ► Rang
- ► Coefficient of Rang

Question No: 10 (Marks: 1) - Please choose one
Standard deviation of the data 7, 7, 7, 7, 7, 7, 7 is

- ► 49
- ► Sqrt (7)
- ► **0 Standard deviation will always be zero if all the values in data are same**
- ► 7

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Question No: 11 (Marks: 1) - Please choose one
Which one is the poor measure of dispersion in open-end distribution?

- **Range**
- Standard deviation
- Mean deviation
- Variance

Question No: 12 (Marks: 1) - Please choose one
Men tend to marry women who are slightly younger than themselves. Suppose that every man married a woman who was exactly 5 years younger than themselves. Which of the following is correct:

- The correlation is −5
- The correlation is 5
- **The correlation is 1** Click here for detail
- The correlation is 0

Question No: 13 (Marks: 1) - Please choose one
Sum of absolute deviations of the values is least when deviations are taken from:

- **Mean**
- Median
- Mode
- Geometric mean

Question No: 14 (Marks: 1) - Please choose one
Which of the following measures of central location is affected most by extreme values:

- Geometric Mean
- Median
- **Mean** Click here for detail
- Mode

Question No: 15 (Marks: 1) - Please choose one
Which of the following is a critical value of Z when $1-\alpha = 95\%$ for one tailed test:

- 2.58
- 1.645
- 1.25
- **1.96**

بہترین تجربہ وہ ہے جس سے نصیحت حاصل ہو

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Question No: 16 (Marks: 1) - Please choose one
In normal distribution Q.D =

- 0.5σ
- 0.75σ
- 0.7979σ
- 0.6745σ Click here for detail rep

Question No: 17 (Marks: 1) - Please choose one
The difference between expected value of statistic and parameter is called:

- Non-sampling error
- Sampling error
- Standard error
- **Bias** Click here for detail

Question No: 18 (Marks: 1) - Please choose one
In Statistics, we have MSE which is abbreviation of……

- **Mean square error (Page 330) rep**
- Measured square error
- Medical screening exam
- Major sampling error

Question No: 19 (Marks: 1) - Please choose one
The following data shows the number of hours worked by 200 statistics students.

Number of Hours	Frequency
0 - 9	40
10 - 19	50
20 - 29	70
30 - 39	40

What is its class interval?

- 9
- **10**
- 11
- Varies from class to class

خوبصورتی علم وادب سے ہوتی ہے لباس وحسن سے نہیں

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Question No: 20 (Marks: 1) - Please choose one
Which one is the formula of range:

- $x_m - x_0$
- $x_0 - x_m$
- $\frac{x_0 - x_m}{2}$
- $\frac{x_0 + x_m}{2}$ **(Page 80)**

Question No: 21 (Marks: 1) - Please choose one
Which one is the formula of Geometric mean for group data:

- $anti\log\left[\frac{\sum f \log X}{n}\right]$
- $anti\log\left[\frac{\sum \log X}{n}\right]$ **(Page 75)**
- $anti\log\left[\frac{\sum \log f X}{n}\right]$
- $anti\log\left[\frac{\sum \log f X}{\sum_{i=1}^{n} f}\right]$

Question No: 22 (Marks: 1) - Please choose one
The F-distribution has …… parameter.

- One
- No
- **Two (Page 312)**
- Three

تم اچھا کرو زمانہ تم کو برا سمجھے یہ اس سے بہتر ہے کہ تم برا کرو اور زمانہ تم کو اچھا سمجھے

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Question No: 23 (Marks: 1) - Please choose one

The sample mean $\bar{X} = \frac{\Sigma X}{n}$ is an unbiased estimator of population mean (μ), because:

- $E(\bar{X}) = 0$
- $E(\bar{X}) = \mu$ **(Page 259)**
- $E(\bar{X}) > \mu$
- $E(\bar{X}) < \mu$

Question No: 24 (Marks: 1) - Please choose one

For a particular hypothesis test, $\alpha = 0.05$ and $\beta = 0.10$. The power of test equals to:

- 0.95
- 0.25
- **0.90**
- 0.14

Question No: 25 (Marks: 1) - Please choose one

The degrees of freedom for a t-test with sample size 10 is:

- 5
- 8
- **9 (Page 298) rep**
- 10

Question No: 26 (Marks: 1) - Please choose one

The degrees of freedom for a t-test with sample size 14 is:

- 14
- **13 (Page 341) rep**
- 7
- 0

Question No: 27 (Marks: 1) - Please choose one

The degrees of freedom for a t-test with sample size 6 is:

- 1
- 3
- **5 (Page 341)**
- 7

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Question No: 28 (Marks: 1) - Please choose one
In testing of hypothesis, we always begin it with assuming that:

- **Null hypothesis is true (Page 277) rep**
- Alternative hypothesis is true
- Sample size is large
- Population is normal

Question No: 29 (Marks: 1) - Please choose one
If a continuous probability distribution has $\beta_2 = 2.14$ then what will be peakedness of the distribution?

- **Platykurtic (Page 119) rep**
- Mesokurtic
- Leptokutic
- Moderately skewed

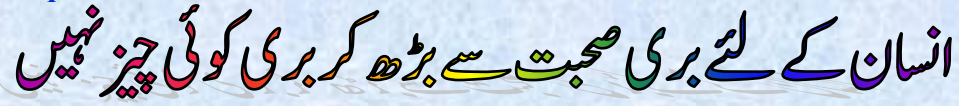
Question No: 30 (Marks: 1) - Please choose one
The difference between statistic and parameter is called:

- **Bias** Click here for detail **rep**
- Standard error
- Sampling error
- None of above

FINAL TERM EXAMINATION
Fall 2009
STA301- Statistics and Probability (Session - 4)

Question No: 1 (Marks: 1) - Please choose one
Mean deviation is always:

- Less than S.D
- Greater than S.D
- Greater or equal to S.D
- **Less or equal to S.D** Click here for detail



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Question No: 2 (Marks: 1) - Please choose one
The value of χ^2 can never be :

- Zero
- Less than 1
- Greater than 1
- **Negative (Page 307) rep**

Question No: 3 (Marks: 1) - Please choose one
The mean of the F-distribution is:

- $\frac{v_1}{v_1 - 2} \quad for v_1 \rangle 2$
- $\frac{v_2}{v_2 - 2} \quad for v_2 \rangle 2$ **(Page 312) rep**
- $\frac{v_1}{v_1 - 2} \quad for v_1 \geq 2$
- $\frac{v_2}{v_2 - 2} \quad for v_1 \leq 2$

Question No: 4 (Marks: 1) - Please choose one
If X and Y are random variables, then $E(X-Y)$ is equal to:

- $E(X)+E(Y)$
- $E(X)-E(Y)$ **(Page 202) rep**
- $X-E(Y)$
- $E(X)-Y$

خاموشی غصے کا بہترین علاج ہے

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Question No: 5 (Marks: 1) - Please choose one
Evaluate: (9-4)!

- ► 362880
- ► **120**
- ► 24
- ► 6

Question No: 6 (Marks: 1) - Please choose one
Which formula represents the probability of the complement of event A:

- ► 1 + *P* (*A*)
- ► **1 - *P* (*A*) (Page 156)**
- ► *P* (*A*)
- ► *P* (*A*) -1

Question No: 7 (Marks: 1) - Please choose one
Ideally the width of confidence interval should be:

- ► **0 (Page 270)**
- ► 1
- ► 99
- ► 100

Question No: 8 (Marks: 1) - Please choose one
If the sampling distribution of $\bar{X}$ is normal, the interval $\mu_{\bar{x}} \pm 3\sigma_{\bar{x}}$ includes:

- ► 99% of the sample means
- ► **99.73% of the sample means (Page 228)**
- ► 98% of the sample means
- ► 95% of the sample means

Question No: 9 (Marks: 1) - Please choose one
The probability distribution of a statistic is called the:

- ► Population distribution
- ► Frequency distribution
- ► **Sampling distribution** click here for detail
- ► Sample distribution

جھوٹ رزق کو کھا جاتا ہے

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Question No: 10 (Marks: 1) - Please choose one

An estimator T is said to be unbiased estimator of θ if

- **E (T) = θ (Page 258) rep**
- E (T) =T
- E (T) =0
- E (T) =1

Question No: 11 (Marks: 1) - Please choose one

If the following is a probability distribution, then what is the value of 'a':

X	1	2	3
P(X)	0.1	a	0.1

- 0.6
- **0.8**
- 0.2
- 0.4

Question No: 12 (Marks: 1) - Please choose one

A discrete probability function f(x) is always:

- Non-negative
- Negative
- **One (Page 168)**
- Zero

Question No: 13 (Marks: 1) - Please choose one

An expected value of a random variable is equal to:

- Variance
- **Mean (Page 191)**
- Standard deviation
- Covariance

افضل انسان وہ ہے جو اپنی اصلاح کی کوشش کرتا ہے

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Question No: 14 (Marks: 1) - Please choose one

$f(x|1) =$ The _____:

- $f(1,1)$
- $f(x,1)$
- $\frac{f(x,1)}{h(1)}$ **(Page 198)**
- $\frac{f(x,1)}{h(x)}$

Question No: 15 (Marks: 1) - Please choose one

The area under a normal curve between 0 and -1.75 is

- .0401
- .5500
- **.4599 (Page 230)**
- .9599

Question No: 16 (Marks: 1) - Please choose one

The continuity correction factor is used when:

- The sample size is at least 5
- Both *nP* and *n (1-P)* are at least 30
- **A continuous distribution is used to approximate a discrete distribution** Click here for detail
- The standard normal distribution is applied

Question No: 17 (Marks: 1) - Please choose one

Which of the following is impossible in sampling:

- Destructive tests
- **Heterogeneous**
- To make voters list
- None of these

اطمینان قلب چاہتے ہو تو حسد سے دور رہو

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Question No: 18 (Marks: 1) - Please choose one
Which of the following is a systematic arrangement of data into rows and columns?

- ► Classification
- ► **Tabulation rep**
- ► Bar chart
- ► Component bar chart

Question No: 19 (Marks: 1) - Please choose one
Which one of the following statements is true regarding a sample?

- ► **It is a part of population (Page 13)**
- ► It must contain at least five observations
- ► It refers to descriptive statistics
- ► It produces True value

Question No: 20 (Marks: 1) - Please choose one
The data for an ogive is found in which distribution?

- ► A relative frequency distribution
- ► A frequency distribution
- ► A joint frequency distribution
- ► **A cumulative frequency distribution (Page 43)**

FINAL TERM EXAMINATION
Fall 2009
STA301- Statistics and Probability

Question No: 1 (Marks: 1) - Please choose one
10! =……

- ► 362880
- ► **3628800**
- ► 362280
- ► 362800



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Question No: 2 (Marks: 1) - Please choose one
When E is an impossible event, then P(E) is:

► 2
► **0 (Page 146) rep**
► 0.5
► 1

Question No: 3 (Marks: 1) - Please choose one
The value of χ^2 can never be :

► Zero
► Less than 1
► Greater than 1
► **Negative (Page 307) rep**

Question No: 4 (Marks: 1) - Please choose one
The curve of the F- distribution depends upon:

► **Degrees of freedom (Page 312)**
► Sample size
► Mean
► Variance

Question No: 5 (Marks: 1) - Please choose one
If X and Y are random variables, then $E(X-Y)$ is equal to:

► $E(X)+E(Y)$
► $E(X)-E(Y)$ **(Page 202) rep**
► $X-E(Y)$
► $E(X)-Y$

Question No: 6 (Marks: 1) - Please choose one
In testing hypothesis, we always begin it with assuming that:

► **Null hypothesis is true (Page 277) rep**
► Alternative hypothesis is true
► Sample size is large
► Population is normal

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Question No: 7 (Marks: 1) - Please choose one

$$\frac{\ell^{-0.135}\ 0.135^{1}}{1!}$$

For the Poisson distribution P(x) = the mean value is :

► 2
► 5
► 10
► **0.135 (Page 222) rep**

Question No: 8 (Marks: 1) - Please choose one

When two coins are tossed simultaneously, P (one head) is:

► $\frac{1}{4}$
► $\frac{1}{2}$
► $\frac{3}{4}$
► 1

Question No: 9 (Marks: 1) - Please choose one

From point estimation, we always get:

► **Single value (Page 257) rep**
► Two values
► Range of values
► Zero

Question No: 10 (Marks: 1) - Please choose one

$$S^2 = \frac{\sum(x-\bar{x})^2}{n}$$

The sample variance is:

► Unbiased estimator of σ^2
► **Biased estimator of σ^2 (Page 260) rep**
► Unbiased estimator of μ
► None of these

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Question No: 11 (Marks: 1) - Please choose one
Var(4X + 5) =_____

► 16 Var (X)
► **16 Var (X) + 5 rep**
► 4 Var (X) + 5
► 12 Var (X)

Question No: 12 (Marks: 1) - Please choose one
When f (x, y) is bivariate probability density function of continuous r.v.'s X and Y, then

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\,dx\,dy$$

is equal to:

► **1 rep**
► 0
► -1
► ∞

Question No: 13 (Marks: 1) - Please choose one
The area under a normal curve between 0 and -1.75 is

► .0401
► .5500
► **.4599 (Page 230) rep**
► .9599

Question No: 14 (Marks: 1) - Please choose one
When a fair die is rolled, the sample space consists of:
► 2 outcomes
► 6 outcomes
► **36 outcomes (Page 145)**
►16 outcomes

Question No: 15 (Marks: 1) - Please choose one
When testing for independence in a contingency table with 3 rows and 4 columns, there are _____ degrees of freedom.
► 5
► **6 (Page 341)**
► 7
► 12

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Question No: 16 (Marks: 1) - Please choose one
The F- test statistic in one-way ANOVA is:

- ► SSW / SSE
- ► MSW / MSE
- ► SSE / SSW
- ► **MSE / MSW (Not sure)**

Question No: 17 (Marks: 1) - Please choose one
The continuity correction factor is used when:

- ► The sample size is at least 5
- ► Both *nP* and *n (1-P)* are at least 30
- ► **A continuous distribution is used to approximate a discrete distribution rep**
- ► The standard normal distribution is applied

Question No: 18 (Marks: 1) - Please choose one
A uniform distribution is defined by:

- ► **Its largest and smallest value** click here for detail
- ► Smallest value
- ► Largest value
- ► Mid value

Question No: 19 (Marks: 1) - Please choose one
Which graph is made by plotting the mid point and frequencies?

- ► **Frequency polygon (Page 34)**
- ► Ogive
- ► Histogram
- ► Frequency curve

Question No: 20 (Marks: 1) - Please choose one
In a set of 20 values all the values are 10, what is the value of median?

- ► 2
- ► 5
- ► **10**
- ► 20

ہر چیز کی ایک پہچان ہوتی ہے اور عقلمند کی پہچان غور و فکر کرنا ہے اور غور و فکر کی پہچان خاموشی ہے

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FINAL TERM EXAMINATION
Fall 2009
STA301- Statistics and Probability

Question No: 1 (Marks: 1) - Please choose one
For a particular data the value of Pearson's coefficient of skewness is greater then zero. What will be the shape of distribution?

- ► **Negatively skewed**
- ► J-shaped
- ► Symmetrical
- ► **Positively skewed (Page 110)**

Question No: 2 (Marks: 1) - Please choose one
In measures of relative dispersion unit of measurement is:

- ► **Changed**
- ► **Vanish Rep**
- ► Does not changed
- ► Dependent

Question No: 3 (Marks: 1) - Please choose one
The F-distribution always ranges from:

- ► 0 to 1
- ► 0 to -∞
- ► -∞ to +∞
- ► **0 to +∞ (Page 312) rep**

Question No: 4 (Marks: 1) - Please choose one
In chi-square test of independence the degrees of freedom are:

- ► **n - p**
- ► n - p-1
- ► n - p- 2
- ► n – 2



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Question No: 5 (Marks: 1) - Please choose one
The Chi- Square distribution is continuous distribution ranging from:

- $-\infty \leq \chi^2 \leq \infty$
- $-\infty \leq \chi^2 \leq 1$
- $-\infty \leq \chi^2 \leq 0$
- **$0 \leq \chi^2 \leq \infty$ (Page 307) rep**

Question No: 6 (Marks: 1) - Please choose one
If X and Y are random variables, then $E(X-Y)$ is equal to:

- $E(X)+E(Y)$
- $E(X)-E(Y)$ **(Page 202) rep**
- $X-E(Y)$
- $E(X)-Y$

Question No: 7 (Marks: 1) - Please choose one
If ŷ is the predicted value for a given x-value and b is the y-intercept then the equation of a regression line for an independent variable x and a dependent variable y is:

- **ŷ = mx + b, where m = slope (Page 121) rep**
- x = ŷ + mb, where m = slope
- ŷ = x/m + b, where m = slope
- ŷ = x + mb, where m = slope

Question No: 8 (Marks: 1) - Please choose one
The location of the critical region depends upon:

- **Null hypothesis**
- **Alternative hypothesis (Page 281) rep**
- Value of alpha
- Value of test-statistic

وہ لوگ مبارک ہیں جو الفاظ سے نصیحت نہیں کرتے بلکہ عمل سے کرتے ہیں

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Question No: 9 (Marks: 1) - Please choose one
The variance of the t-distribution is give by the formula:

$$\sigma^2 = \sqrt{\frac{\nu}{\nu - 2}}$$

▶

$$\sigma^2 = \frac{\nu^2}{\nu - 2}$$

▶

$$\sigma^2 = \frac{\nu}{\nu - 1}$$

▶

$$\sigma^2 = \frac{\nu}{\nu - 2}$$ **(Page 293)**

▶

Question No: 10 (Marks: 1) - Please choose one
Which one is the correct formula for finding desired sample size?

$$n = \left(\frac{Z_{\alpha/2}.\sigma}{e} \right)^2$$ **(Page 276)**

▶

$$n = \left(\frac{Z_{\alpha/2}.\sqrt{\sigma}}{e} \right)^2$$

▶

$$n = \left(\frac{Z_{\alpha/2}.\bar{X}}{e} \right)^2$$

▶

$$n = \frac{Z_{\alpha/2}.\sigma}{e}$$

▶



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Question No: 11 (Marks: 1) - Please choose one
A discrete probability function f(x) is always:

- ► Non-negative
- ► Negative
- ► **One (Page 168) rep**
- ► Zero

Question No: 12 (Marks: 1) - Please choose one
E(4X + 5) =_____

- ► 12 E (X)
- ► **4 E (X) + 5**
- ► 16 E (X) + 5
- ► 16 E (X)

Question No: 13 (Marks: 1) - Please choose one
How P(X + Y < 1) can be find:

- ► f(0, 0) + f(0, 1) + f(1, 2)
- ► f(2, 0) + f(0, 1) + f(1, 0)
- ► f(0, 0) + f(1, 1) + f(1, 0)
- ► f(0, 0) + f(0, 1) + f(1, 0)

Question No: 14 (Marks: 1) - Please choose one
$f(x|1) =$
The _____:

- ► $f(1,1)$
- ► $f(x,1)$
- ► $\frac{f(x,1)}{h(1)}$ **(Page 198) rep**
- ► $\frac{f(x,1)}{h(x)}$
- ►

ہر کسی کی روٹی نہ کھا بلکہ ہر شخص کو اپنی روٹی کھلا

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Question No: 15 (Marks: 1) - Please choose one
The area under a normal curve between 0 and -1.75 is

- .0401
- .5500
- **.4599 (Page 230) rep**
- .9599

Question No: 16 (Marks: 1) - Please choose one
In normal distribution M.D. =

- 0.5σ
- 0.75σ
- 0.7979σ Click here for detail **rep**
- 0.6445σ

Question No: 17 (Marks: 1) - Please choose one
In an ANOVA test there are 5 observations in each of three treatments. The degrees of freedom in the numerator and denominator respectively are.....

- 2, 4
- 3, 15
- 3, 12
- **2, 12**

Question No: 18 (Marks: 1) - Please choose one
A set that contains all possible outcomes of a system is known as

- Finite Set
- Infinite Set
- **Universal Set (Page 134)**
- **No of these**

Question No: 19 (Marks: 1) - Please choose one
Stem and leaf is more informative when data is :

- Equal to 100
- Greater Than 100
- **Less than 100** click here for detail rep
- In all situations



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Question No: 20 (Marks: 1) - Please choose one
A population that can be defined as the aggregate of all the conceivable ways in which a specified event can happen is known as:

- ► Infinite population
- ► Finite population
- ► Concrete population
- ► **Hypothetical population (Page 12)**

STA 301 – Quizzes

Question # 1 of 10 (Total Marks: 1) Select correct option:
The number of telephone calls that pass through a switchboard has a Poison distribution with mean equal to 2 per minute. The probability that no telephone calls pass through the switchboard in two consecutive minutes is:

0.2707
0.0517
0.0183
0.0366

Question # 2 of 10 (Total Marks: 1) Select correct option:
The range of the binomial distribution is:
0, 1, 2, ... , 100
0, 1, 2, … , n Click here for detail
0, 1, 2, … , x
1, 2, … , n

Question # 3 of 10 (Total Marks: 1) Select correct option:
Which of the following is NOT CORRECT about a standard normal distribution?

$P(0 \leq Z \leq 1.50) = .4332$
$P(Z \geq 2.0) = .0228$
$P(Z \geq -2.5) = .4938$ (Page 230)
$P(Z \leq -1.0) = .1587$

دنیا حقیر نظر آتی ہے، جب غم یا خوشی کی انتہاء ہو جائے

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Question # 4 of 10 (Total Marks: 1) Select correct option:
The distribution function (df) is also known as cumulative distribution function (cdf).
Yes (Page 173)
No

Question # 5 of 10 (Total Marks: 1) Select correct option:
Which of the following pairs of events are mutually exclusive?

A:the numbers above100;B: the numbers less than-200
A:the odd numbers; B:the number 5
A:the even numbers; B:the numbers greater than 10
A:the numbers less than 5; B:all the negative numbers

Question # 6 of 10 (Total Marks: 1) Select correct option:
Two events A & B are said to be independent if...
P(A) + P(B)
P(B/A) = P(B)
P(A) * P(B) (Page 162)
P(A/B) = P(A)

Question # 7 of 10 (Total Marks: 1) Select correct option:
The collection of all outcomes for an experiment is called:

A sample space Click here for detail
Joint probability simple event
The intersection of events
Random experiment

Question # 8 of 10 (Total Marks: 1) Select correct option:
Symbolically, a conditional probability is:
P(AB)
P(A/B) (Page 159)
P(A)
P(AUB)

Question # 9 of 10 (Total Marks: 1) Select correct option:
Which of the following statements is INCORRECT about the sampling distribution of the sample mean?

The standard error of the sample mean will decrease as the sample size increases
The standard error of the sample mean is measure of the variability of the sample mean among repeated samples
The sample mean is unbiased for the true (unknown) population mean
The sampling distribution shows how the sample was distributed around the sample mean
Click here for detail

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Question # 10 of 10 (Total Marks: 1) Select correct option:
If one event is unaffected by the outcome of another event the two events are said to be:

1. Dependent
2. Not Mutually Exclusive
3. Mutually Exclusive

Independent Click here for detail

Question # 1 of 10 (Total Marks: 1) Select correct option:
Let X be a random variable with binomial distribution, that is (X=0,1,…, n). The expected value E[X] is

p
np (Page 214)
np(1-p)
Xnp

Question # 2 of 10 (Total Marks: 1) Select correct option:
The sample mean is an unbiased estimator for the population mean. This means:

The sample mean has a normal distribution
The average sample mean, over all possible samples, equals the population mean (Page 259)
The sample mean is always very close to the population mean
The sample mean will only vary a little from the population mean

Question # 4 of 10 (Total Marks: 1) Select correct option:
Probability of an impossible event is always:

Less than one
Greater than one
Between one and zero
Zero (Page 146)

Question # 5 of 10 (Total Marks: 1) Select correct option:
The function abbreviated to d.f. is also called the.....

Probability density function
Probability distribution function
Commutative distribution function (Page 173)
Discrete function



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Question # 6 of 10 (Total Marks: 1) Select correct option:
The total area under the normal curve is:

0
1 (Page 186)
0.5
0.75

Question # 7 of 10 (Total Marks: 1) Select correct option:
Two events A & B are said to be independent if....

P (A) + P (B)
P (B\A) = P (B)
P (A) * P (B) (Page 162) rep
P (A\B) = P (A)

Question # 8 of 10 (Total Marks: 1) Select correct option:
When two coins are tossed the probability of at most one head is:

1/4
2/4
3/4 rep
1

Question # 9 of 10 (Total Marks: 1) Select correct option:
For exhaustive events, the P(AUBUC) is equal to:

P(A)
P(S) (Page 147)
P(A) * P(B)* P(C)
P(B)

Question # 10 of 10 (Total Marks: 1) Select correct option:
One card is drawn from a standard 52 card deck. In describing the occurrence of two possible events, an Ace and a King, these two events are said to be:
Select correct option:
independent
randomly independent
random variables
mutually exclusive click here for detail

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STA 301 – Quizzes

Question # 1 of 10 (Total Marks: 1) Select correct option:
The number of parameters in hypergeometric distribution is (are):

1
2
3 (Page 219)
4

Question # 2 of 10 (Total Marks: 1) Select correct option:
If Y=bX, then variance of Y is

b*2 var(x)
var(x)
b var(x)
b square root var(x)

Question # 3 of 10 (Total Marks: 1) Select correct option:
If f(x) is a continuous probability function, then P(X = 2) is:

1
0 (Page 186)
1/2
2

Question # 4 of 10 (Total Marks: 1) Select correct option:
In regression line Y=a+bX, Y is called:

Dependent variable (Page 121)
Independent variable
Explanatory variable
Regressor

پہلے وقتوں میں تعلیم کم تھی اور علم زیادہ، آج کل علم کم ہے اور تعلیم زیادہ

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Question # 5 of 10 (Total Marks: 1) Select correct option:
If A and B are mutually exclusive events with P (A) =0.25 and P (B) = 0.50, Then P (A or B) =……

0.25
0.75 (Page 154)
0.50
1

Question # 6 of 10 (Total Marks: 1) Select correct option:
In a 52 well shuffled pack of 52 playing cards, the probability of drawing any one diamond card is

1/52
4/52
13/52
52/52

Question # 7 of 10 (Total Marks: 1) Select correct option:
Probability of a sure event is

8
1 (Page 154)
0
0.5

Question # 8 of 10 (Total Marks: 1) Select correct option:
If Y=3X+5,then S.D of Y is equal to

9 s.d(x)
3 s.d(x)
s.d(x)+5
3s.d(x)+5

Question # 9 of 10 (Total Marks: 1) Select correct option:
The probability of drawing a red queen card from well-shuffled pack of 52 playing cards is

4/52
2/52
13/52
26/52

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Question # 10 of 10 (Total Marks: 1) Select correct option:
If P (B|A) = 0.25 and P (A and B) =0.20, then P (A) is

0.05
0.80 (Page 159)
0.95
0.75

Question # 1 of 10 (Total Marks: 1) Select correct option:
A student solved 25 questions from first 50 questions of a book to be solved. The probability that he will solve the remaining all questions is:

0.25
0.5
1
0

Question # 1 of 10 (Total Marks: 1) Select correct option:
The classical definition of probability assumes:

Exhaustive events
Mutually exclusive events
Equally likely evens (Page 151)
Independent evens

Question # 2 of 10 (Total Marks: 1) Select correct option:
In scatter diagram, the variable plotted along Y-axis is:
Independent variable
Dependent variable
Continuous variable
Discrete variable

Question # 3 of 10 (Total Marks: 1) Select correct option:
Which of the following measures of dispersion are based on deviations from the mean?

Variance
Standard deviation
Mean deviation
All of the these

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Question # 4 of 10 (Total Marks: 1) Select correct option:
What does it mean when a data set has a standard deviation equal to zero?
All values of the data appear with the same frequency.
The mean of the data is also zero.
All of the data have the same value. Click here for detail
There are no data to begin with.

Question # 5 of 10 (Total Marks: 1) Select correct option:
A set of possible values that a random variable can assume and their associated probabilities of occurrence are referred to as _____.

Probability distribution
The expected return
The standard deviation
Coefficient of variation

Question # 6 of 10 (Total Marks: 1) Select correct option:
Which of the following can never be probability of an event?

0
1
0.5
-0.5

Question # 7 of 10 (Total Marks: 1) Select correct option:
The standard deviation of -1, -1, -1, -1 will be

1
-1
0
Does not exist

Question # 8 of 10 (Total Marks: 1) Select correct option:
The Special Rule of Addition is used to combine:

Independent Events
Mutually Exclusive Events click here for detail
Events that total more than 1.00
Events based on subjective probabilities

بہادر آدمی ایک مرتبہ جبکہ بزدل آدمی کئی مرتبہ مرتا ہے

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Question # 9 of 10 (Total Marks: 1) Select correct option:
set which is the sub-set of every set is

Empty Set (Page 134)
Power Set
Universal Set
Super Set

Question # 10 of 10 (Total Marks: 1) Select correct option:
When two dice are rolled the number of possible sample points is :

6
12
24
36 (Page 145)

Question # 1 of 10 (Total Marks: 1) Select correct option:
If two events A and B are not mutually exclusive then

P (A or B) = P (A) + P (B) – P (A and B) (Page 157)
P (A or B) = P (A) + P (B)
P (A or B) = P (A) x P (B)
P (A or B) = P (A) + P (B)

Question # 2 of 10 (Total Marks: 1) Select correct option:
Evaluate (10-4)!

1000
720
480
32

Question # 3 of 10 (Total Marks: 1) Select correct option:
When we toss a coin , we get only:

1 outcome
2 outcome
3 outcome
4 outcome



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Question # 4 of 10 (Total Marks: 1) Select correct option:
If we roll a die then probability of getting a '6' will be

2/6
1/6
4/6
1

Question # 5 of 10 (Total Marks: 1) Select correct option:
If P(A) = 0.45, P(B) = 0.35, and P(A and B) = 0.25, then P(A | B) is:

1.4
1.8
0.714 **0.25/0.35 = 0.714** **(Page 159)**
0.556

Question # 6 of 10 (Total Marks: 1) Select correct option:
Which of the following is not a measure of central tendency?

Percentile
Quartile
Standard deviation
Mode

Question # 7 of 10 (Total Marks: 1) Select correct option:
Random experiment can be repeated any no. of times under the…… conditions.
Different
Similar **(Page 144)**

Question # 9 of 10 (Total Marks: 1) Select correct option:
The simultaneous occurrence of two events is called:
Joint probability **(Page 194)**
Subjective probability
Prior probability
Conditional probability

Question # 10 of 10 (Total Marks: 1) Select correct option:
In regression analysis, the variable that is being predicted is the

Dependent variable click here for detail
Independent variable
Intervening variable
None of these

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Question # 1 of 10 (Total Marks: 1) Select correct option:
The probability of continuous random variable x on any particular point is always zero..
Yes (Page 186)
No

Question # 2 of 10 (Total Marks: 1) Select correct option:
P(an event) =no of favorable outcome/total no. of outcomes is the definition of
Subjective approach (Page 149)
Objective approach

Question # 3 of 10 (Total Marks: 1) Select correct option:
If C is a constant ,then E(c)=
0
1
C (Page 180) rep
-c

Question # 4 of 10 (Total Marks: 1) Select correct option:
When we toss a fair coin 4 times, the sample space consists of….points.
4
8
12
16

Question # 5 of 10 (Total Marks: 1) Select correct option:
If we roll three fair dices then the total number of outcomes is:
6
36
216 6*6*6=216
1296

Question # 6 of 10 (Total Marks: 1) Select correct option:
The probability of an event is always:
greater than 0
less than 1
between o and 1 click here for detail
greater than 1

Question # 7 of 10 (Total Marks: 1) Select correct option:
In a multiplication theorem P (A and B) equals:
P(A)
P (B) P(A) + P (B)
P(A) * P (B|A) (Page 159)
P(B\A)*P(B)

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Question # 8 of 10 (Total Marks: 1) Select correct option:
If a die is rolled, what is the probability of getting an even number greater than 2?
1/2
1/3 2/6 = 1/3
2/3
5/6

Question # 9 of 10 (Total Marks: 1) Select correct option:
In a Discrete probability distribution, P (x > 23) is read as:
P (there are more than 23 successes)
P (there are less than 23 successes)
P (there are at least 23 successes)
P (there are at most 23 successes)

Question # 10 of 10 (Total Marks: 1) Select correct option:
A dormitory on campus houses 200 students. 120 are male, 50 are upper division students, and 40 are upper division male students.A student is selected at random. The probability of selecting a lower division student, given the student is a female, is:

7/8
7/20
7/15
1/4

Question # 1 of 10 (Total Marks: 1) Select correct option:
The function F(x) gives the probability of the event that X takes a value .....
Less than x
Greater or equal to x
Less or equal x (Page 173)
Equal to x

Question # 2 of 10 (Total Marks: 1) Select correct option:
In a simple regression line model ,it is assume that the intercept parameter is equal to zero,
The regression line will pass through the origin.
The regression line will pass through the point (0,10)
The regression line will pass through the point (0,-10)
The slope of the line will also be zero.

Question # 3 of 10 (Total Marks: 1) Select correct option:
If p(AnB)=p(A/B).p(B),then A and B are
Independent
Dependant
Equally likely
Mutually exclusively

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Question # 4 of 10 (Total Marks: 1) Select correct option:
A fair coin is tossed three times, the probability that at least one head appears,

1/8
7/8 (HHH,HHT,HTH, HTT, THH THT TTH TTT)
3/8
5/8

Question # 5 of 10 (Total Marks: 1) Select correct option:
In probability distribution, the sum of probabilities is equals to

0
0.1
0.5
1 click here for detail

Question # 6 of 10 (Total Marks: 1) Select correct option:
When a coin is tossed 3 times, the probability of getting 3 tails is

1/8 (HHH,HHT,HTH, HTT, THH THT TTH TTT)
3/8
3/6
2/8

Question # 7 of 10 (Total Marks: 1) Select correct option:
In how many ways can a team of 11 players be chosen from a total of 16 players?

4368
2426
5400
2680

Question # 8 of 10 (Total Marks: 1) Select correct option:
The standard deviation of c (constant) is

c
c square
0 click here for detail
does not exist

اچھائی کرنے کے لئے ہمیشہ کسی بہانے کی تلاش میں رہو

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Question # 9 of 10 (Total Marks: 1) Select correct option:
Let E and F be events associated with the same experiment. Suppose the E and F are independent and that P(E) = 1/4 and P(F) = 1/2 Then P(E U F) is:

1/8
3/4 **(Page 157)**
7/8
5/8

STA 301 – Quizzes

Question # 1 of 10 (Total Marks: 1) Select correct option:
In which of the following situations binomial distributions is approximate to normal distribution?

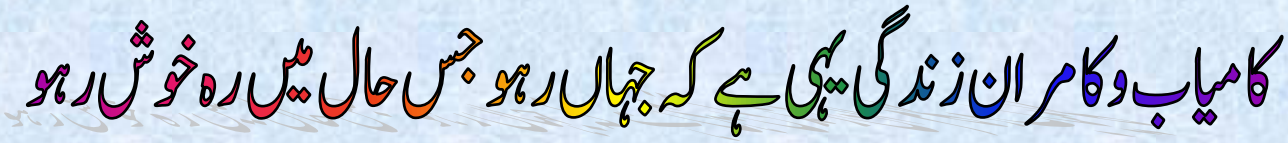
n = 50, p = 0.01
n = 500, p = 0.001
n = 100, p = 0.05
n = 50, p = 0.02 click here for detail

Question # 2 of 10 (Total Marks: 1) Select correct option:
The location and shape of the normal curve is (are) determined by:

Mean
Variance
Mean & variance
Mean & standard deviation click here for detail

Question # 3 of 10 (Total Marks: 1) Select correct option:
A random experiment has five outcomes in its sample space {s1, s2, s3, s4, s5}. If P(s1)=0.2,P(s2)=0.3, P(s3)=0.1 and P(s4)=0.2 then P(s5)=?

1
0.2
0.8
0.5



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Question # 4 of 10 (Total Marks: 1) Select correct option:
The joint density function f(x,y) will be a pdf if

Both of integrals f(x,y) dx dy =1 (Page 200)
Both of integrals f(x,y) dx dy >1
Both of integrals f(x,y) dx dy <1
Both of integrals f(x,y) dx dy =0

Question # 5 of 10 (Total Marks: 1) Select correct option:
Which of the following is correct property for joint probability distribution of X and Y:

Sigma f(X,Y)=1 (Page 194)
Sigma f(Y,X)=1
Both of above
None of above

Question # 6 of 10 (Total Marks: 1) Select correct option:
A random variable that can assume every possible value in an interval [a, b]:

Discrete variable
Continuous variable (Page 9)
Qualitative variable
Categorical variable

Question # 7 of 10 (Total Marks: 1) Select correct option:
Normal approximation to the binomial distribution is used when:

np>5
nq>5
Both of the above (Page 235)
None of the above

Question # 8 of 10 (Total Marks: 1) Select correct option:
The Maximum Likelihood Estimators (MLE) are …… and …… but not necessarily ……
Unbiased, consistent, efficient
Consistent, unbiased, efficient
Unbiased, efficient, consistent
Consistent, efficient, unbiased (Page 264)

کتنا شریف ہے وہ غمزدہ دل جو سب کو خوش کرنے کی کوشش کرتا ہے

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Question # 9 of 10 (Total Marks: 1) Select correct option:
A probability density function 'f(x)' has the following property:

f(x) <= 0
f(x) < 0
f(x) > 0 (Page 186)
f(x) >= 0

Question # 10 of 10 (Total Marks: 1) Select correct option:
For a continuous random variable X, P(X = x) is:
Select correct option:

0 (Page 186)
0.5
1
undefined

Question # 1 of 10 (Total Marks: 1) Select correct option:
An urn contains 4 red balls and 6 green balls. A sample of 4 balls is selected from the urn without replacement. It is the example of:

Binomial distribution
Hypergrometric distribution (Page 219)
Poisson distribution
Exponential distribution

Question # 2 of 10 (Total Marks: 1) Select correct option:
The probability distribution of the proportion of successes in all possible samples is called the:

Sampling distribution (Page 237)
Sampling probability distribution
Sampling distribution of sample proportions
Sampling distribution of Population proportions

Question # 3 of 10 (Total Marks: 1) Select correct option:
For good approximation of Poisson distribution to the binomial distribution, which of the following condition (s) is/are required:

The population size is large relative to the sample size
The probability, p, is close to .5 and the population size is large
The probability, p, is small and the sample size is large (Page 222)
The probability, p, is close to .5 and the sample size is large

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Question # 4 of 10 (Total Marks: 1) Select correct option:
If an estimator is more efficient then the other estimator, its shape of the sampling distribution will be

Flattered
Normal
Highly peaked (Page 261)
Skewed to right

Question # 6 of 10 (Total Marks: 1) Select correct option:
Match the binomial probability P(x< 23) with the correct statement.

P(there are at most 23 successes)
P (there are fewer than 23 successes)
P (there are more than 23 successes)
P (there are at least 23 successes)

Question # 7 of 10 (Total Marks: 1) Select correct option:
In statistics, the term 'expected value' implies the ___ value.

Independent
Normal
Standard
Mean (Page 191)

Question # 8 of 10 (Total Marks: 1) Select correct option:
A quantity obtained by applying certain rule or formula is known as

Estimate (Page 264)
Estimator
Parameter
Proportion

Question # 9 of 10 (Total Marks: 1) Select correct option:
Total no. of possible samples of size 2 (without replacement) from the population of size 6, will be:

20
15
10
18

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Question # 10 of 10 (Total Marks: 1) Select correct option:
When a coin is tossed 3 times, the probability of getting 3 or less tails is

1/2 (not sure)
0
1
3/5

Question # 1 of 10 (Total Marks: 1) Select correct option:
Total no. of possible samples of size 3 (with replacement) from the population of size 6, will be:

256
196
216 6^3
325

Question # 2 of 10 (Total Marks: 1) Select correct option:
Using the normal approximation to the binomial distribution with n= 3 and p= 0.0571 the value of mean is:

0.1713 (Page 235)
0.2132
0.5133
0.1923

Question # 3 of 10 (Total Marks: 1) Select correct option:
Suppose 60% of a herd of cattle is infected with a particular disease. Let Y = the number of non-diseased cattle in a sample of size 5. the distribution of Y is:

Binomial with n = 5 and p = 0.6
Binomial with n = 5 and p = 0.4
Binomial with n = 5 and p = 0.5
Poisson with u = .6

Question # 4 of 10 (Total Marks: 1) Select correct option:
The probability of success changes from trial to trial, is the property of:

Binomial experiment (Page 211)
Hypergeometric experiment
Both binomial & hypergeometric experiment
Poisson experiment

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Question # 5 of 10 (Total Marks: 1) Select correct option:
If a statistic used as an estimator, has its expected value equal to the true value of the population parameter being estimated then it is called.....

Consistent
Unbiased (Page 258)
Efficient
Sufficient

Question # 6 of 10 (Total Marks: 1) Select correct option:
In moments method, how many equations are needed to find the 2 unknown parameters?

2 (Page 263)
3
n/2

Question # 7 of 10 (Total Marks: 1) Select correct option:
Which of the following is desirable for a good point estimator?

Consistency
Unbiasedness
Efficiency
All of these (Page 258)

Question # 8 of 10 (Total Marks: 1) Select correct option:
The distribution function (df) is also known as

Probability distribution
Probability mass function
Probability density function
Cumulative distribution function (Page 173)

Question # 9 of 10 (Total Marks: 1) Select correct option:
If X and Y are two discrete r.v's with joint probability function f(x,y), then the conditional probability function Y given X, f(y|x) is given by

f(yj | xi) = f(xi, yj) / g(xi) (Page 195)
f(yj | xi) = f(xi, yj) / h(yj)
f(yj | xi) = f(xi, yj) / Sum of g(xi)
f(yj | xi) = f(xi, yj) / sum of h(yj)

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Question # 1 of 10 (Total Marks: 1) Select correct option:
Which of the probability distributions has three parameters?

Binomial distribution
Normal distribution
Hypergeometric distribution (Page 219)
Poisson distribution

Question # 2 of 10 (Total Marks: 1) Select correct option:
A standard deviation obtained from sampling distribution of sample statistics is known as

Sampling error
Standard error (Page 240)
Minimum error
Universal error

Question # 3 of 10 (Total Marks: 1) Select correct option:
A parameter is a ..... quantity.

Constant
Variable
Sample
Random

Question # 4 of 10 (Total Marks: 1) Select correct option:
How can you define statistical inference?

A decision, estimate, prediction or generalization about the population based on information contained in a sample click here for detail

A statement made about a sample based on the measurements in that sample
A set of data selected from a larger set of data
A decision, estimate, prediction or generalization about sample based on information contained in a population

Question # 5 of 10 (Total Marks: 1) Select correct option:
Which of the following is most important and most widely used method in point estimation?

The method of moments
The method of fractional moments
The method of leas square (Page 263)
The method of maximum likelihood

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Question # 6 of 10 (Total Marks: 1) Select correct option:
Binomial distribution is skewed to the right if:

p=q
P<q (Page 215)
p>q
p=n

Question # 7 of 10 (Total Marks: 1) Select correct option:
In a discrete distribution function, F(23) can be stated as:

P (there are at most 23 successes)
P (there are at least 23 successes)
P (there are less than 23 successes)
P (there are more than 23 successes)

Question # 8 of 10 (Total Marks: 1) Select correct option:
The Probabilities of the various values of the sample satistic can be computed using the..... definition of probability.

Subjective
Objective
Classical (Page 149)
None of the baove

Question # 9 of 10 (Total Marks: 1) Select correct option:
E(10X +3) =_____

10 E(X)
E(X)+3
10 E(X)+3
100E(X)

Question # 10 of 10 (Total Marks: 1) Select correct option:
If p is very small and n is considerably large then we shall apply the:

Binomial distribution
Hypergeometric distribution
Poisson distribution (Page 222)
Exponential distribution

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Question # 1 of 10 (Total Marks: 1) Select correct option:
Which of the following is NOT applicable to a Poisson distribution?

IF P = 0.5 & n = 19
IF P = 0.01 & n = 200
IF P = 0.02 & n = 300
IF P = 0.03 & n = 500

Question # 2 of 10 (Total Marks: 1) Select correct option:
The normal distribution has points of infection which are equidistance from the:

Median
Mean (Page 229)
Mode
Mean,Median & Mode

Question # 3 of 10 (Total Marks: 1) Select correct option:
If a random variable X denotes the number of heads when we toss a fair coin 5 times, the X assumed the values:

0,1,2,3
1, 2,3,4,5
0, 1, 2,3,4,5
1, 5, 5

Question # 4 of 10 (Total Marks: 1) Select correct option:
As a rule of thumb, when n>=30, then we can assume that……is normally distributed:

Probability distribution
Sampling distribution (Page 243)
Binomial distribution
Sampling distribution of sample mean

Question # 5 of 10 (Total Marks: 1) Select correct option:
If b(x, 7, 0.30), the variance of this distribution is:

1.77
1.74
1.44
1.47 (Page 333)

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Question # 6 of 10 (Total Marks: 1) Select correct option:
Which of the following is a characteristic of a binomial probability experiment?

Each trial has more than two possible outcomes
P(success) = P(failure)
Probability of success changes for each trail
The result of one trial does not affect the probability of success on any other trial (Page 211)

Question # 7 of 10 (Total Marks: 1) Select correct option:
The number of parameters in uniform distribution is (are):

1
2 (Page 225)
3
4

Question # 8 of 10 (Total Marks: 1) Select correct option:
The probability can never be:

1
1/2
1
-1/2

Question # 9 of 10 (Total Marks: 1) Select correct option:
The conditional probability P (A\B) is:

P(A n B)/P(B) (Page 159)
P(A n B)/P(A)
P(A U B)/P(B)
P(A U B)/P(A)

Question # 10 of 10 (Total Marks: 1) Select correct option:
A random sample of n=25 values gives sample mean 83. Can this sample be regarded as drawn from a normal population with μ= 80 and s= 7? In this question the alternative hypothesis will be:

H1: μ = 80
H1: μ ≠ 80
H1: μ > 80
H1: μ <80

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Question # 1 of 10 (Total Marks: 1) Select correct option:
The binomial distribution is negatively skewed when:
p>q (Page 215)
p<q
p=q
p=q=1/2

Question # 2 of 10 (Total Marks: 1) Select correct option:
When we draw the sample with replacement, the probability distribution to be used is:
Binomial
Hypergeometric
Binomial & hypergeometric
Poisson

Question # 3 of 10 (Total Marks: 1) Select correct option:
The moment ratios of normal distribution come out to be:
0 and 1
0 and 2
0 and 3 (Page 227)
0 and 4

Question # 4 of 10 (Total Marks: 1) Select correct option:
Suppose the test scores of 600 students are normally distributed with a mean of 76 and standard deviation of 8. The number of students scoring between 70 and 82 is:
272
164
260
328 click here for detail

Question # 5 of 10 (Total Marks: 1) Select correct option:
If P (A) = 0.3 and P (B) = 0.5, find P (A/B) where 'A' and 'B' are independent:

0.3
0.5
0.8
0.15 (Page 162)

Question # 6 of 10 (Total Marks: 1) Select correct option:
If the second moment ratio is less than 3 the distribution will be:
Mesokurtic
Leptokurtic
Platykurtic
None of these

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Question # 7 of 10 (Total Marks: 1) Select correct option:
For the independent events A and B if P (A) = 0.25, P (B) =0.40 then P (A and B) =……
Select correct option:

0.65
0.1 (Page 162)
0.50
0.15

Question # 8 of 10 (Total Marks: 1) Select correct option:
A random variable X has a probability distribution as follows: X | 0 1 2 3 P(X) | 2k 3k 13k 2k What is the possible value of k:

0.01
0.03
0.05
0.07

Question # 9 of 10 (Total Marks: 1) Select correct option:
The probability of drawing any one spade card is:
1/52
4/52
13/52
52/52

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STA301- Statistics and Probability

Final Term Examination - Spring 2012

1- A sample of size n=3 drawn without replacement experiment from a population N=5 items whose values 0,2,3,6,7 draw possible samples

Answer:-

Sample without replacement = $\binom{N}{n} = \binom{5}{3} = 10$

(0,2,3), (0,2,6), (0,2,7), (0,3,6), (0,3,7), (0,6,7), (2,3,6), (2,3,7), (2,6,7), (3,6,7).

2- Calculate sampling error if sample mean is 102 and population mean is 100

Answer:- (Page 14)

Sampling error $= \overline{X} - \mu$

$= 102 - 100$

$= 2$

3- Define Unbiased estimator

Answer:- (Page 258)

An estimator is defined to be unbiased if the statistic used as an estimator has its expected value equal to the true value of the population parameter being estimated. $E(\hat{\theta}) = \theta$

4- Quartile deviation

Answer:- (Page 84)

The quartile deviation is defined as half of the difference between the third and first quartiles.

$$Q.D = \frac{Q_3 - Q_1}{2}$$

دنیا میں سب سے مشکل کام اپنی اصلاح اور سب سے آسان کام دوسروں پر نکتہ چینی کرنا ہے

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STA301- Statistics and Probability

Final Term Examination - Spring 2012

Q1.decribe about significance level

Answer:

The significance level is the criterion used for rejecting the null hypothesis. It show you how likely a result is due to chance. The most frequently used values of α, the significance level, are 0.05 and 0.01, i.e. 5 percent and 1 percent.

Q2 .discuss about center limit theorem

Answer:- (Page 243)

"If a variable X from a population has mean μ and finite variance σ/2, then the sampling distribution of the sample mean−X approaches a normal distribution with mean μ and variance σ2/n as the sample size n approaches infinity.**"**

Q3. What is difference between constant and random variable

Answer:

A variable whose value cannot be changed once it has been assigned a value while those variable whose values changes when we assign value to it.

STA301- Statistics and Probability

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Q: Define an Un-Biased estimator? 2 Marks

Answer: Rep

Q: Briefly Explain an experiment design? 2 Marks

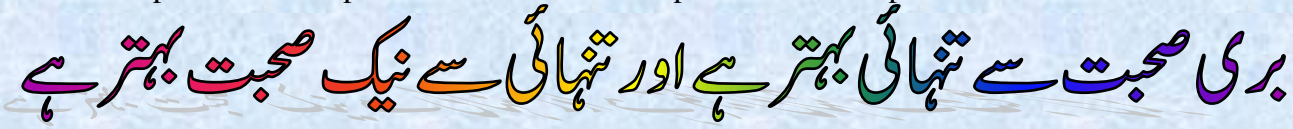
Answer:- (Page 320)

By an experimental design, we mean a plan used to collect the data relevant to the problem under study in such a way as to provide a basis for valid and objective inference about the stated problem.

Q: Define an independent and dependent variable in regression? 2 Marks

Answer:- (Page 121)

In regression Y represents the dependent variable and X represents the independent variable



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Find the mean and variance for the sampling distribution given below. 5 Marks

$(\hat{p})$	No. of Samples	Probability $f(\hat{p})$
0	1	1/20
1/3	9	9/20
2/3	9	9/20
1	1	1/20
Σ	20	1

Answer:- (Page 247)

$\hat{p}$	No. of samples	$f(\hat{p})$	$\hat{p}.f(\hat{p})$	$\hat{p}^2.f(\hat{p})$
0	1	$\frac{1}{20}$	0	0
$\frac{1}{3}$	9	$\frac{9}{20}$	$\frac{3}{20}$	$\frac{1}{20}$
$\frac{2}{3}$	9	$\frac{9}{20}$	$\frac{3}{10}$	$\frac{1}{5}$
1	1	$\frac{1}{20}$	$\frac{1}{20}$	$\frac{1}{20}$
$\sum$	20	1	$\sum \hat{p}.f(\hat{p}) = \frac{1}{2}$	$\sum \hat{p}^2.f(\hat{p}) = \frac{3}{10}$

$$\mu_{\hat{p}} = \sum \hat{p}.f(\hat{p}) = \frac{1}{2} = 0.5$$

$$\sigma^2_{\hat{p}} = \sum \hat{p}^2.f(\hat{p}) - \left[\sum \hat{p}.f(\hat{p})\right]^2$$

$$= \frac{3}{10} - \left[\frac{1}{2}\right]^2$$

$$= \frac{3}{10} - \frac{1}{4}$$

$$= \frac{1}{20} = 0.05$$



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What is the difference between an outcome and an event? (2)

Answer:- (Page 145)

An outcome is a result of a single trial of an experiment while an event is an individual outcome or any number of outcomes.

The mean of a Poisson distribution is 5 while its standard deviation is 4. Comment on it (2)

Answer:- (Page 223)

In Poisson distribution mean and variance are always equal but this statement is not satisfying this property of Poisson distribution.

If an automobile is driven on the average no more than 16000 Km per year, then formulate the null and alternative hypothesis. (2)

Answer:

$H \leq 16000$

$H1 > 16000$

Q: Discuss three properties of normal distribution? 3 Marks

Answer:- (Page 227)

Normal distribution is absolutely symmetrical, hence , μ_3 the third moment about the mean is zero

The normal curve is asymptotic to the x-axis as $x \rightarrow \pm \infty$.

For the normal distribution, it can be mathematically proved that $\mu_4 = 3\sigma_4$

Q: The 90% confidence interval for the population mean is 11 to 20, interpret this result? 3 Marks

Answer: - We are 90% sure that our Population mean lie between 11 – 20.

Q: Define LSD test? 3 Marks

Answer:- (Page 330)

According to this procedure, we compute the smallest difference that would be judged significant, and compare the absolute values of all differences of means with it. This smallest difference is called the least significant difference or

LSD. And is given by

$$LSD = t_{\frac{\alpha}{2},(v)} \sqrt{\frac{2(MSE)}{r}}$$

ایماندار کو غصہ دیر سے آتا ہے اور جلدی دور ہو جاتا ہے

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How many parameters are associated with F- distribution and what is the range of the distribution? (3)
Answer:- (Page 312)
It has two parameters v1 and v2 which are known as the degrees of freedom and it ranging from zero to plus infinity.

Which of the following statement represents continuous data and discrete data? (5)
i) Number of shopes in a plaza. **Discrete data**
ii) Hourly temperature recorded by whether bureau. **Continuous data**
iii) Inches of rainfall in a city. **Continuous data**
iv) Number of passengers carried by rail every year. **Discrete data**
v) Height measurements of boys studying in a college. **Discrete data**

If the population proportions are gives as: P1 = 0.4, P2 = 0.20
find sigma^2 P-hat 1 - P-hat 2 , where n = 12.
Answer:- (Page 256)

$$\sigma_{\hat{p}1-\hat{p}2} = \frac{p_1 q_1}{n_1} + \frac{p_2 q_2}{n_2}$$

$$q_1 = 1 - p_1 \Rightarrow 1 - 0.40 = 0.6$$

$$q_2 = 1 - p_2 \Rightarrow 1 - 0.20 = 0.8$$

$$\sigma_{\hat{p}1-\hat{p}2} = \frac{(0.4)(0.6)}{10} + \frac{(0.20)(0.80)}{10}$$

$$= 0.024 + 0.016$$

$$= 0.04$$

STA301- Statistics and Probability

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what are steps involved in statistical research
Answer: (Page 11)
STEPS INVOLVED IN ANY STATISTICAL RESEARCH
• Topic and significance of the study
• Objective of your study
• Methodology for data-collection
 Source of your data
 Sampling methodology
 Instrument for collecting data

زندگی میں کامیابی کا یہی راز ہے کہ پریشانیوں سے پریشان مت بنو

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STA301- Statistics and Probability

Final Term Examination - Spring 2012

Q. (Marks 5)Two different dice are thrown. Make the sample space and also find the number of sample points in an even A that the sum is 7.

Answer:-

$$S = \begin{pmatrix} (1,1) & (2,1) & (3,1) & (4,1) & (5,1) & (6,1) \\ (1,2) & (2,2) & (3,2) & (4,2) & (5,2) & (6,2) \\ (1,3) & (2,3) & (3,3) & (4,3) & (5,3) & (6,3) \\ (1,4) & (2,4) & (3,4) & (4,4) & (5,4) & (6,4) \\ (1,5) & (2,5) & (3,5) & (4,5) & (5,5) & (6,5) \\ (1,6) & (2,6) & (3,6) & (4,6) & (5,6) & (6,6) \end{pmatrix}$$

$$A = \{(1,6) \quad (2,5) \quad (3,4) \quad (4,3) \quad (5,2) \quad (6,1)\}$$

Q. (Marks 2) Define the error sample and also write its formula.

Answer:- (Page 13)

The difference between the estimate derived from the sample (i.e. the statistic) and the true population value (i.e. the
parameter) is technically called the sampling error

$$\text{Sampling error} = \overline{X} - \mu$$

Q. (Marks 2) Properties of hyper geometric distribution.

Answer:- (Page 218)

- The outcomes of each trial may be classified into one of two categories, success and failure.
- The probability of success changes on each trial.
- The successive trials are not independent.
- The experiment is repeated a fixed number of times

5 number summery

Answer:- (Page 97)

A five-number summary consists of X0,Q1, Median, Q3, and Xm ; It provides us quite a good idea about the shape of the distribution.



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STA301- Statistics and Probability

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1.what s bias?(2)

Answer:- (Page 258)

$E(\hat{\theta}) \neq \theta$, the statistic is said to be a biased estimator

2.advantages and disadvantages of median?(3)

Answer:- Click here for detail

Advantages

(1) It is very simple to understand and easy to calculate. In some cases it is obtained simply by inspection.
(2) Median lies at the middle part of the series and hence it is not affected by the extreme values.
(3) In grouped frequency distribution it can be graphically located by drawing ogives.
(4) It is especially useful in open-ended distributions.

Disadvantages

(1) In simple series, the item values have to be arranged. If the series contains large number of items, then the process becomes tedious.
(2) It is a less representative average because it does not depend on all the items in the series.

3. Mathematical expectation of discrete random variable?(3)

Answer:- (Page 179)

In probability theory the expected value (or mathematical expectation) of a random variable is the sum of the product of the values within the range of the discrete random variable and their respective probabilities of occurrence.

$$E(X) = \sum_{i=1}^{n} x_i f(x_i)$$

4.any two properties of mathematical expectation?(2)

Answer:- (Page 202)

The important properties of the expected values of a random variable are as follows:

• If c is a constant, then E(c) = c. Thus the expected value of a constant is constant itself. This point can be understood easily by considering the following interesting example: Suppose that a very difficult test was given to students by a professor, and that every student obtained 2 marks out of 20! It is obvious that the mean mark is also 2. Since the variable 'marks' was a constant, therefore its expected value was equal to itself.

• If X is a discrete random variable and if a and b are constants, then E(aX + b) = a E(X) + b.

5.what s statistical test?(2)

Answer:- (Page 279)

A statistic, which provides a basis for testing a null hypothesis, is called a test-statistic. Every test-statistic has a probability distribution (i.e. sampling distribution) which gives the probability that our test-statistic will assume a value greater than or equal to a specified value OR a value less than or equal to a specified value when the null hypothesis is true.

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6. Decide a small sample and large sample?(2)

Answer:-

If the sample size 'n' is less than or equal to 30 (<=30) then it is said to be small sample, and it the sample size is larger than 30 then it is said to be large population.

10.tell the null and alternative hypothesis of 150?(2)

Answer:-

$H_0 = \mu = 150$

$H_1 = \mu \neq 150$

STA301- Statistics and Probability

Final Term Examination - Spring 2012

35. In which condition, Poisson distribution is used to approximate the hyper geometric distribution?

Answer:- (Page 224)

The Poisson distribution can be used to approximate the hyper geometric distribution when n < 0.05N, n > 20, and p < 0.05

36. Elaborate the Least Significant Difference (LSD) test.

Answer:- Rep

37. Write down the formula of combined or pooled proportion of two samples.

Answer:- (Page 290)

$$\overline{p_c} = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2}$$

38. If approximate value of class interval is 2.96 and range = 14.8 then find the number of classes.

Answer:- (Page 29)

$$Class\ interval = \frac{range}{number\ of\ classes}$$

$$2.96 = \frac{14.8}{number\ of\ classes}$$

$$number\ of\ classes = \frac{14.8}{2.96} = 5$$

جھوٹ انسان اور ایمان دونوں کا دشمن ہے

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40. Find the coefficient of variation (C.V) for the following price of a commodity.
Price (X): 8, 13, 18, 23, 30

Answer:-

X	**X^2**
8	**64**
13	**169**
18	**324**
23	**529**
30	**900**
$\sum X = 92$	$\sum X^2 = 1986$

$$\overline{X} = \frac{\sum X}{n} = \frac{92}{5} = 18.4$$

$$S = \sqrt{\frac{\sum X^2}{n} - \left(\frac{\sum X}{n}\right)^2}$$

$$S = \sqrt{\frac{1986}{5} - \left(\frac{92}{5}\right)^2}$$

$$S = \sqrt{397.2 - 338.56} = \sqrt{58.64} = 7.66$$

$$C.V = \frac{S}{\overline{X}} \times 100$$

$$C.V = \frac{7.66}{18.4} \times 100 = 41.63\%$$

41. Flaws in plywood occur at random with an average of one flaw per 50 square feet. What is the probability that 32 square feet will have no flaws?

عقل مند کہتا ہے میں کچھ نہیں جانتا جبکہ بے وقوف کہتا ہے کہ میں سب کچھ جانتا ہوں

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2- Probability of one electric device failure is 0.01. If a sample is chosen of 400, what is probability that exactly 2 are defective (Marks 5)

Answer:- (Page 218)

$p = 0.01$

$q = 1 - p \Rightarrow q = 1 - 0.01 \Rightarrow q = 0.99$

$$P(X = 2) = \binom{400}{2}(0.01)^2 (0.99)^{400-2}$$

$$= 79800 * 0.0001 * 0.02$$

$$= 0.1596$$

4- Two dices are rolled. Find probability that outcome is equal or more than 11. (Marks 5)

Answer:-

$$S = \begin{pmatrix} (1,1) & (2,1) & (3,1) & (4,1) & (5,1) & (6,1) \\ (1,2) & (2,2) & (3,2) & (4,2) & (5,2) & (6,2) \\ (1,3) & (2,3) & (3,3) & (4,3) & (5,3) & (6,3) \\ (1,4) & (2,4) & (3,4) & (4,4) & (5,4) & (6,4) \\ (1,5) & (2,5) & (3,5) & (4,5) & (5,5) & (6,5) \\ (1,6) & (2,6) & (3,6) & (4,6) & (5,6) & (6,6) \end{pmatrix}$$

$$P = (5,6),(6,5),(6,6)$$

$$= \frac{3}{36} = \frac{1}{12} = 0.83$$

5- If X is binomial distribution n=5, p=0.5, q=0.5 then find SD(X) (Marks 3)

Answer:- (Page 214)

$$S.D(X) = \sqrt{npq}$$

$$= \sqrt{5 \times 0.5 \times 0.5} = \sqrt{1.25}$$

$$= 1.12$$

خود کو تمھیں سے بڑھ کر کوئی اچھا مشورہ نہیں دے سکتا

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6- n=24, Mean=33, s=15, x=40.4. Compute t statistic (Marks 3)

7- 90% confidence interval of population mean is 11 to 20. Interpret result (Marks 3)
Answer:- Rep

8- E(XY)=421, E(X)=42, E(Y)=15. Check independence (Marks 3)
Answer:-

$$Cov(X,Y) = E(XY) - E(X)E(Y)$$
$$= 421 - (42)(15)$$
$$= -209$$

It is not independent.

9- Difference between statistics and statistic (Marks 3)
Answer:-
Statistics is the plural of statistic, the science that deals with the collection, classification, analysis, and interpretation of numerical facts or data. While the statistic is a single value or piece of data.

10- Why we call standard deviation the standard error (Marks 2)
Answer:- (Page 240)
The square root of the variance is the standard deviation, and the standard deviation of a sampling distribution is termed as its standard error.

11- if E(Y)=0.5 then find E(10.5+2Y) ... (Marks 2)
Answer:-

$$= E(10.5 + 2Y)$$
$$= 10.5 + 2E(Y)$$
$$= 10.5 + 2 * 0.5$$
$$= 11.5$$

12- What is the meaning of 'b' in Y=a+bX (Marks 2)
Answer:- (Page 121)
b represents the slope of the line

13- Explain level of significance (Marks 2)
Answer:- rep

جو شخص ناکامیوں سے ڈر کر بھاگتا ہے کامیابی اُس سے ڈر کر بھاگتی ہے

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Q# 28: can all deciles be expressed as percentile? Explain (2)
Answer: Yes all deciles cab be expressed as percentiles
As the D1 is 10 percent of data and P10 is 10% of data.
D2 is 20% of data whereas in percentile it is P20 and so on …

Q# 29: what is meant by sampling distribution? (2)
Answer: Page 237
The probability distribution of any statistic (such as the mean, the standard deviation, the proportion of successes in a sample, etc.) is known as its sampling distribution.

Q# 30: The department claims that the exceeds Rs. 2500 at the 0.05 level, then formulate null alternative hypothesis?(2)
Answer: Page 283
$H_0 : \mu < 2500$
$H_1 : \mu > 2500$

Q# 31: How we decide that the drawn sample is "Small Sample " or a "Large Sample"?(2)
Answer: rep

Q# 32: If E(X)=0.7 then find E(2X)? (2)
Answer:
$E(2X) = 2E(X)$
$= 2 * 0.7$
$= 1.4$

Q# 33: State the Baye's theorem? (3)
Answer: Page 166
If events A1, A2… Ak form a PARTITION of a sample space S (that is, the events Ai are mutually exclusive and exhaustive (i.e. their union is S)),
and if B is any other event of S such that it can occur ONLY IF ONE OF THE Ai OCCURS, then for any i,

$$P(A_i / B) = \frac{P(A_i)P(B / A_i)}{\sum_{i=1}^{k} P(A_i)P(B / A_i)}$$

Q# 35: "The 95% confidence interval for population mean is 1.3 to 4.7". Interrupt this result (3)
Answer: We are 95% sure that the population mean lies between 1.3 to 4.7

جو لوگوں کے سامنے فخر کرتا ہے وہ لوگوں کی نظروں سے گر جاتا ہے

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Q# 37: If E(XY)=045 E(X)=0.50 and E(Y)=0.90 then X and Y are independent?(3)

Answer:

$$Cov(X,Y) = E(XY) - E(X)E(Y)$$
$$= 0.45 - (0.50)(0.90)$$
$$= 0.45 - 0.45$$
$$= 0(\text{Therefore it is independent})$$

STA301- Statistics and Probability

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2 Marks

2) If we draw a card from an ordinary deck of 52 cards. Can king and diamond be mutually exclusive events? Give reason to support your answer.

Answer:- (Page 146)

No, because If the two events *can* occur at the same time, they are not mutually exclusive, Therefore, kings and diamonds are not mutually exclusive.

3) Suppose that we toss a fair coin three times. What is its sample space?

Answer:- {(H,H,H),(H,T,H),(H,H,T),(T,H,H),(T,T,H),(T,H,T),(H,T,T),(T,T,T)}

4) Why we take B_2 =3 as the criteria for measuring the kurtosis of any distribution?

Answer:- (Page 228)

For the normal distribution, β2 comes out to be 3, this is why this value has been taken as a criterion for measuring the kurtosis of any distribution.

5) Define level of significance?

Answer:- rep

6) How we decide that the drawn sample is "Small sample" or a "large sample"?

Answer:- Rep

عقل مند اپنے عیب خود دیکھتا ہے اور بیوقوفوں کے عیب دنیا دیکھتی ہے

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3 Marks

7) If mean of a distribution is 0.925 and standard deviation is 0.132. Find out the limits by applying chebychev's inequality for k=2. How much fraction of data will lie between the two limits?

Answer:- (Page 95)

$$\left(\overline{X}-kS,\overline{X}+kS\right)$$

$$\left(0.925-2(0.132),0.925+2(0.132)\right)$$

$$\left(0.925-0.264,0.925+0.264\right)$$

$$\left(0.661,1.189\right)$$

95% of the measurements will fall within these limits.

8) If E(XY)=7.5, E(X)=2.4 and E(Y)=4.3, calculate covariance of X and Y.

Answer:-

$$\begin{aligned} Cov(X,Y) &= E(XY)-E(X)E(Y) \\ &= 7.5-(2.4)(4.3) \\ &= 7.5-10.32 \\ &= -2.82 \end{aligned}$$

11) Write down the two properties of sampling distribution of proportion p', when the sampling is performed with replacement?

Answer:- (Page 241)

Property 1: $\mu_{\overline{X}} = \mu$

Property 2: In case of sampling with replacement:

$$\sigma_X = \frac{\sigma}{n}$$

12) What is natural pairing in observations? Give example.

Answer:- (Page 302)

Natural pairing occurs whenever measurement is taken on the same unit or individual at two different times. For example, suppose ten young recruits are given a strenuous physical training programme by the Army. Their weights are recorded before they begin and after they complete the training. The two observations obtained for each recruit i.e. the before-and-after measurement constitute natural pairing. The above is natural pairing.

بد صورت چہرہ بد صورت دماغ سے بہتر ہے

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5 Marks

13) Interpret the concept of five numbers summary and also explain the purpose of five number summaries.

Answer:- (Page 97)

A five-number summary consists of X0,Q1, Median, Q3, and Xm ; It provides us quite a good idea about the shape of
the distribution
If the data were perfectly symmetrical, the following would be true:
The distance from Q1 to the median would be equal to the distance from the median to Q3
The distance from X0 to Q1 would be equal to the distance from Q3 to Xm.
The median, the mid-quartile range, and the midrange would all be equal. All these measures would also be equal to the arithmetic mean of the data

On the other hand, for non-symmetrical distributions, the following would be true:
In right-skewed distributions the distance from Q3 to Xm greatly exceeds the distance from X0 to Q1
in right-skewed distributions,
median < mid-quartile range < midrange:
THE POSITIVELY SKEWED CURVE

14) What is the probability that a poker hand of 5 cards contain exactly 2 aces (hyper geometric distribution)?

Answer:-

$$P(X=2)=\frac{\binom{4}{2}\binom{48}{3}}{\binom{52}{5}}$$

$$=\frac{6\times 17296}{2598960}=\frac{103776}{2598960}=0.0399$$

16)From the table given below find the value of chi square

Obervation Frequency Oij	Expected Frequency Eij
120	100
130	150
80	100
170	150

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the mean of a population is 6 a sample of size '20' is taken with replacement from this population.which one of the following is the mean of sampling distribution of mean.
1) 2 2) 8 3) 6 4) 3
Answer:-
6

Q no 2 Write down the significance of experimental design? marks :2
Answer:- Rep

Q No: 03 Explain the formula of hyper geometric probability distribution. marks :2
Answer:- (Page 219)

$$P(X=x)=\frac{\binom{k}{n}\binom{N-k}{n-x}}{\binom{N}{n}}$$

Where
N = number of units in the population,
n = number of units in the sample, and
k = number of successes in the population.
The hyper geometric probability distribution has three parameters N, n and k.

Q no 5
Suppose that we toss a fair coin three times. What is the sample space? marks :2
Answer:- Rep

Compute the following formula which is used to find the sample size.
Answer:- For finding sample space of coin the formula is: $2^n = 2^3 = 8$

If x is a passion random variable with parameter value '4.5', find p(x=1)? Marks: 03
Answer:- (Page 222)

$$P(X=1)=\frac{e^{-4.5}(4.5)^1}{1!}$$
$$=0.0499$$

عقل مند آدمی اس وقت تک نہیں بولتا جب تک خاموشی نہیں ہو جاتی

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Q no: 09
When the value of chi-square is equal to zero? Marks : 03
Answer:- (Page 338)
If the observed frequencies are exactly equal to the expected ones, then chi-square will be exactly equal to zero.

Q no: 10 If x= 255, n=634, Po=0.60 then find the z-test statistics for probability. Marks: 03
Answer:- (Page 289)

$$Z = \frac{X \pm \frac{1}{2} - np_0}{\sqrt{np_0(1-p_0)}}$$

$$np_0 = 634 * 0.60 = 380.4$$

$$here\ X < np_0\ therefore\ we\ use\ X + \frac{1}{2}$$

$$Z = \frac{X + \frac{1}{2} - np_0}{\sqrt{np_0(1-p_0)}}$$

$$Z = \frac{225 + \frac{1}{2} - 380.4}{\sqrt{380.4(1-0.60)}}$$

$$Z = \frac{-154.9}{12.34}$$

$$Z = -12.55$$

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Describe the formula for hypergeometric distribution? 2 marks
Answer:- Rep
What is graphical representation? 2 marks
Answer:-
The plot of the points in the plane which constitute the graph of a given real function or a pictorial diagram depicting interdependence of variables.

انسان دکھ نہیں دیتے بلکہ انسانوں سے وابستہ امیدیں دکھ دیتی ہیں

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What is the t formula for paired distribution? 3marks.

Answer:- (Page 302)

$$t = \frac{\bar{d} - \mu_d}{s_d / \sqrt{n}}$$

What is the probability that at least one head occurs if a coin tossed 6 times successively? 5 marks

Answer:-

$$P(X = 1) = \binom{6}{1} \cdot \left(\frac{1}{2}\right)^1 \cdot \left(\frac{1}{2}\right)^{6-1}$$

$$= \binom{6}{1} \cdot \left(\frac{1}{2}\right)^1 \cdot \left(\frac{1}{2}\right)^5$$

$$= 3.03$$

Find first 2 moments for function with respect to mean? 5 marks

Answer:- (Page 192)

16)From the table given below find the value of chi square

Obervation Frequency Oij	**Expected Frequency Eij**
120	**100**
130	**150**
80	**100**
170	150

Answer:- Rep

STA301- Statistics and Probability

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Q no 5

Suppose that we toss a fair coin three times. what is the sample space? marks :2

Answer: Rep

بہترین تجربہ وہ ہے جس سے نصیحت حاصل ہو

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Compute the following formula which is used to find the sample size.

$$n = \left(\frac{Z_{\alpha/2}}{e} \right)$$

Page 276

FINALTERM EXAMINATION
Spring 2010
STA301- Statistics and Probability (Session - 4)

Question No: 31 (Marks: 2)
What is the mean and variance of Poisson distribution?
Answer:- (Page 233)
If the random variable X has a Poisson distribution with parameter μ, then its mean and variance are given by E(X) = μ and Var(X) = μ.

Question No: 32 (Marks: 2)
Explain the Chi-square test of goodness of fit.
Answer:- (Page 332)
The chi-square test of goodness-of-fit is a test of hypothesis concerned with the comparison of observed frequencies of a sample, and the corresponding expected frequencies based on a theoretical distribution.

Question No: 33 (Marks: 2)
Define level of significance?
Answer:- Rep

Question No: 34 (Marks: 3)
Find the value of F(table value), when $n_1 = 7$, $n_2 = 10$ and $\alpha = 0.05$
Answer:- (Page 316)

Question No: 35 (Marks: 3)
Define Null and Alternative hypothesis.
Answer:- Rep

خوبصورتی علم وادب سے ہوتی ہے لباس و حسن سے نہیں

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Question No: 36 (Marks: 3)
A random sample of 100 is taken from a population with mean 30 and standard deviation 5. The probability distribution of the parent population is unknown, find the mean and standard error of the sampling distribution of $\bar{X}$.

Question No: 37 (Marks: 5)
What is the probability that a poker hand of 5 cards contain exactly 2 aces (hypergeometric distribution)?
Answer:- Rep

Question No: 38 (Marks: 5)
A random sample of size n is drawn from normal population with mean 5 and variance σ^2.
If n=25, s=10 and t=2, what is the values of $\bar{x}$?

Question No: 39 (Marks: 5)
Describe the main steps of General Procedure for Testing Hypothesis.
Answer:- (Page 281)
Step-1:
Formulation of the Null and Alternative Hypotheses:
Step-2:
Decision Regarding the Level of Significance
Step-3:
Test Statistic (that statistic that will enable us to test our hypothesis):
Step-4:
Calculations:
Step-5:
Critical Region (that portion of the X-axis which compels us to reject the null hypothesis):
Step-6:
Conclusion:

FINALTERM EXAMINATION
Spring 2010
STA301- Statistics and Probability (Session - 4)

Question No: 31 (Marks: 2)
How many parameters are involved in hyper geometric distribution?
Answer:- (Page 291)
The hyper geometric probability distribution has three parameters N, n and k.

تم اچھا کرو زمانہ تم کو برا سمجھے یہ اس سے بہتر ہے کہ تم برا کرو اور زمانہ تم کو اچھا سمجھے

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Question No: 32 (Marks: 2)
If an automobile is driven on the average no more than 16000 Km per year, then formulate the null and alternative hypothesis.
Answer:- Rep

Question No: 33 (Marks: 2)
Write down the test statistic when chi- square goodness of fit test is performed.
Answer:- (Page 334)

$$\chi^2 = \sum_i \frac{(o_i - e_i)^2}{e_i}$$

Question No: 34 (Marks: 3)

Find the value of F(table value), when $n_1 = 7$, $n_2 = 10$ and $\alpha = 0.05$
Answer:- (Page 316)
F0.05(n1-1, n2-1) = (6,9) =3.37
F0.05(n2-1, n1-1) = (9,6)
V2=6 , v1=8 => 4.15,
V2=6, v1=12 => 4.00
Mean = (4.15+4)/2 = 4.075
Hence value of F0.05(9,6) = 4.075

Question No: 35 (Marks: 3)

If X = 327, n = 634, $p_0 =$ 0.50 then find the z-test statistic for proportion.
Answer:- (Page 289)

$$Z = \frac{X \pm \frac{1}{2} - np_o}{\sqrt{np_o(1-p_o)}}$$

$$np_o = 634 * 0.50 = 317$$

$$here\ X > np_o\ therefore\ we\ use\ X - \frac{1}{2}$$

$$Z = \frac{X - \frac{1}{2} - np_o}{\sqrt{np_o(1-p_o)}}$$

$$Z = \frac{327 - \frac{1}{2} - 317}{\sqrt{317(1-0.50)}}$$

$$Z = \frac{9.5}{12.58}$$

$$Z = 0.75$$

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Question No: 36 (Marks: 3)
If population proportions are given as:

$$P_1 = 0.30,\ P_2 = 0.20.$$

Find $\sigma^2_{\hat{p}_1 - \hat{p}_2}$,where n = 10

Answer:- (Page 256)

$$\sigma_{\hat{p}1-\hat{p}2} = \frac{p_1 q_1}{n_1} + \frac{p_2 q_2}{n_2}$$

$$q_1 = 1 - p_1 \Rightarrow 1 - 0.30 = 0.7$$

$$q_2 = 1 - p_2 \Rightarrow 1 - 0.20 = 0.8$$

$$\sigma_{\hat{p}1-\hat{p}2} = \frac{(0.30)(0.7)}{10} + \frac{(0.20)(0.80)}{10}$$

$$= 0.021 + 0.016$$

$$= 0.037$$

Question No: 37 (Marks: 5)
A candidate for mayor in a large city believes that he appeals to at least 10 per cent more of the educated voters than the uneducated voters. He hires the services of a poll-taking organization, and they find that 62 of 100 educated voters interviewed support the candidate, and 69 of 150 uneducated voters support him at the 0.05 significance level.
Answer:- (Page 292)
Step-1:
The null and alternative hypothesis is
H0 : p1 – p2 > 0.10, and
H1 : p1 – p2 < 0.10, where p1 = proportion of educated voters, and p2 = proportion of uneducated voters.

Step-2:
Level of Significance:
α = 0.05.

انسان کے لئے بری صحبت سے بڑھ کر بری کوئی چیز نہیں

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Step-3:

Test Statistic:

$$Z = \frac{(\hat{p}_1 - \hat{p}_2) - 0.10}{\sqrt{\frac{\hat{p}_1\hat{q}_1}{n_1} + \frac{\hat{p}_2\hat{q}_2}{n_2}}}$$

which for large sample sizes, is approximately standard normal.

Step-4:

Computation:

$$\text{Here} \quad \hat{p}_1 = \frac{62}{100} = 0.62, \text{ so that } \hat{q}_1 = 0.38,$$

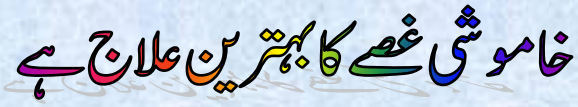
$$\hat{p}_2 = \frac{69}{150} = 0.46, \text{ so that } \hat{q}_2 = 0.54.$$

$$\text{Thus} \quad z = \frac{(0.62 - 0.46) - 0.10}{\sqrt{\frac{(0.62)(0.38)}{100} + \frac{(0.46)(0.54)}{150}}}$$

$$= \frac{0.06}{\sqrt{0.002356 + 0.001656}} = \frac{0.06}{0.063} = 0.95.$$

Step-5:
Critical Region:
As this is a one-tailed test, therefore the critical region is given by
Z < -z0.05 = -1.645

Step-6:
Conclusion:
Since the calculated value z = 0.95 does not fall in the critical region, so we accept the null hypothesis H0 : p1 – p2 > 0.10.The data seems to support the candidate's view.
Until now, we have discussed in considerable detail interval estimation and hypothesis-testing based on the standard normal distribution and the Z-statistic.
Next, we begin the discussion of interval estimation hypothesis-testing based on the t-distribution.



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Question No: 38 (Marks: 5)
If we have RCBD with MSE=3.19, no.of.treatments = 4, no.of.blocks = 5; then find the value of LSD (least significant difference) for treatments by using α=0.05 and error degrees of freedom is 12.
Answer:- Rep

Question No: 39 (Marks: 5)
Find the mean and variance for the sampling distribution given below.

$(\hat{p})$	No. of Samples	Probability $f(\hat{p})$
0	1	1/20
1/3	9	9/20
2/3	9	9/20
1	1	1/20
Σ	20	1

Answer:- Rep

FINALTERM EXAMINATION
Spring 2010
STA301- Statistics and Probability (Session - 3)

Question No: 31 (Marks: 2)
Define normal distribution.
Answer:- (Page 226)
A continuous random variable is said to be normally distributed with mean μ and standard deviation σ if its probability
density function is given by

$$f(x)=\frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{1}{2}\left[\frac{x-\mu}{\sigma}\right]^2}$$

Question No: 32 (Marks: 2)
Fill up the missing values in the formula.

$$n=\frac{(\quad_{\alpha/2})^2\ \hat{p}\hat{q}}{\quad}$$

جھوٹ رزق کو کھا جاتا ہے

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Answer:- (Page 277)

$$n = \frac{\left(z_{\alpha/2}\right)^2 \hat{p}\,\hat{q}}{e^2}$$

Question No: 33 (Marks: 2)
Explain the Chi-square test of goodness of fit.
Answer:- Rep

Question No: 34 (Marks: 3)
What is natural pairing in observations? Also give an example.
Answer:- Rep

Question No: 35 (Marks: 3)
If population proportions are given as:

$$P_1 = 0.30,\ P_2 = 0.20.$$

Find $\sigma^2_{\hat{p}_1 - \hat{p}_2}$,where n = 10

Answer:- Rep

Question No: 36 (Marks: 3)
Explain the method of Maximum Likelihood in Point Estimation?
Answer:- Page 263
The method of maximum likelihood is regarded as the *MOST important* method of estimation, and is the *most* widely used method. This method was introduced in 1922 by Sir Ronald A. Fisher (1890-1962).The mathematical technique of finding Maximum Likelihood Estimators is a bit *advanced*, and involves the concept of the Likelihood Function.

Question No: 37 (Marks: 5)
The following data was obtained for a randomized block design involving five treatments and three blocks SST=430, SSTR=310, SSB=85, Setup the ANOVA table.
Answer:- (Page 332)

Source	d.f	SS	MS
Between treatments	4	430	107.5
Between Blocks	2	85	42.5
Error	8	310	38.75
Total	14	825	188.75

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Question No: 38 (Marks: 5)
In a Standard Normal Distribution, find:
Mean and Standard Deviation
Lower Quartile
Upper quartile
Inter-quartile range
Mean Deviation

Answer:-
In normal distribution

i. Mean and Standard Deviation
Mean is 0 and S.D is 1

ii. Lower Quartile:
$Q_1 = \mu - 0.6745\sigma$
=0-0.6745(1)
Q_1 **= -0.6745**

iii. Upper Quartile:
$Q_3 = \mu + 0.6745\sigma$
=0.6745

iv. Inter-quartile range
$Q_3 - Q_1/2$ **= 0.6745 - (-0.6745)/2 = 0.6745**

v. Mean Deviation
= 0.7979 σ
= 0.7979(1)
= 0.7979

Question No: 39 (Marks: 5)
If $n = 13, \bar{x} = 34, \sigma^2 = 70, \alpha = 0.10$
Test the hypothesis $\mu > 31$

افضل انسان وہ ہے جو اپنی اصلاح کی کوشش کرتا ہے

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Answer:-

$H_0 = \mu = 30$

$H_1 = \mu > 30$

Level of significance α =0.10

α /2=0.05

Critical region=z>1.645

$$z = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

$$Z = \frac{34 - 31}{2.32}$$

Z=1.29

Conclusion: Since the value Z=1.29 is less than the value of critical region so we accepted H_0 at 0.10 Level of significance.

FINALTERM EXAMINATION
Fall 2009
STA301- Statistics and Probability (Session - 4)

Question No: 21 (Marks: 1)
Write down the formula for binomial distribution.

Answer:- (Page 212)

$$P(X = x) = \binom{n}{x} p^x q^{n-x}$$

Question No: 22 (Marks: 2)
Write down the formula for testing the equality of two population proportions.

Answer:- (Page 290)

$$Z = \frac{(\hat{p}_1 - \hat{p}_2) - 0}{\sqrt{\hat{p}_c \hat{q}_c \left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

اطمینان قلب چاہتے ہو تو حسد سے دور رہو

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Question No: 23 (Marks: 3)
Define moment ratios. In which unit they are expressed?

Answer:- (Page 119)

$$b_1 = \frac{(m_3)^2}{(m_2)^3} \quad and \quad b_2 = \frac{m_4}{(m_2)^2}$$

They are independent of origin and units of measurement, i.e. they are pure numbers.

ہر چیز کی ایک پہچان ہوتی ہے اور عقلمند کی پہچان غور و فکر کرنا ہے اور غور و فکر کی پہچان خاموشی ہے

اس سے پہلے کہ تمہیں شہوت فتنے میں ڈالے نکاح کر لو

اپنی مرضی اور اللہ کی مرضی میں فرق کا نام غم ہے

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